Autonomous Mobile Robot
USER MANUAL

P200

Integration Development Guidelines

Context

PREFACE

P200 is an indoor autonomous mobile robot with high degree of flexibility, which can be integrated with collaborative robots, lift deck modules, roller conveyor and other actuators. The robot is designed with a long battery runtime, easy deployment and the flexibility to scale up.

P200 Appearance Main View

- **THE TARGET AUDIENCE**

  This manual is targeted for:
- Electrical engineer
- Installation and Commissioning Engineer
- Technical Support Engineer
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➢ DOCUMENT HISTORY

REVISON DATE	CONTENT	REVISION VERSION	HARDWARE VERSION	YOUIFLEET	YOUIPilot
2022-06-27	Created	V2.0	2.0.1	YOUICompass4.3	3.3
This document is created based on the content of P200-2 200kg Differential Mobile Robot Chassis Quick Start Manual V3.1 and adds content related to integrated design guidance in Chapter 8					
2022-07-01	Revision	V2.1	2.0.1	YOUICompass4.3	3.3
Revise the content of Chapter 8 and add relevant wiring reference circuit diagrams					
2022-07-10	Revision	V2.2	2.0.5-1	YOUICompass4.7	3.7
Revise the hardware corresponding content, software version, and supplement the "Register" related content in Chapter 8					
2022-07-20	Revision	V2.3	2.0.5-1	YOUICompass4.7	3.7
Added content related to integrated reference examples in Chapter 9, supplemented content related to emergency stop status output points in Chapter 8, and corrected known errors.					
2022-09-29	Revision	V2.4	2.0.6-1	YOUICompass4.8	3.8
Fix some known errors					
2022-10-17	Revision	V2.5	2.0.6-1	YOUICompas	3.8

				s4.8	
Rewrite section 8.3.4 of Chapter 8 to fix serious errors and fix other known errors.					
2022-10-21	Revision	V2.6	2.0.6-1	YOUICompass4.8	3.8
Corrected the erroneous content in section 3.6.2 in Chapter 3, and added the content about brake release in section 3.6.4					
2022-11-29	Revision	V2.7	2.0.6-1	YOUICompass4.8	3.8
Corrected pictures and descriptions related to battery replacement, and corrected known errors					
2023-03-06	Revision	V2.8	2.0.6-1	YOUICompass4.8	3.8
Corrected 2.4 light color, corrected 3.6.3 remote control emergency stop operation, corrected known errors					
2023-12-15	Revision	V2.9	2.0.6-1	YOUICompass4.8	3.8
Corrected 1.4 software version description, corrected 4.1 content, added radar division area description, corrected 7.1 network description, corrected known errors, corrected 8.3.2 external network segment configuration, corrected document appearance picture					

CHAPTER1
PRODUCT INTRODUCTION

1.1 CONCEPT DEFINITION

AMR/AGV/Robot - P200 mobile robot

To simplify the introduction, the P200 mobile robot is defined as the robot.

Platform – Platform of P200 mobile robot

The AMR is equipped with a load-bearing cover plate on top. Top compartment is provided with screw holes for quick mounting, integration and safety interfaces under a cover plate. See Figure 1.1.

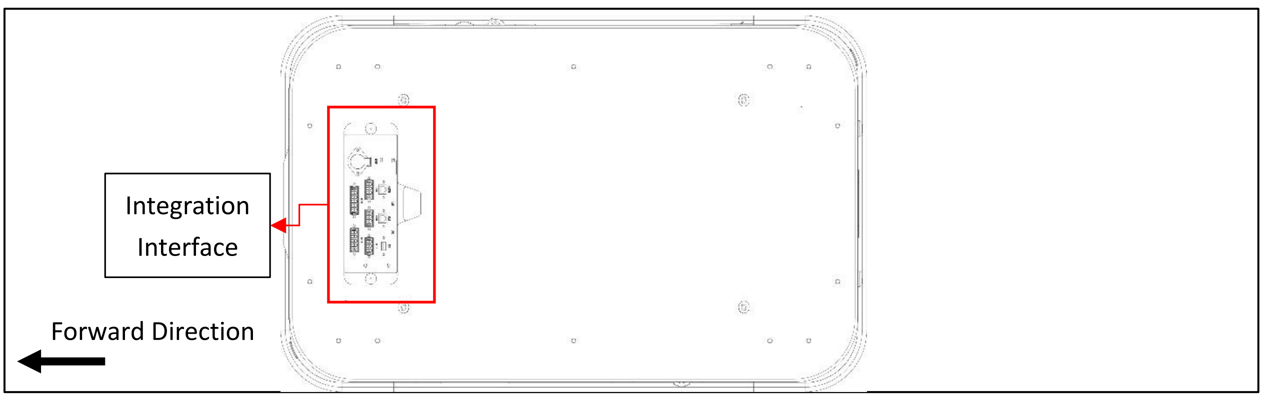


Figure 1.1 Top view of P200 mobile robot platform

Top module integration – top modules and their control components mounted on the platform

All the components installed on the platform including but not limited to the cobots, roller conveyor, lifting module, gripper and other modules and their control components, collectively referred to as the top modules.

1.2 BASIC PERFORMANCE LIST

Category	Item	Parameter	Description
Basic Specification	Size	1020*655*280mm	
	Material	Cold Rolled Steel	
	Main Color	White ral9003, black ral9011	

	Weight	180KG (With battery)	
Payload	Payload	≤200KG	
	Payload size limit	Length ≤1014mm、Width≤650mm Height≤1800mm The height of center of gravity ≤600mm	It is recommended that the payload size doesn't exceed the length and width of robot base and the payload height doesn't exceed 1/3 of robot base height.
Mobile Performance	Travers ability (no load)	15mm	Speed needs to be at 0.4m/s when crossing steps, and it will affect the lifespan of transmission parts if the robot passes gullies at a higher speed. You may not be able to cross these steps at a lower speed. Steps are recommended to have a smooth circular outline.
	Ability To Cross Gullies	30mm	
	Climbing Ability	4 °	
	Maximum Speed	1.5 m/s	
	Maximum Rotation Speed	180 °/s	
Power Supply	Battery	Lithium Iron Phosphate Battery	
	Voltage	48V	
	Battery Capacity	60Ah	
	Runtime	11H	
	Charging Time	≤2H	
	Battery Life	Cycle Life ≥2000 Times, After 2000 Times Retain Capacity 80%	
	Charging Method	Automatic/Manual Charging	The battery is replaceable.
	External Power Supply	Max 48V	
Safety Features	Safety Lidar	Two lidars located diagonally on the robot base	

	Protective Bumper	Around The AMR	
	Status Indicator	Located around the robot base	
	Voice Alarm	Optional	
	Emergency Stop Button	Two, on both sides of the AMR	

1.3 ELECTRICAL INTERFACES

The electrical interfaces in the top compartment include the power supply interface, network interface, and other functional interfaces as shown in Figure 1.2. Its functional definition is summarized in Table 1.1.

Table 1.1 Overview of extension interface function definitions

Port	Type	Mark	Description
POWER	Aviation Connector	Power Supply Interface of Top module	It supplies power to the top module, including 1 channel 48VDC and 1 channel 24VDC
CANOPEN	Rj45	CAN Bus Interface	Robot base CAN bus communication interface, support system diagnosis
WLAN	Rj45	Ethernet Interface (External Network)	Robot base external network segment Ethernet communication interface
LAN1	Rj45	Ethernet Interface (Intranet)	Robot base internal network segment Ethernet communication port, vertical direction, can be noticed from the side.
LAN2	Rj45	Ethernet Interface (Intranet)	Robot base internal network segment Ethernet communication port, vertical direction, can be noticed from the side.
USB	USB 3.0 Type - A	USB Serial Port	Robot base serial communication interface
1XT10	TE Connectivity Connector	10p	System safety loop Serial interface, system output terminal
2XT10	TE Connectivity Connector	10p	System safety loop serial interface, input terminal of top module
1XT12	TE Connectivity Connector	12p	System status extension output interface-1
1XT16	TE Connectivity Connector	16P	System status extension output interface-2
1XT20	TE Connectivity Connector	20P	Register extension interface

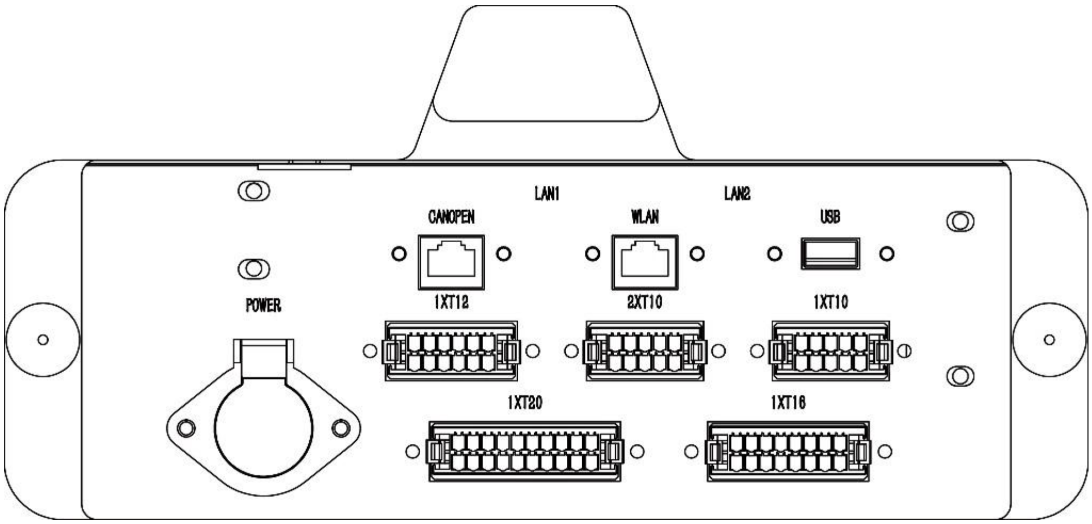


Figure 1.2 Top view of the extension interface

1.4 SYSTEM INSTALLED LIST

Some systems and functions are already included with the chassis. The system list and corresponding versions are shown in Table 1.2.

Table 1.2 List of software included with the chassis

System Name	Version	Description
YOUIPilot	3.8	Single robot control system, including robot navigation, etc.
YOUICompass	4.8	Single robot control system, including robot configuration, map recording, etc.

1.5 LIST OF COMPLEMENTARY SOFTWARE FOR AMR

Some systems and functions are built in with the AMR. The system list and corresponding versions are shown in Table 1.2.

Table 1.3 List of required engineer

Name	The Required Level	Note
Electrical Engineer	Intermediate level or above	
Mechanical Design Engineer	Intermediate level or above	

Network Engineer	Elementary level or above	

1.6 ENVIRONMENT REQUIRED FOR NORMAL OPERATION

In order for the robot to operate normally, the operating environment must meet certain conditions, as shown in **Table 1.4.**

Table 1.4 Operating environment requirements table

Category	Item	Parameter	Description
Basic Conditions	Operating Temperature	0 to 40 ° C	
	Environmental Humidity	Relative humidity 10 to 90% (No condensation)	
	Operating Environment	Only for indoor use, no excessive dust, no corrosive gas	
Floor Conditions	Slip Coefficient	⩾0. 5	
	Floor Condition	Concrete level ground (free of water, oil and dust)	
	Minimum Floor Flatness	FF25 (*ACI 117 standard)	ACI 117 is the American Concrete Institute standard for concrete flooring. FF is flatness and FL is levelness. Flatness figure is bigger, floor is more level off.
Network Conditions	Wireless Network	802.11a/b/g/n 802.11i Wireless Security	Support the 802.11 a/b/g/n protocol authenticated by the Wi-Fi alliance

CHAPTER2 SAFETY MEASURES AND EMERGENCY TREATMENT

2.1 SAFETY WARNING

Improper operation may damage the robot, or may even lead to injuries. Please read the operation manual carefully before operating the robot.

Extension ports for safety controls such as emergency stop button and safety status indicators are reserved for top module integration. Please make sure that security-related interfaces are used correctly and properly.

2.2 EMERGENCY RESPONSE

In the event of an emergency, pressing the emergency stop button will stop all movements of the robot. Emergency shutdowns are not intended to be used as risk reduction measures, but as secondary protection devices.

There are emergency stop buttons on both sides of the chassis, which must be pressed in case of danger or emergency, as shown in Figure 2 1. Rotate the button in the direction shown by the arrow on the button to release it. After releasing it, you need to press the reset button to reset it before it can be used normally.

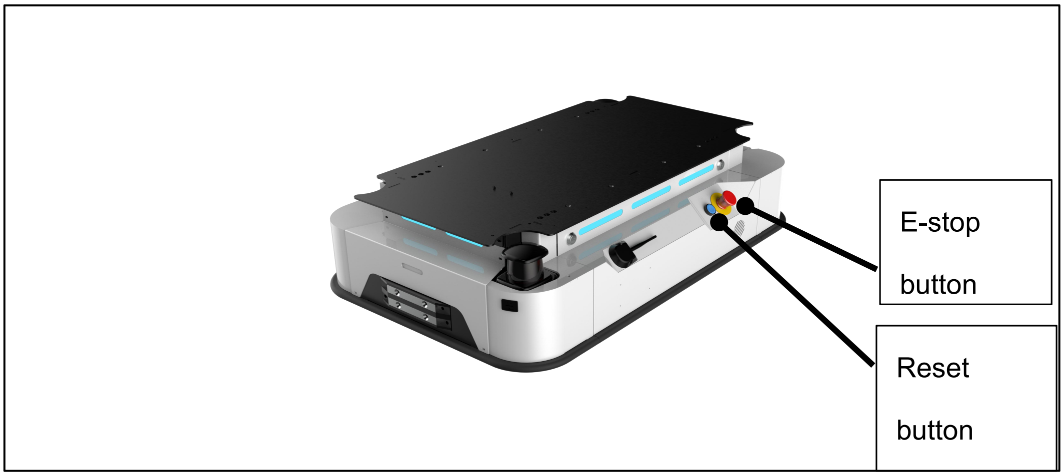


Figure 2.1 Emergency buttons on AMR

The emergency stop buttons are located on both sides of the robot in the shape of a mushroom with distinctive red color. In case of emergency, the button can be tapped to trigger an immediate stop.

2.3 APPLICATIONS

P200 is an industrial mobile robot designed for indoor applications. It is intended to be used for industrial logistics purposes such as material handling.

Any uses fall outside of its design should be strictly prohibited, including but not limited to:

- Usage in potentially explosive environments;
- Medical and life-related application;
- Applications carried out without risk assessment;
- Applications beyond its design specification;
- Applications involving carriage of passengers;
- Operation beyond the allowed operating specification

2.4 SAFETY RANGE

The robot is equipped with remote operation function.

During the operation of a robot program, all personnel should stay within the safety zone and stay clear of the robot movement/operational area. Leave the robot movement space (in orange) as shown in Figure 2.2, and keep worker or obstacles clear from the space. Operators should not start the robot program when there are still people in the orange space.

Operational status of the robot can be indicated by the color of the status indicator. Indicator light is green when safety zone is clear. Obstacle avoidance function is triggered when safety lidar detects obstacles or people in the deceleration zone. Robot decelerates and the indicator light turns purple as continue to proceed at a slow speed. When the obstacle is detected in the stop zone, the robot decelerate to a full stop and the indicator light is purple. When the obstacle is detected in the collision avoidance zone, the indicator light turns red and robot will come to a full stop immediately.

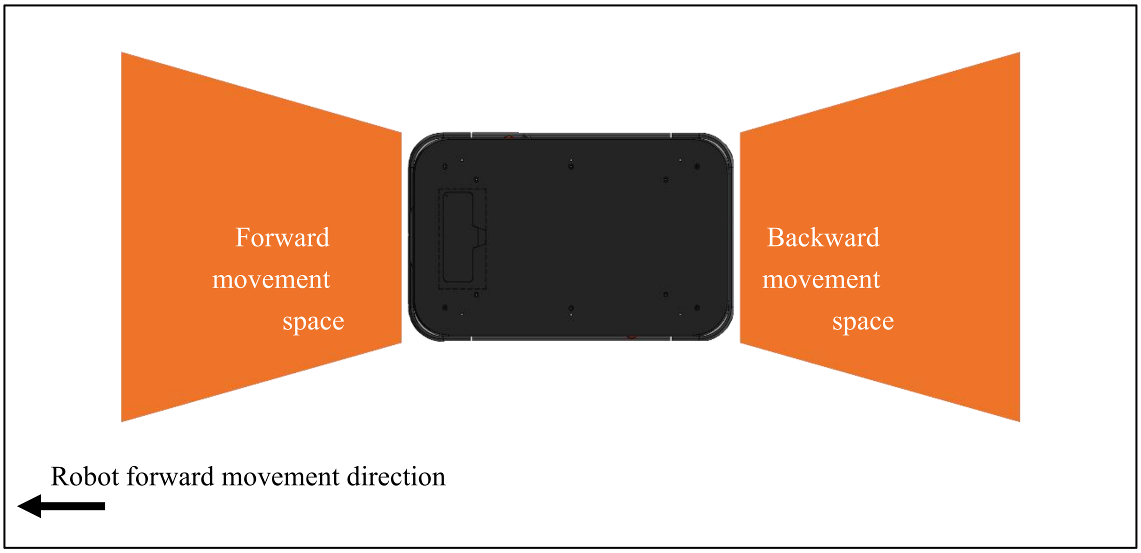


Figure 2.2 Robot movement space should be clear

2.5 PRECAUTIONS

Please make sure that operating environment of the robot within 0 to 40 degrees Celsius and humidity is within the range of 10-90%. Operation outside of the given conditions may lead to serious safety hazard.

Please strictly follow the precautions and instructions when operating the robot. Unauthorized operation may result in serious safety hazard.

The robot system may use a high voltage, it is not recommended for unauthorized personnel to disassemble the robot.

When the robot is operating at a high speed, please take good safety precautions and stand clear of its operational area. Workers should stand at a safe distance of at least 2 meters away from the robot to avoid any risk of injuries.

If any modifications are made to the robot software or hardware, please conduct a detailed safety assessment and take protective measures before deploying the modified robot.

2.6 OCCUPATIONAL SAFETY

Keep the robot clean and in good working condition. Perform regular maintenance check according to maintenance instructions to avoid disoperation.

- If any errors or defects are found that may affect the safe operation of the robot, please stop the robot until such errors or defects have been resolved;
- If error occurs, please switch power off and notify the site supervisor;
- Do not repair any parts of the robot without authorization from distributor or manufacturer. Do not repair any parts of the robot without the permission of the site supervisor.
- When using a robot, it is necessary to be responsible for the safety of the operator and the environment in which the operator is located.
- If any accident or injury occurs, please notify the site supervisor. In this case, please keep the robot at the scene of the accident.

2.7 OVERLOAD SAFETY PROTECTION

The robot has overload safety protection function. When the robot is overloaded or experienced a big impact, the motor driver will trigger an overcurrent protection. Then the robot functions will be suspended and the it can be recovered with a restart.

2.8 HANDLING AND PRECAUTIONS

When it is required to move the robot for a short distance, it is not recommended to push the robot directly. The robot motor has a self-locking function (even when power is switched off) and moving the robot with external forces may damage the motor and its related modules.

When moving the robot for a short distance, operator may use the wireless controller or the virtual controller toggles in the YOUICompass web portal. When operating the robot in manual mode, you need to ensure that the robot is within the line of sight to prevent any damage to the robot and other items due to disoperation.

Operator may also use the robot's brake releasing knob to release the motor self-locking (when robot is powered on). After the motor self-locking is released, the robot can be pushed to move around for short distance.

Before transport of a long distance, the power protection switch inside the debug interface compartment of the robot should be switched off to completely power off the robot. Make sure that all bolts on the robot have been tightened before packing and shipping.

CHAPTER3
QUICK GUIDE

3.1 PACKING LIST

A complete packing list of P200 robot is listed as follows:

Item	Quantity
AMR	1
Charger	1
Remote Handle	1
Product Qualification Certificate	1

The product appearance is shown in Figure 3.1.

Figure 3.1 Main view of product appearance

3.2 PACKAGING AND TRANSPORTATION

During the transportation, the robot should be properly placed in the aviation standard shipping crate, as shown in Figure 3.2. There are metal corners on the shipping crate to enhance the strength of the crate. There are pallet tracks at the bottom, which can be used to transport with a forklift.

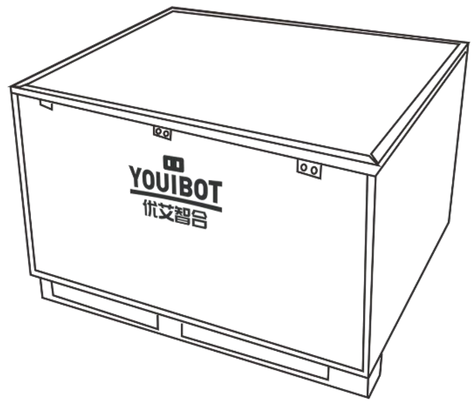


Figure 3.2 Robot base transport aviation wooden case

Transportation should avoid strong vibration, placing crate upside down, direct contact of or damp environment. When transporting the robot base, please use the listed packaging materials. The packing materials are shown in the following table

Name	Number
Bottom Of Shipping Crate (Pallet)	1
Shipping Crate Cover (Slope)	1
Shipping Crate Board	4
Wheel Paddle	4
Protective Foam Block	Several
Protective Knee Brace	4
Lock	24

The forklift is used to fork into the bottom of the box.

CAUTION

The AMR must be properly secured and transported in an upright position. Damages from negligence during transport is not covered by product warranty.

3.3 UNPACKING

Before unpacking, place the crate on flat and open ground. When unpacking, use tools to wedge the locks around the cover, as shown in Figure 3.3. After the locks are wedged, remove the cover of the crate and use it as a slope. Then carefully remove all panels and take out the AMR.

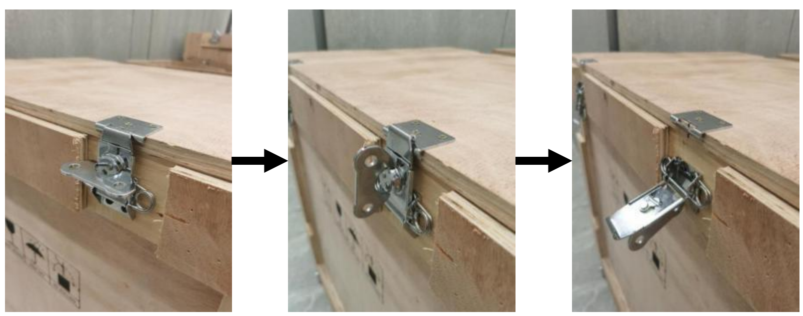


Figure 3.3 Process diagram of packing case lock buckle warping

3.4 INVENTORY CHECK

After all the goods are received and unpacked, check the packing list. If there is any problem, contact the distributor or manufacturer immediately.

3.5 PREPARATION

Switching on the power protection switch for the first time. The power protection switch is located inside the AMR. Open the side compartment of the AMR to operate the protection switch.

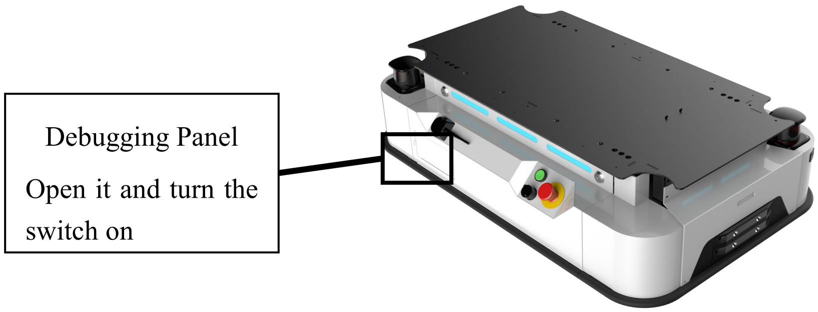


Figure 3.4 Debugging interface compartment

Turn the switch upwards to switch on the power. Operators can keep the switch on during the daily operation. But it is always recommended to switch it off during the transport or if plans to store AMR away for some time.

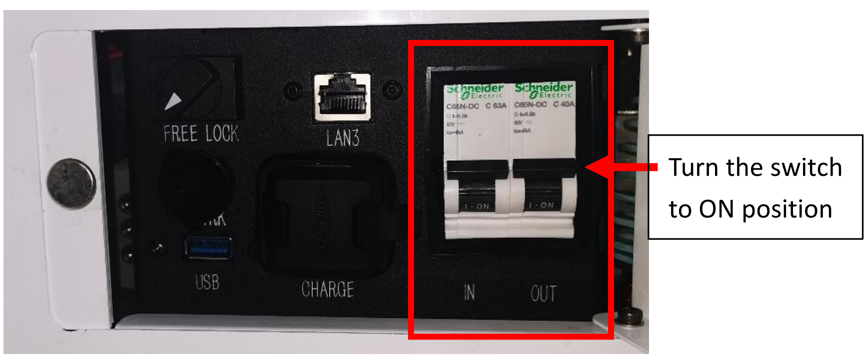


Figure 3.5 Power protection switch in the debug interface compartment

When connected to power, check and turn all emergency stop buttons on the robot base clockwise to ensure AMR is released from emergency brake.

3.6 QUICK USE

3.6.1 POWER ON &OFF

1) POWER ON

After switching on the power supply, press and hold start button and the green indicator light will illuminate. About 5 seconds later, you can hear the crisp clutching sound of the relay, while the green light in the button remains on after releasing the button. The status indicators around the AMR will illuminate, as shown in Figure 3.6, indicating that the AMR has been started. If the status indicator is red, check if the emergency stop buttons are released.

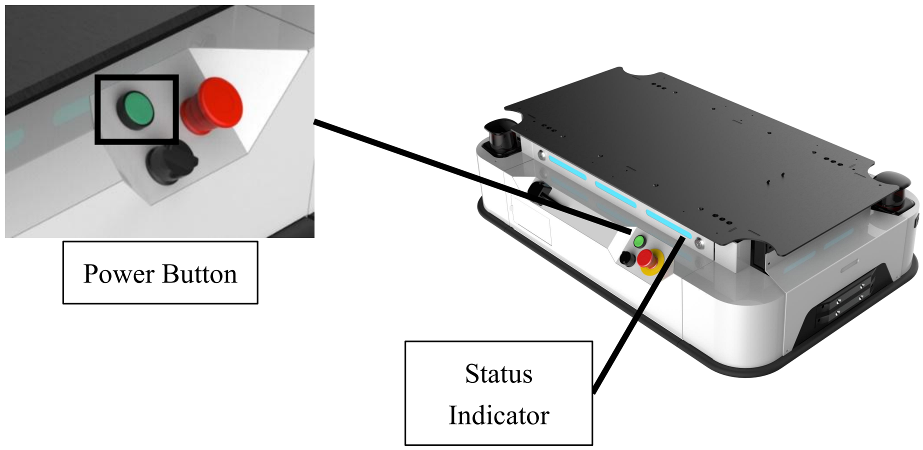


Figure 3.6 Schematic diagram of start button and status indicator

2) POWER OFF

Long press the power button until the status indicator extinguishes from breathing yellow. Release the power button, the green indicator light on the start button goes out, indicating that the AMR has been shut down.

3.6.2 AUTOMATIC/MANUAL MODE SWITCH

When the robot is powered on, the running mode of robot base can be switched to manual or automatic mode. When the robot goes with manual mode, the robot can be operated by wireless controller or the virtual controller toggles on the YOUICOMPASS system. **Note!** In this mode, the obstacle avoidance function can't work but emergency stop function still works. Thus, when the robot detects the obstacle, it will not decelerate or stop, but when the obstacle enters into the emergency stop zone of the lidar, the anti-collision bumper gets touched, or the emergency stop button is pressed, the emergency stop function still gets activated. When the robot goes with automatic mode, the robot receives the task sent by robot management system, the robot can't be operated by wireless controller or virtual controller toggles of the robot management system. Manual and automatic mode can be changed by turning over the manual and automatic mode switch located on the side of the robot base. The manual and automatic mode switch is usually located near the power button, as shown in Figure3-7.

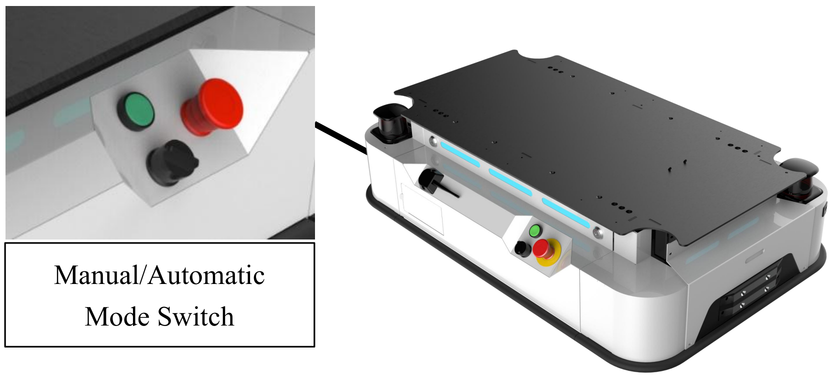


Figure 3.7 Manual/Automatic mode switch

3.6.3 REMOTE HANDLE

Insert 2 sets AA new battery for the remote controller (as shown in figure 3.8), and insert the receiver in any USB port in the top compartment or debugging interface compartment of the AMR. Press Mode to keep the indicator light on and the controller will be successfully connected to the AMR. Then the controller is able to drive the robot.

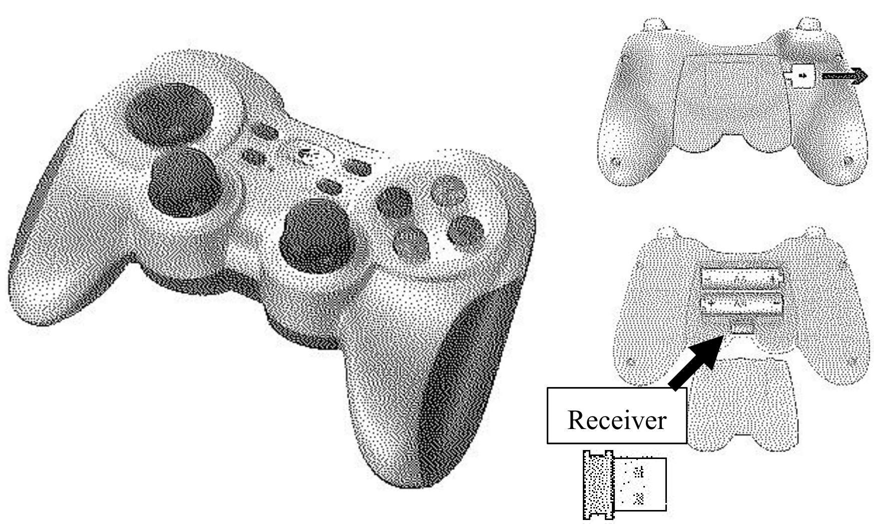


Figure 3.8 Wireless controller, battery compartment and receiver

Press and hold both RB and the low-speed mode button (blue X key) or the high-speed mode button (Red B key) simultaneously and then use the directional button to drive the robot in any directions (forward, backward, left or right). The maximum forward speed is 0.8m/s and the maximum turning speed is 1.0 rad/s using the controller to navigate AMR movements.

When the robot base status light turns red (emergency stop), make sure that there are no obstacles within its emergency brake zone. Operator may press LB key and RB key at the same time to release AMR from emergency stop status.

Figure 3.9 Buttons of Wireless Controller

Kind reminder: When stopping the robot, please remove the left hand first. Then please remove the right hand after the robot stops moving.

3.6.4 BRAKE RELEASE KNOB

When the robot can't move autonomously and be operated by the remote controller, please turn the free lock and then push the robot. The free lock is located inside the debugging panel, as shown in Figure 3-11.

Figure 3.10 Free Lock Picture

When the free lock is switched to ON(Free), the motor brake status gets released to push the robot after the robot is turned on. When the free lock is switched to OFF(Lock), the external power supply can be used to release the motor brake status to push the robot as the robot turned off. The external power supply port is located directly below the Free/Lock, its silkscreen is P_VRK. Please use the dedicated 24V brake release power supply to be connected to this port. Power supply plug model is SF1210/S2(two-pin plug) and pin 1 is connected to positive pole and pin 2 to negative pole.

3.6.5 STATUS INDICATOR

The status indicator light strip is built around the robot base to prompt the robot operational status for operator to take necessary actions accordingly. The corresponding messages to each of the indicator status are shown in Table 3.1.

Table 3.1 Indicator effect and corresponding status table

Number	Lighting Effect	State of the Robot
1	Green	AMR is in normal working condition
2	Breathing Yellow	AMR is shutting down
3	Flashing Yellow	Low battery
4	Red	Emergency stop engaged
5	Purple	Slowdown mode engaged

3.6.6 NETWORK CONFIGURATION

Insert network cable into the LAN3 Ethernet port (the internal network segment interface) in the debugging interface compartment, connect to the computer, and set the IP of the computer to 192.168.0.101. Ping the IP address of the robot 192.168.0.100 with command console in Windows system and see if return is normal. If so, you should be able to access the robot scheduling system web portal with a browser (YOUICompass system).

Figure 3.11 Network interface in the debugging interface compartment

If the operator can ping to 192.168.0.100 successfully as shown in Figure 3.12,the operator can enter the IP address in browser to log in YOUICompass. Regarding YOUICompass Quick to Use Instruction, please refer to 《Chapter 4 YOUICompass Quick to Use instruction》. Please refer to 《YOUICompass user manual》for detailed operation instruction.

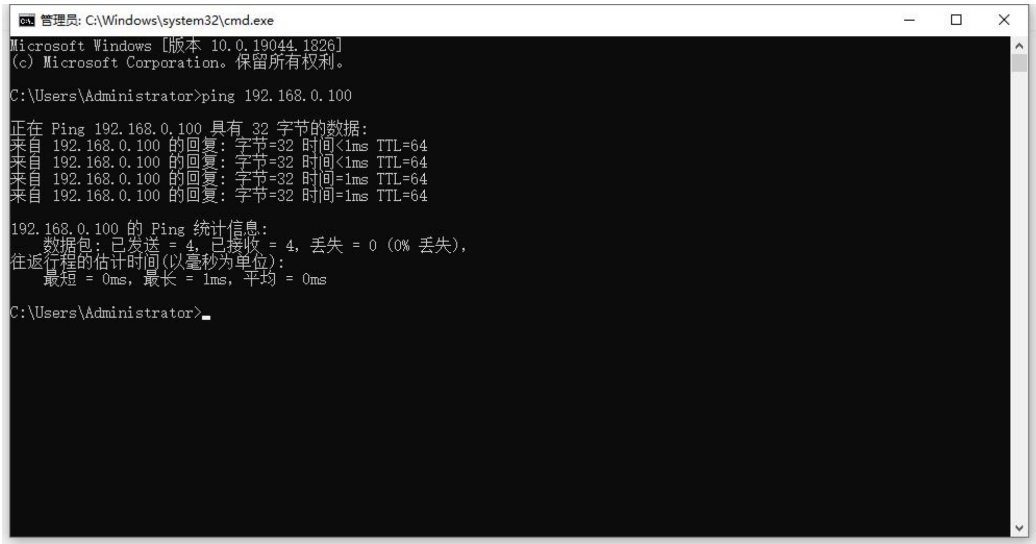


Figure 3.12Ping command example picture

CHAPTER4 YOUICOMPASS INSTRUCTION

4.1.1 YOUICOMPASS INTRODUCTION

The YOUICompass system is a proprietary control portal developed by Youibot. The robot scheduling system adopts B/S architecture, and it is designed to perform autonomous tasks such as map and marker recording, creating and executing the tasks, viewing or returning information to a specified API, and a series of scheduling and mission programming function.

Notice! The actual version may be different from the system version shown in this chapter. The content in this chapter is for reference only. Please refer to the instruction in this chapter for operations as appropriate. If necessary, please directly read the user manual of the correct version of YOUICompass.

After you have successfully established the connection between the computer and the robot, enter the AMR's IP address in web browser to see the YOUICompass login page, as shown in Figure 4.1.

Note:

Default username/password are both 'admin'.

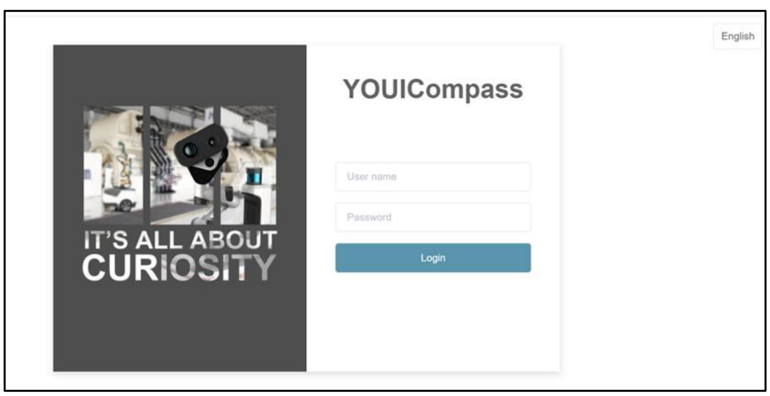


Figure 4.1 YOUICompass login page

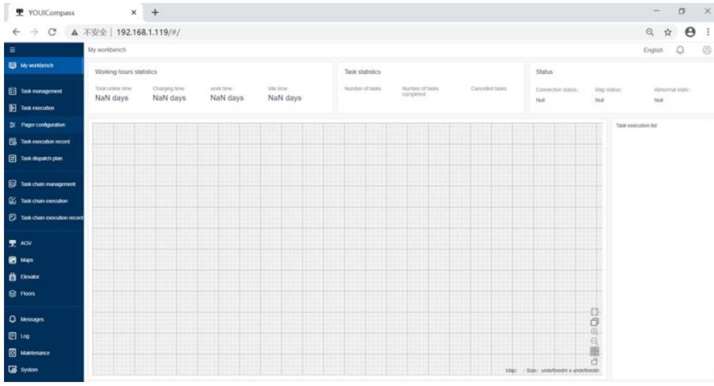


Figure 4.2 Workbench page

4.1.2 CHECK ROBOT STATUS

After entering YOUICompass, click ‘AGV’ in the left menu column to enter the robot status page, as shown in Fig. 4.3.The top right corner of the page shows the current computer and network connection status, control mode, battery level, language and account information. The connection status can be disconnected (yellow) or connected (green). The control modes are Manual Mode ( ), Automatic Mode ( ). The top row of the page is the control area with a list of icons where operator can select to synchronize map, switch control modes, relocate the current position, one-click operation, parameter setting and manual device control operation.

On the right side of the page where it shows the operational details of the robot, and the bottom left corner of the map is a control toggle that can be used in manual mode.

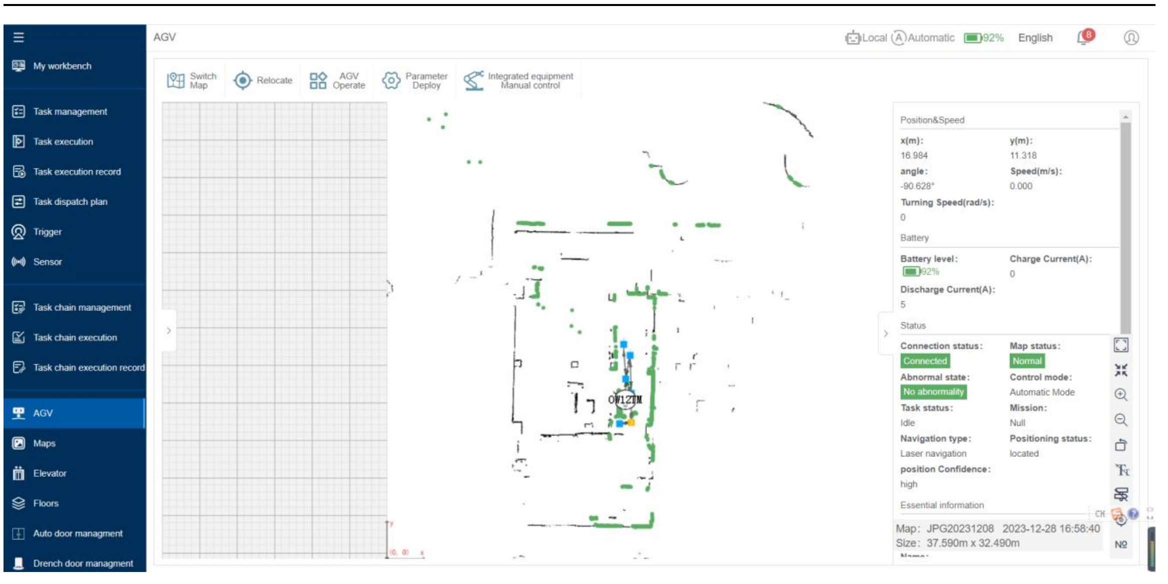


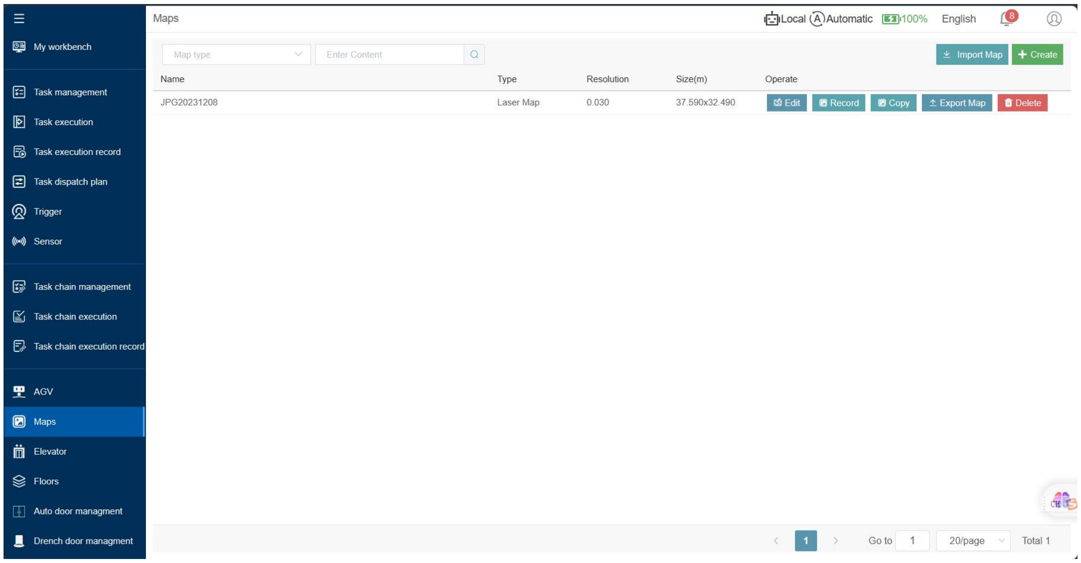
Figure 4.3 Robot status page

4.1.3 MAP

The map stores information such as the robot's points and routes. The quality of the map construction directly affects the operation of the robot and the robot's walking route. If a robot wants to reach a certain destination, it needs to be the same as humans drawing maps. The process of describing and understanding the environment mainly relies on maps.

This module supports creating, recording, editing, deleting, importing, and exporting maps. Click [Map] and the page will jump to the map list.

On the default system, the map list is empty,



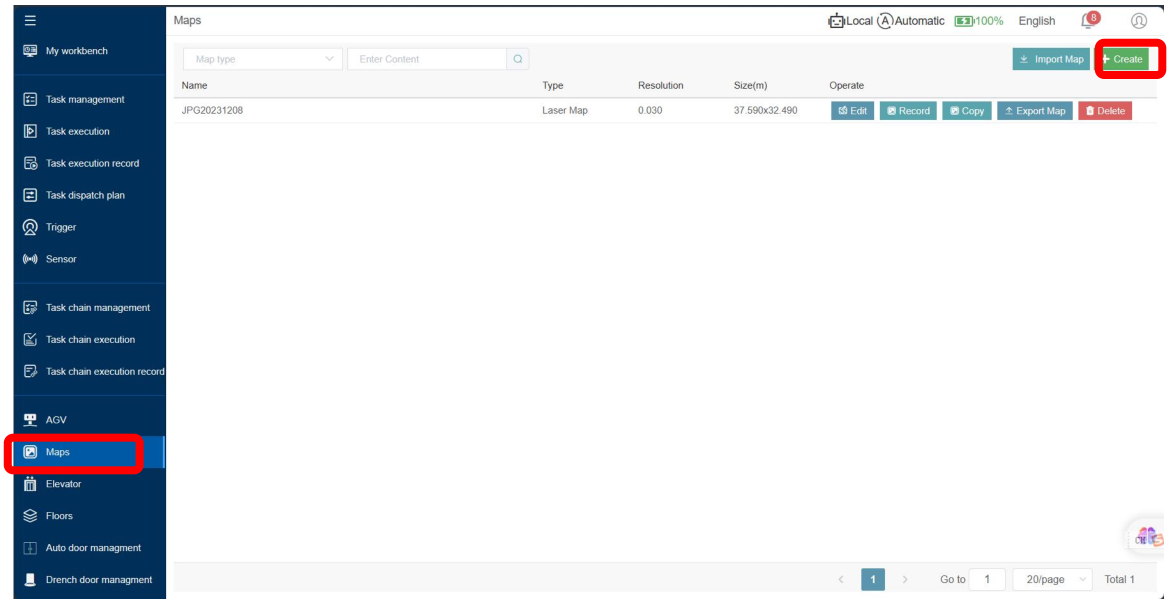
4.1.3.1 CREATE/RECORDING MAP

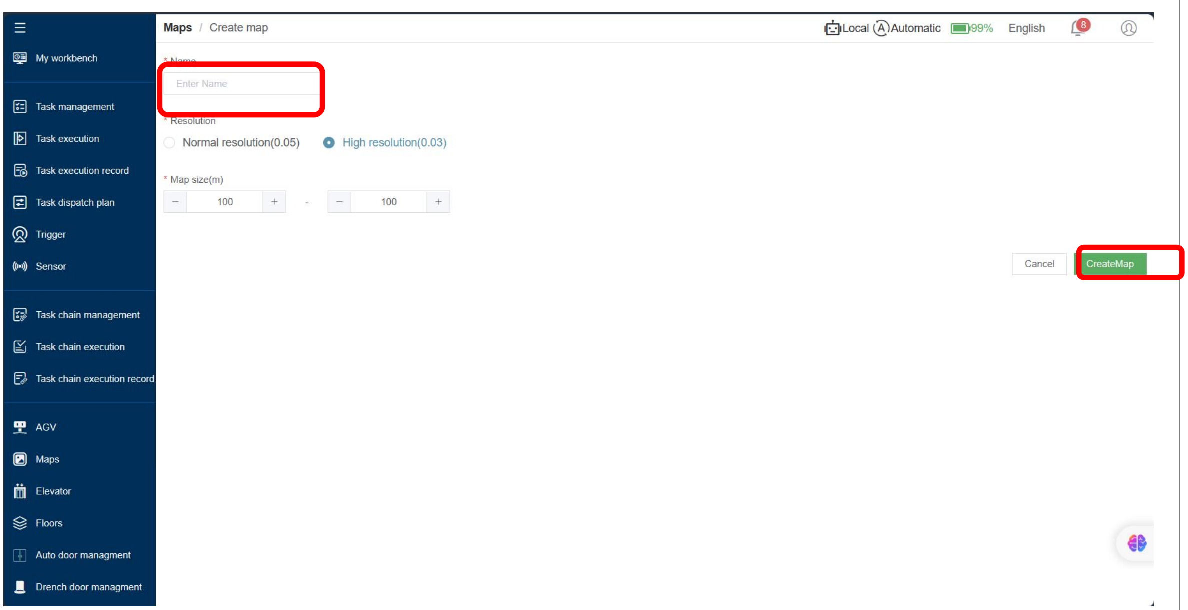
It’s necessary for the user to record a map the first time using the robot. Open the 【Map】, click the 【Create】, the page will turn to the map-creating the interface.

1. Click the 【Create】 button to open the map-creating interface. All items in the interface are required, and 【Resolution】 and 【Map Size】 are selected by default;

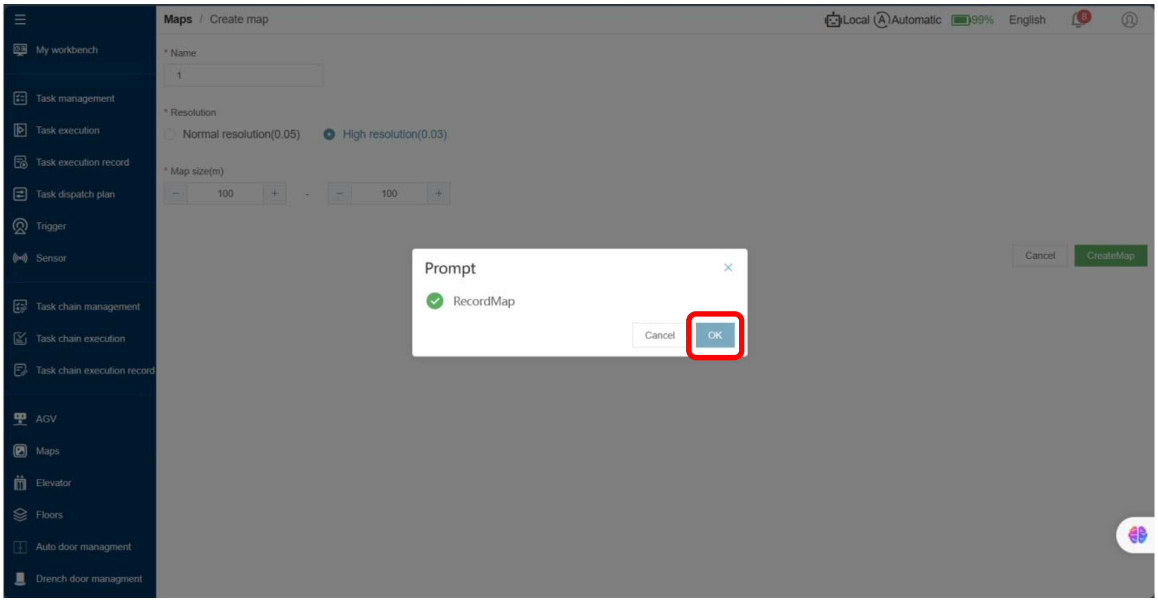
2. Click 【Create】, enter the map 【Name】, English and numbers are supported, Chinese characters, etc. are not allowed to be entered;

3. After filling in all the items, click the【Create】button in the lower right corner to save the new map.





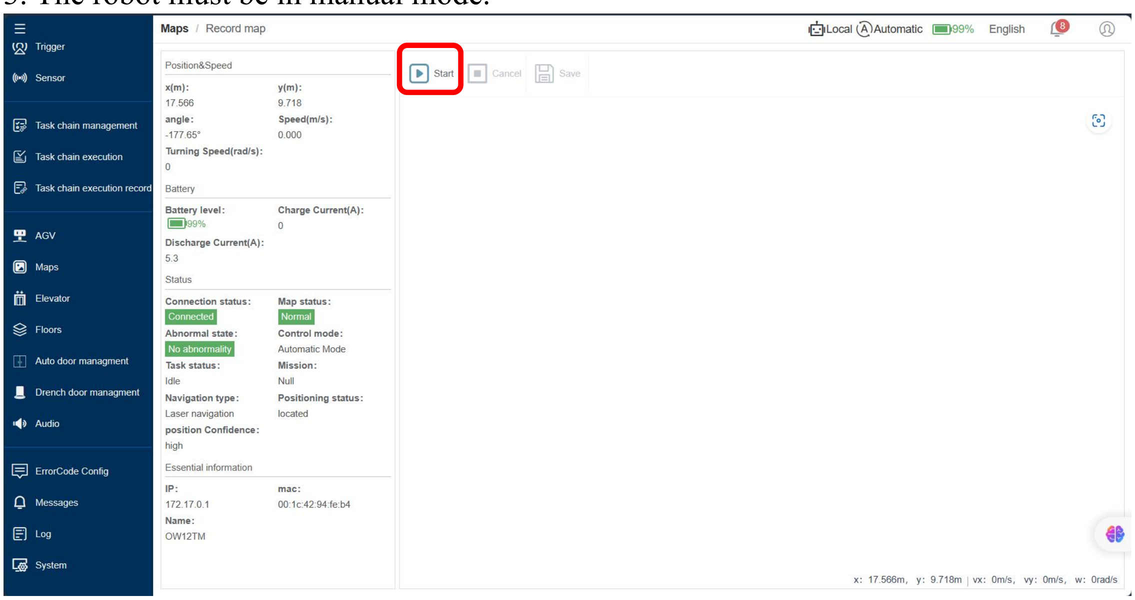
Click【Create】, and the [Record Map] prompt will pop up on the page. Click [Confirm] to jump to the map recording page.



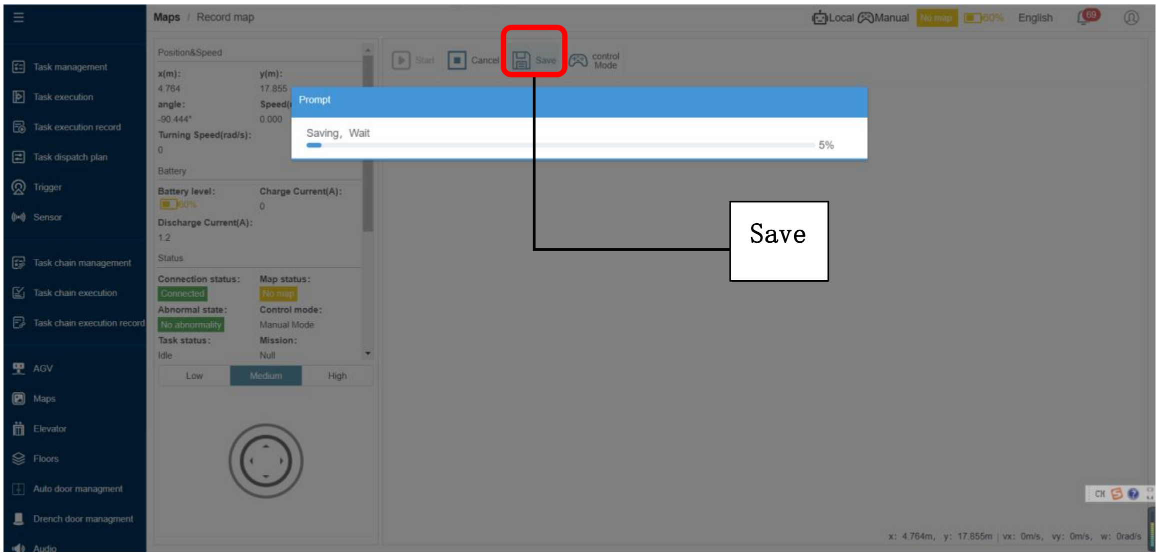
Click the [Start] button, and then use the remote controller or the virtual joystick in manual mode to control the movement of the robot.

Notice:

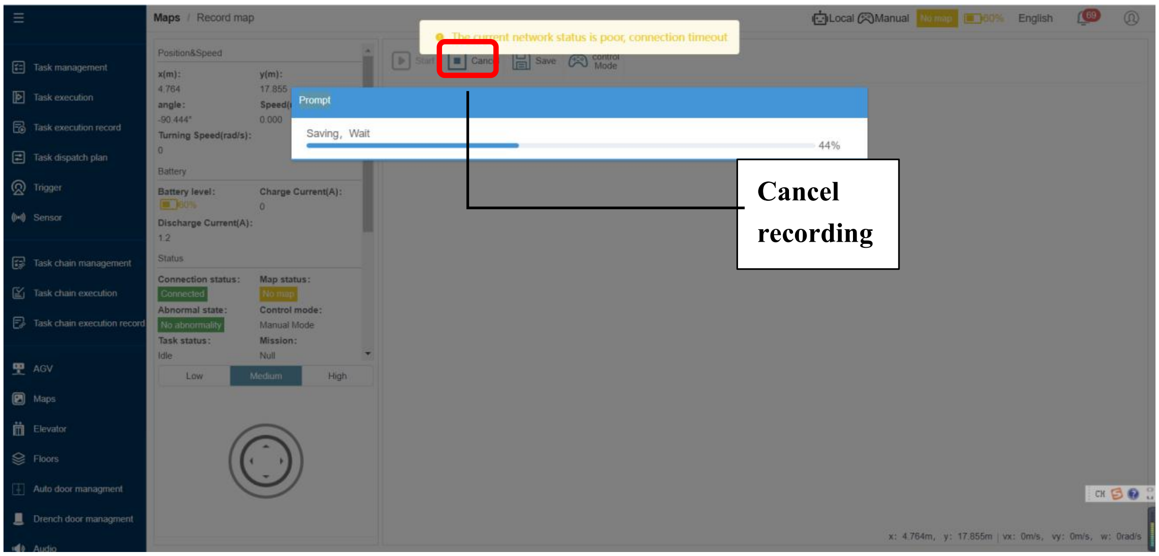
1. The robot tries to keep recording at a constant speed;
2. Try to ensure that the robot's operating path covers all environmental areas;
3. The robot must be in manual mode.



When the robot has finished scanning the environment, click the Save button. The newly created map information will be displayed. After confirming that the map recording is correct, the map recording is completed. If the mapping effect is not good and needs to be recorded again, just repeat the above map recording.



If want to cancel the map recording while recording the map, click the [Cancel] button. At this time, the recording of the map being recorded will stop and the currently recorded map will not be saved.

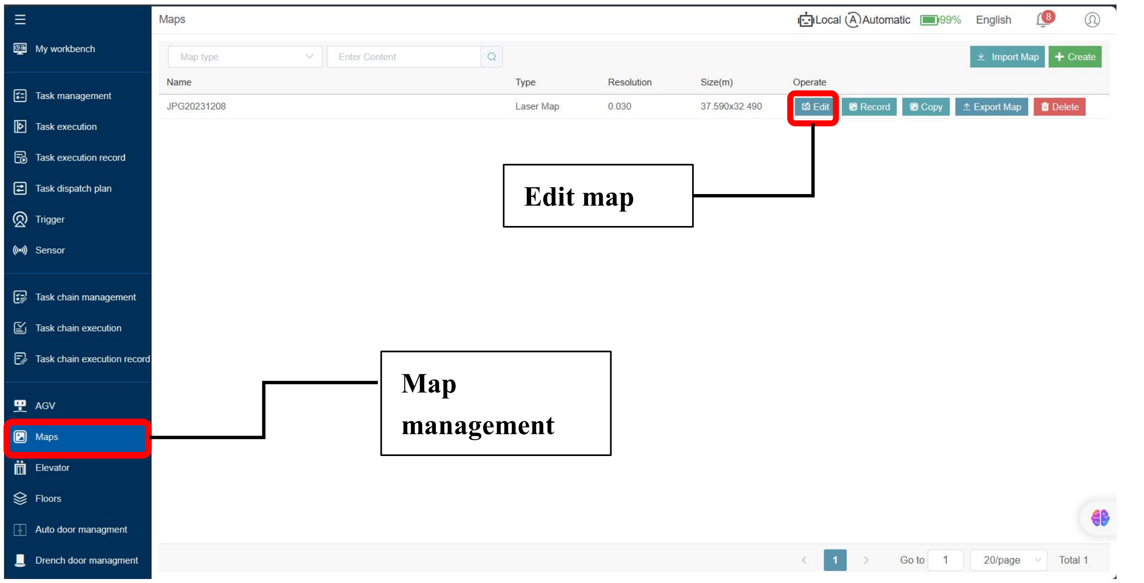


Notice:

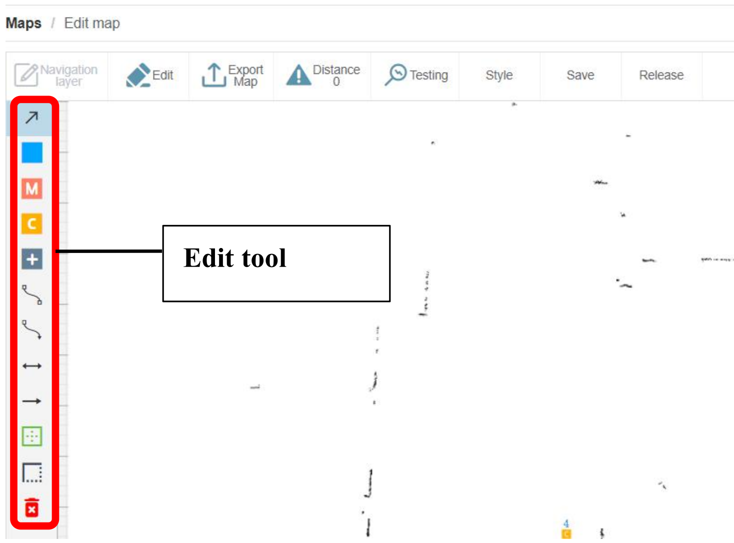
1. When using the virtual joystick to control the robot, it is recommended to plug in a network cable to avoid security risks caused by wireless network delays.
2. When the robot scans the map, it is recommended to keep a path and scan it repeatedly (more than 2 times) to ensure that the scanned map is clear.

4.1.3.2 EDIT MAP

Click Map Management to switch to the map management interface, click the Edit Map button to enter the map editing interface;



1. Map editing includes functions such as planning the robot's moving points and routes on the map;
2. When editing the map for the first time, when entering the [Map Edit] interface, the recorded map will be displayed;
3. The map has been edited, but exited without saving. The system automatically saves the last edited content as a draft. When enter again, there will be prompted whether to continue editing on the last draft. If abandon the last edited content, the system will delete it. If continue the last edit, the last edit will be saved on the map;



4. In the [Map] list interface, click [Edit] to enter the corresponding [Map Edit] interface;
5. The left side of the interface is the operation bar. Place the mouse on the button icon to view the function name of each button;
6. Click the button in the left operation bar to enter the corresponding function operation mode. From top to bottom, the button usage instructions are as follows:

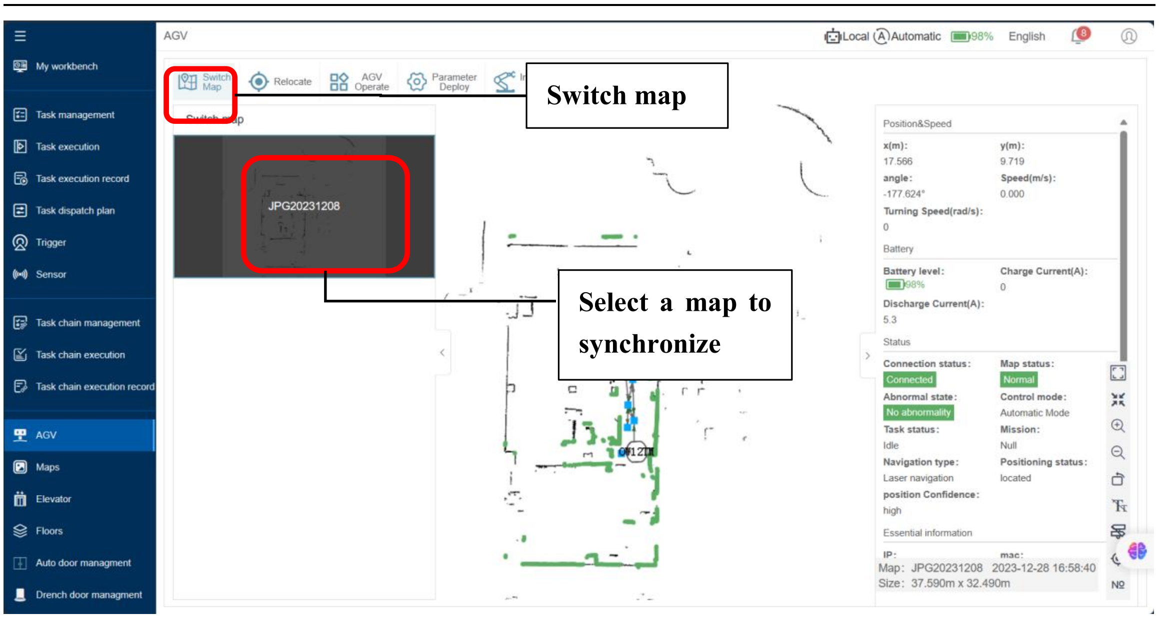
Name	Description
Radio mode	In ratio mode: Place the mouse on the map and click and drag to move the map; Place the mouse on the marked point on the map. When the color of the point changes to indicate that it has been selected, the point can drag and move the point; When the mouse is placed on the bidirectional curve, two gray points will be displayed on the bidirectional curve. Click the gray point, select the gray point, and drag to change the shape of the curve;
Add navigation point	In this operation mode, when the mouse clicks on the map, the clicked point is marked as a navigation point.
Add a work point	In this operation mode, when the mouse clicks on the map, the clicked point is marked as a work point.
New charging point	In this operation mode, when the mouse clicks on the map, the clicked point is marked as a charging point.
New avoidance point	In this operation mode, when the mouse clicks on the map, the clicked point is marked as an avoidance point.
Add bidirectional curve	In this operation mode, click the mouse on the map to select a point, and then click to select another point to connect two points. Two points connected by a two-way curve. User can change to ratio mode to set the curve line.

Add unidirectional curve	In this operation mode, click the mouse on the map to select a point, and then click to select another point to connect two points. Two points connected by a one-way curve. User can change to ratio mode to set the curve line.
Add a two-way straight line	In this operation mode, click the mouse on the map to select a point, and then click to select another point to connect the two points. Two points connected by a two-way straight line
Add one-way line	In this operation mode, click the mouse on the map to select a point, and then click to select another point to connect the two points. Two points connected by a one-way straight line
Add a single machine area	In this operation mode, click the mouse on the map to draw a closed area, which is used in YOUIFLEET system to limit the robot’s moving space.
Batch operation mode	In this operation mode, use the mouse to select an area with points, and all points in the area will be selected. All selected points can be dragged and moved as a whole.
Delete element	In this operation mode, click the mouse to select a point, and the point will be deleted; click the mouse to select a stand-alone area, and the area will be deleted.

4.1.3.3 SYNCHRONIZE MAP

Turn the mode switch knob to the Manual to change the robot to manual mode.

Click Sync Map and select the current map to synchronize;



4.1.3.4 ROBOT RELOCATION

When using the system for the first time, after synchronizing the map, repositioning is required for the robot icon to be displayed on the map; after the synchronization of the map is completed, the robot is required to perform repositioning of the robot's position, so that the robot knows where it is in the map.

1)This function has the same function as in [Record marker] and the same usage method, both are used to locate and calibrate the robot's position in the map;

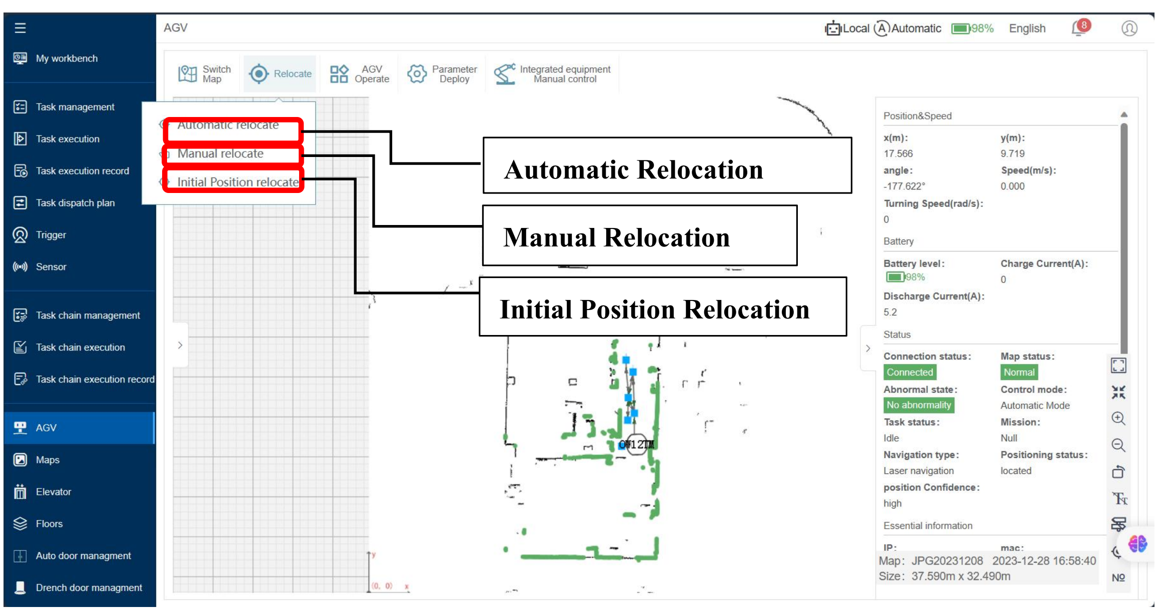
2)The robot can only perform the task after the robot's position on the map is positioned; after successful positioning, the robot's positioning status is “Located” status.

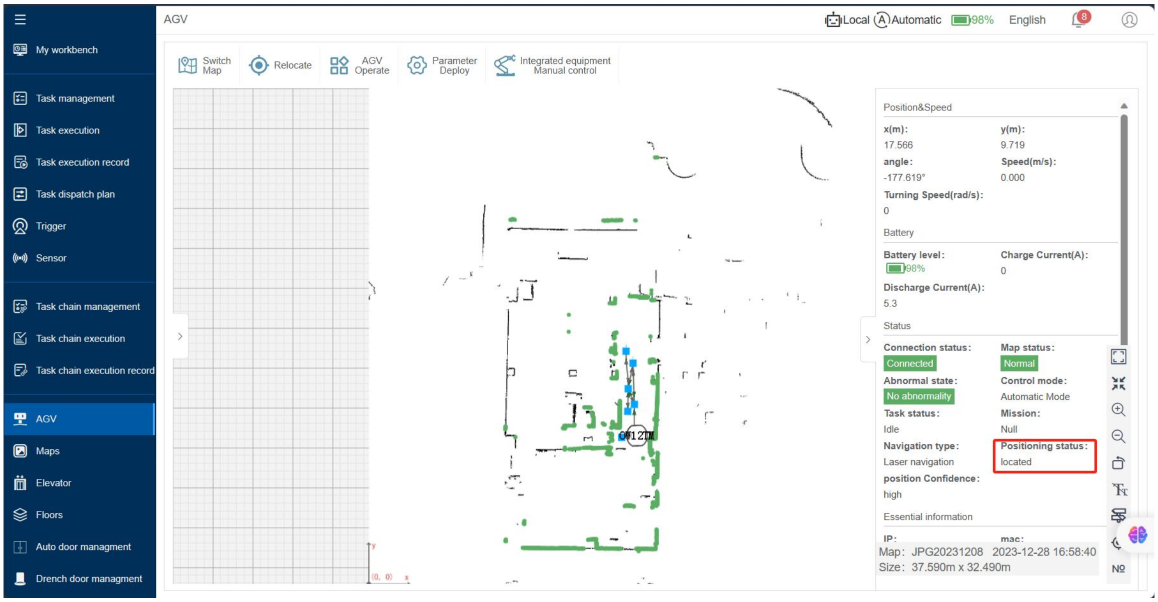
There are three relocation methods: automatic relocation, manual relocation and initial point relocation.

- Automatic relocation: If the environment has not varied much since the recording of the map (compared with the environment the operator records), operator can use the automatic relocation. First use the manual relocation mode to give a rough position on the map (error tolerance is 1/10 of the map size): Click manual relocation, a robot will appear on the cursor. Move the cursor/robot to the approximate position on the map, left click and hold to drag out the forward direction, then click

‘Automatic Relocation’, the dashboard will pop up “relocation succeeds” message.

- Initial Position Relocation: use the navigation point recording function, record an initial position on the map (instructions for recording navigation points are covered in the following pages). Move the robot to the initial position with controller, and make sure the robot is facing the same direction as recording of the initial position. Click on ‘Initial Position Relocation’, in the pop-up box input initial position ID, and click ok.
- Manual Relocation: Select on the ‘Manual Relocation’. According to the robot’s current position on the floor, operator will need to point out the robot’s corresponding location on the map. When pointing out the cursor/robot on map, operator will need to left click and hold on the mouse to select a close spot of the exact robot location on map, and drag out a forward direction of the robot. After placing robot on the map, please use the controller to spin the robot for 1-2 laps on the same spot for it to scan and confirm the surrounding environment.
- Mark of successful relocation: The green lines composed by LiDAR point cloud (a series of green dots) overlap actual obstacles and layout on map. Upon completion of relocation, operator may move robot to the navigation point or preset route on map to be ready for a task.

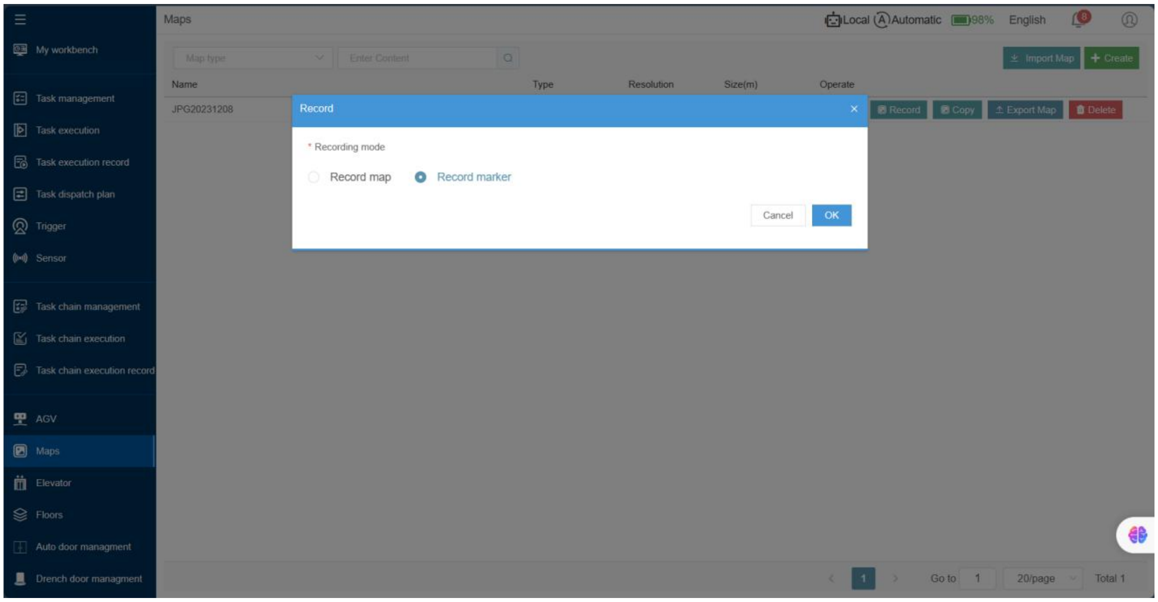




4.1.3.5 VIRTUAL POINT EDITING

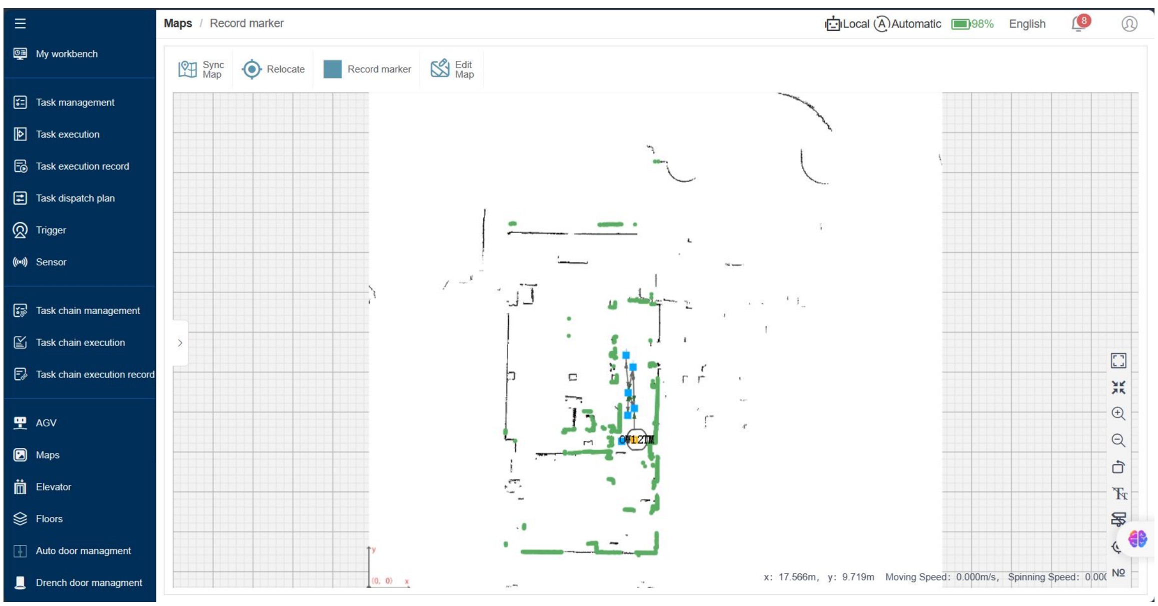
The actions of robots are all based on points. YOUICompass has three types of points: navigation points, work points, and charging points (laser maps have added initial points).

Click the 【Map】 management button, the page switches to the map page. Click the "Record" button to open the recording window, select 【Record Marker】 and the robot, then click the 【confirmation】 button to enter the recording page.

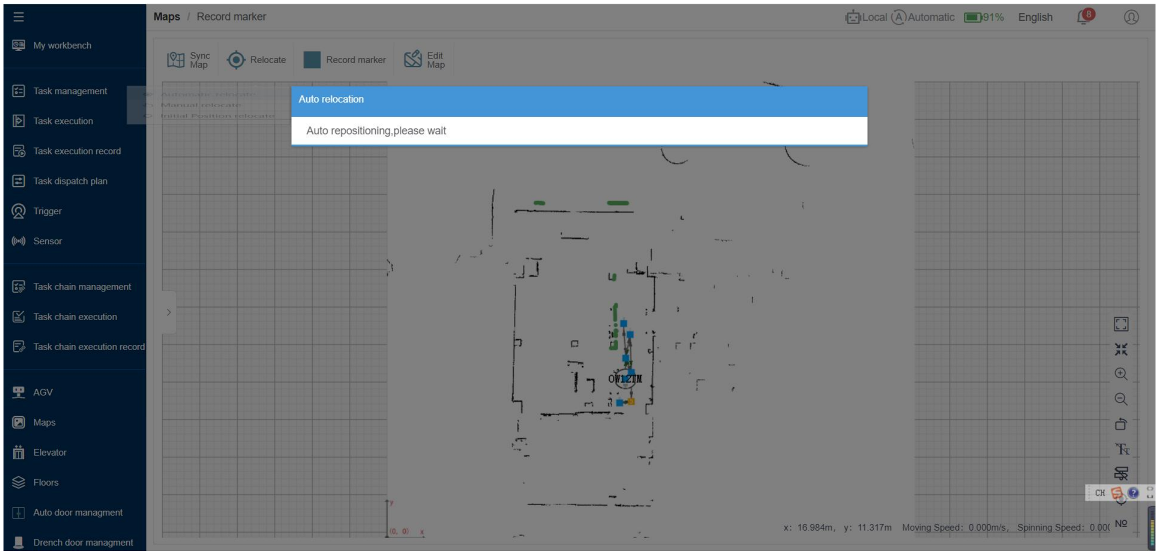


1、Procedure of marker recording:

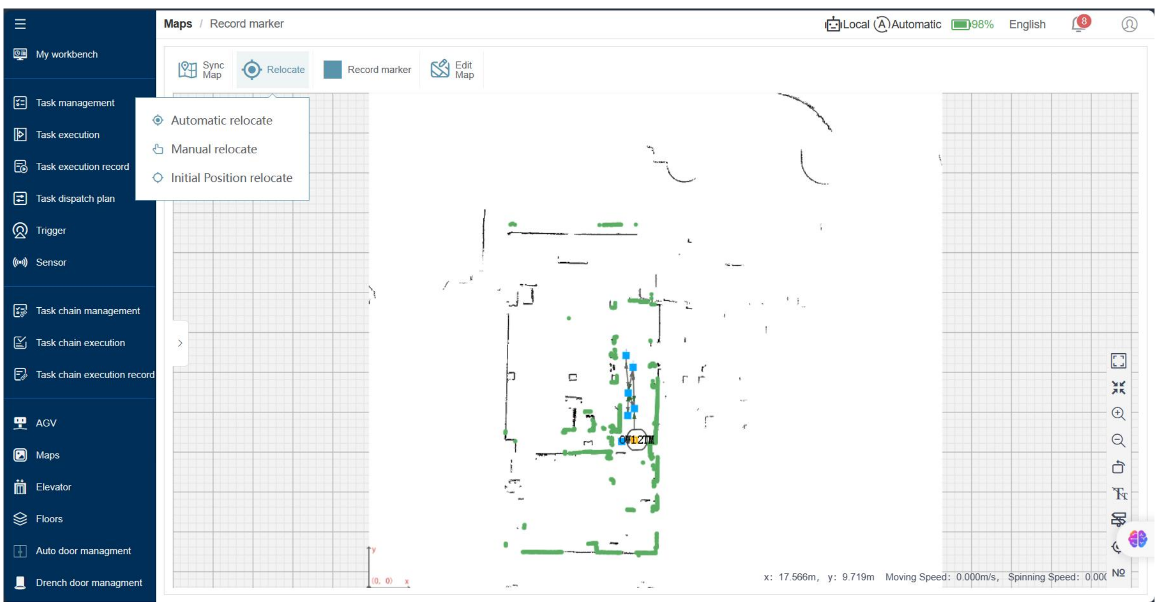
1) Switch the robot to manual mode;



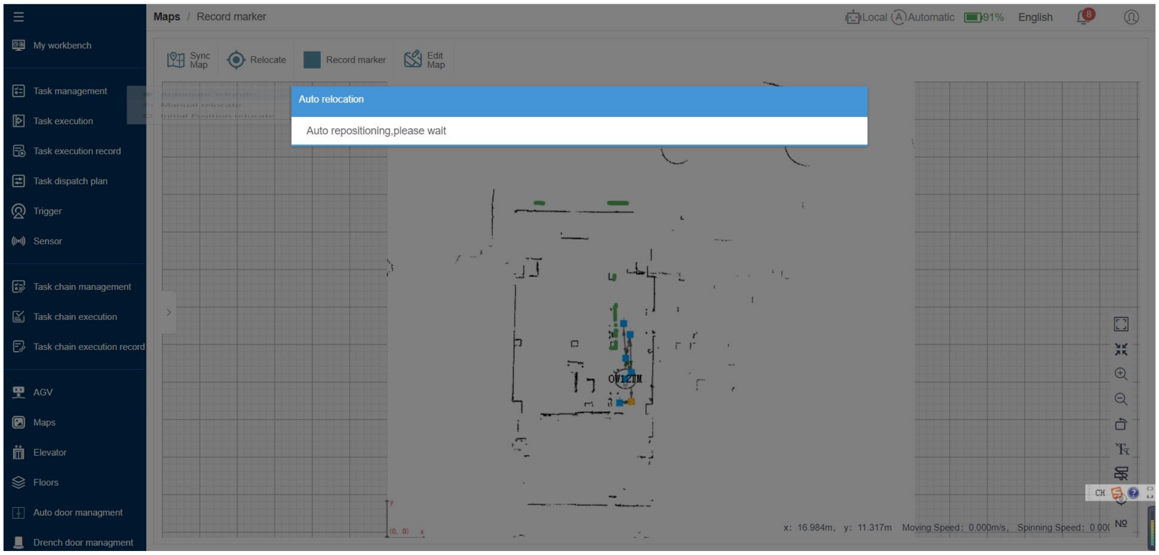
2) Click【Switch Map】;



3) Click【Relocate】and then select【Manual Relocation】; Estimate the robot’s location and place the robot’s icon at the position.



4) Click【Relocate】 and then select【automatic relocation】,The system calibrates the robot's assigned point on the map, adjusting it to the actual positions of the robot on the map.



5) Click on the【Recording marker】button, choose the functional point to be recorded. Once selected, the robot's current position becomes the chosen functional point, and this point is displayed in the corresponding color for the selected function.

6) After completing the recording in the【Recording marker】interface, it is necessary to enter the【Edit Map】section to perform the【Save】or【Publish】operation.【Publish】function only used when linked with YOUIFLEET system. Only then will the marked points recorded during the【Recording marker】be saved.

7) In the【Recording marker】section, you can directly click on "Edit Map" to enter the map editing interface for map editing.

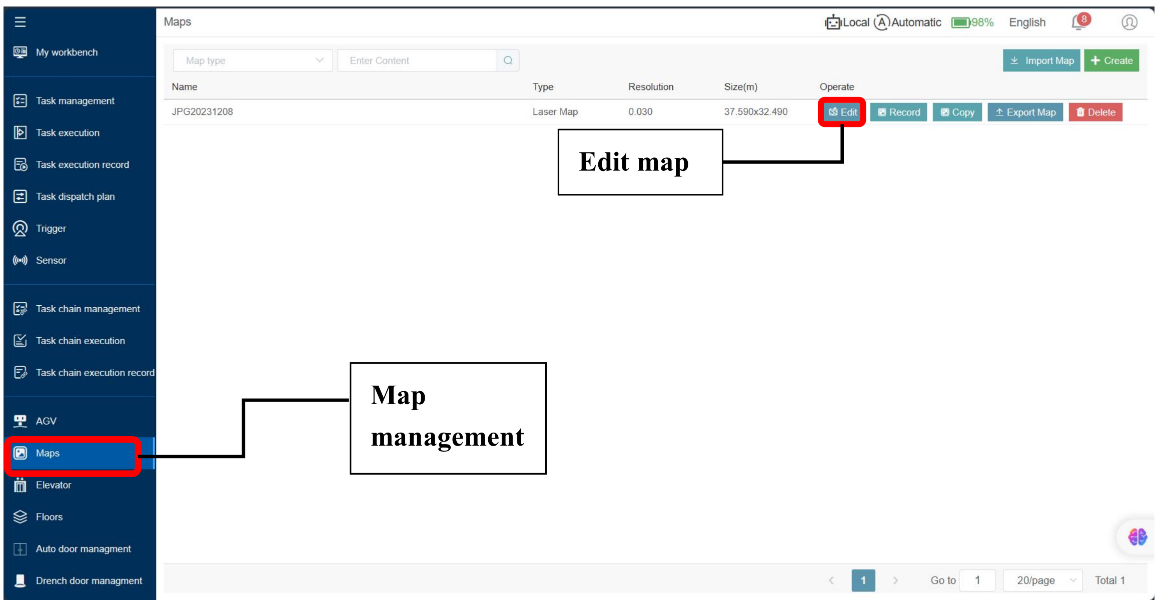
- Note:
- The AGV recording point pose should align with the charging station (which needs to be fixed), with a distance between 0.5-1.2m.
- After recording the initial point, it is essential to note the position of the initial point and the robot's pose during point recording. When using the initial point for repositioning in the future, ensure that the robot's pose is consistent with the recorded pose.

PATH PLANNING

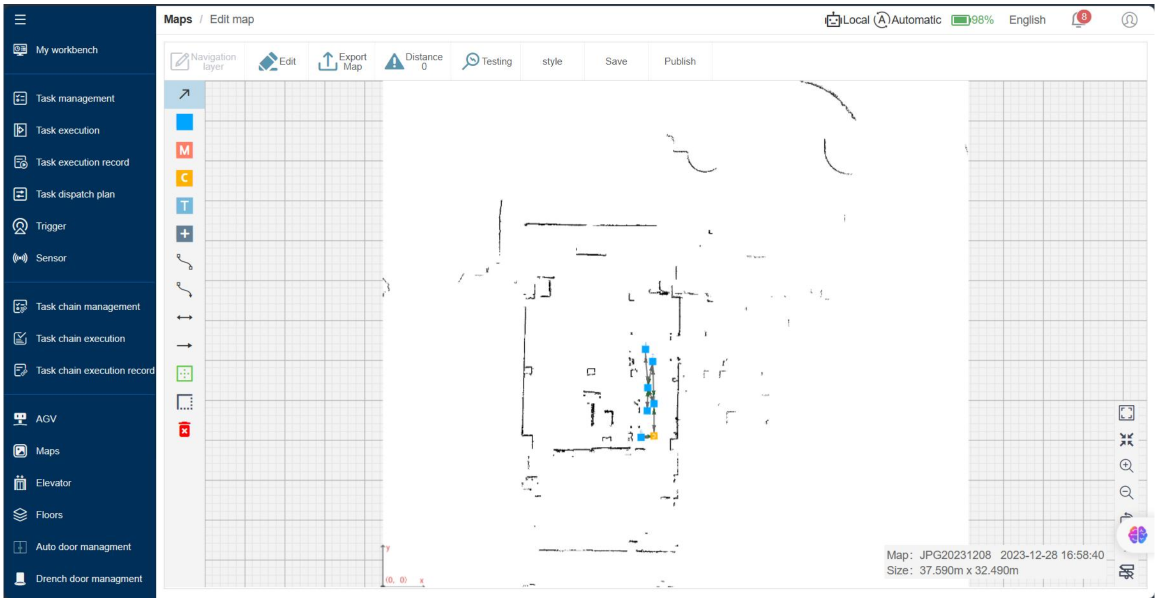
In order to ensure the efficiency and safety of robot task execution, we need to plan the routes between points and points for the robot, as well as the robot movement parameters (speed, acceleration, obstacle avoidance area, obstacle avoidance switch) of each route.

Click the 'edit' button in the map bar to jump to the map edit page.

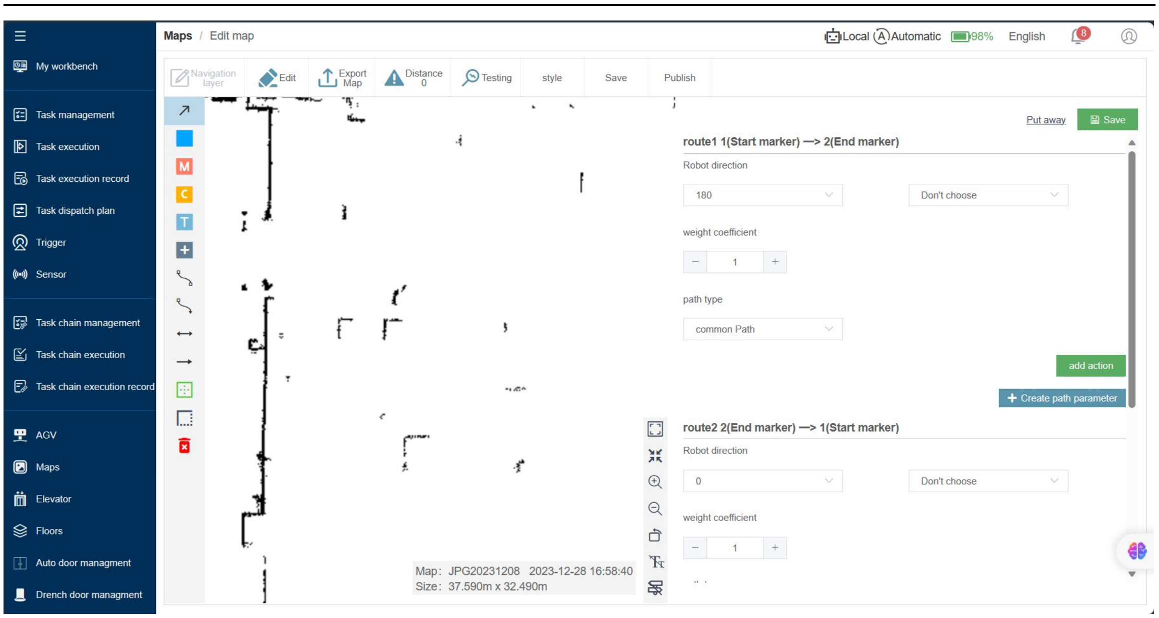
Long press the 'Curved Path' button or 'Straight Path' button on the left side of the map, the page will pop up bidirectional or one-way options, select the required type of line, and connect all points as desire.



Left-click the 【ratio mode】 button on the left side of the map using the mouse. Choose the desired type of route. Click on a point on the map to select it, then click on another point to connect the two points. Connect all points based on the path planning



After connecting the points, double-click the line on the map with the left mouse button. The route parameter page will pop up on the right side of the page. Modify the corresponding data according to actual needs and click Save.

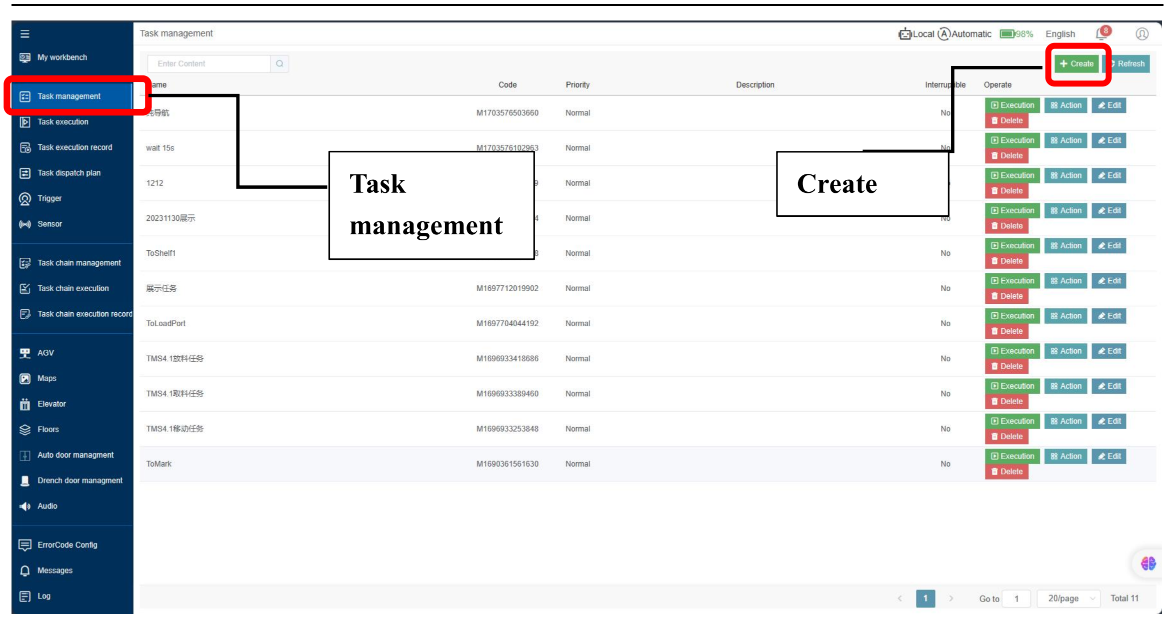


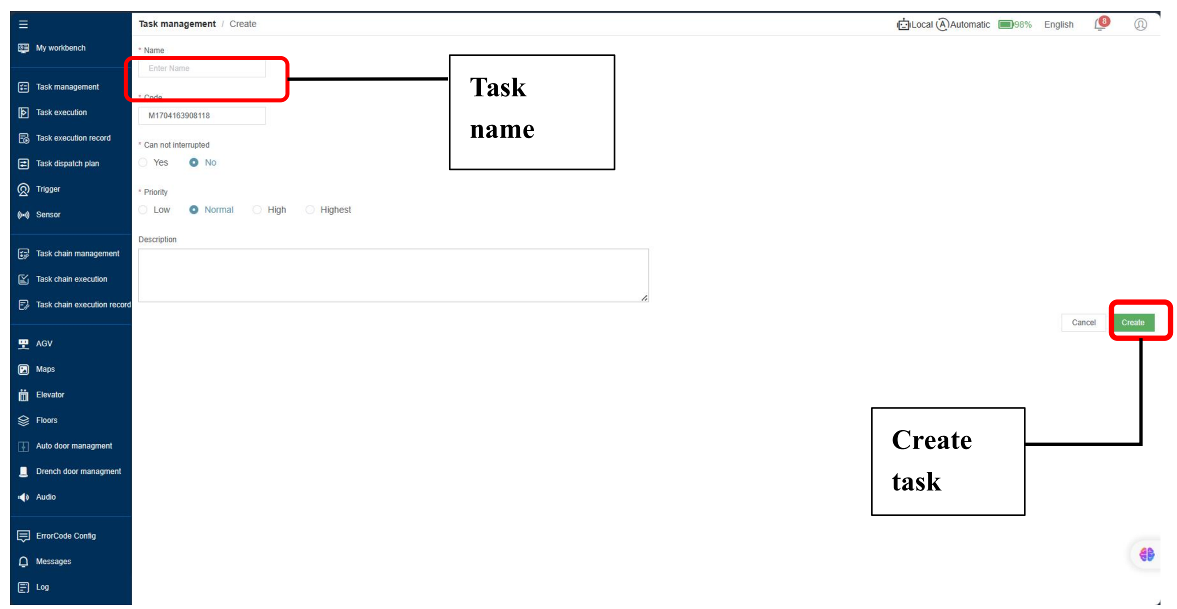
4.1.4 TASK MANAGEMENT

A task consists of one or more actions, and the robot will perform the task according to the predefined actions.

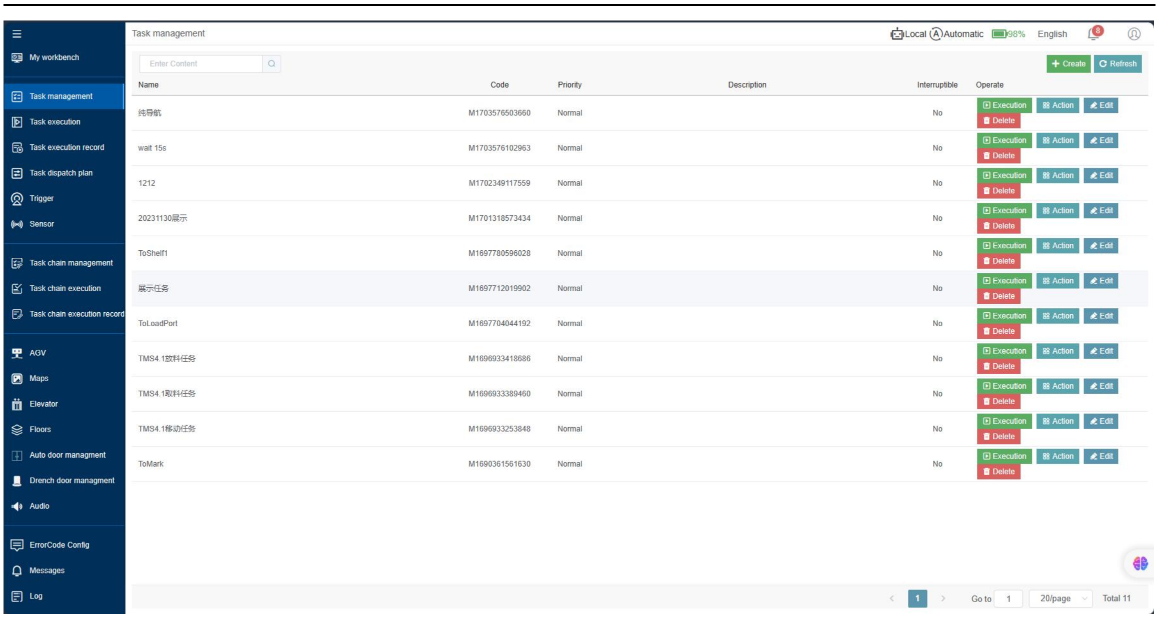
4.1.4.1 CREATE TASK

Click Task Management to switch to the task management dashboard. Click ‘New Task’ button to create a task and fill in the task information.

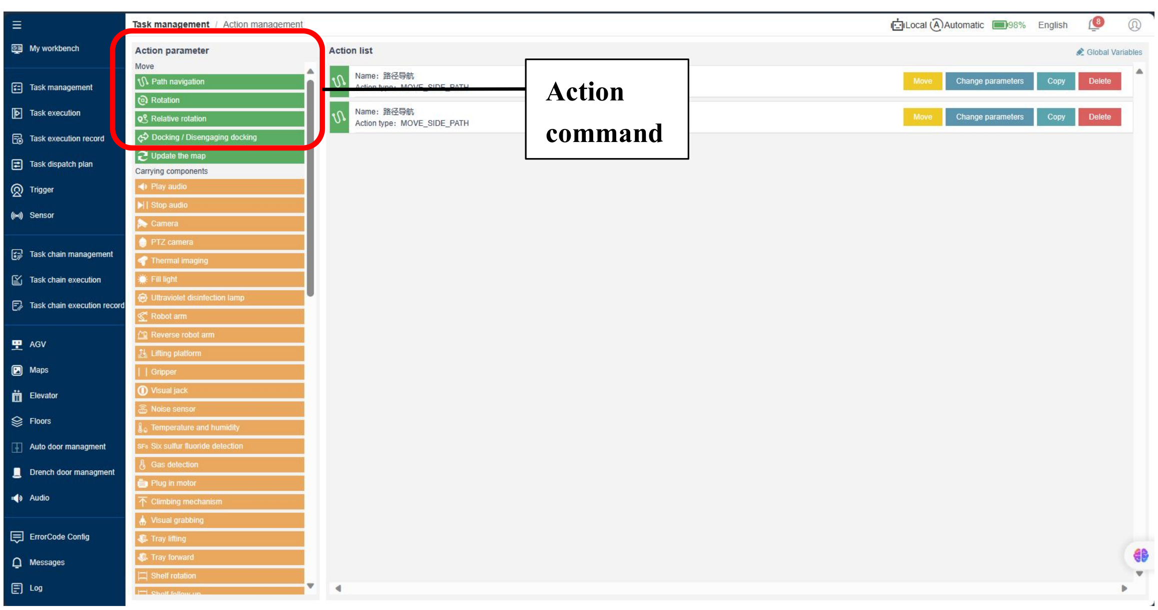




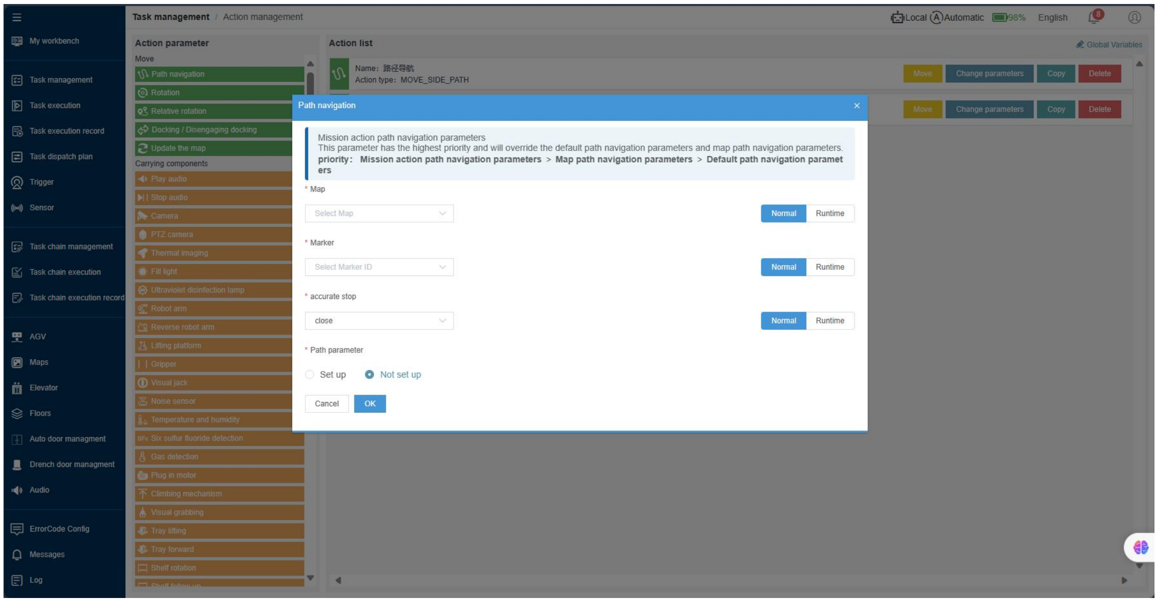
After the task is successful created, click ‘Action’ in the task bar to jump to the action programming dashboard.

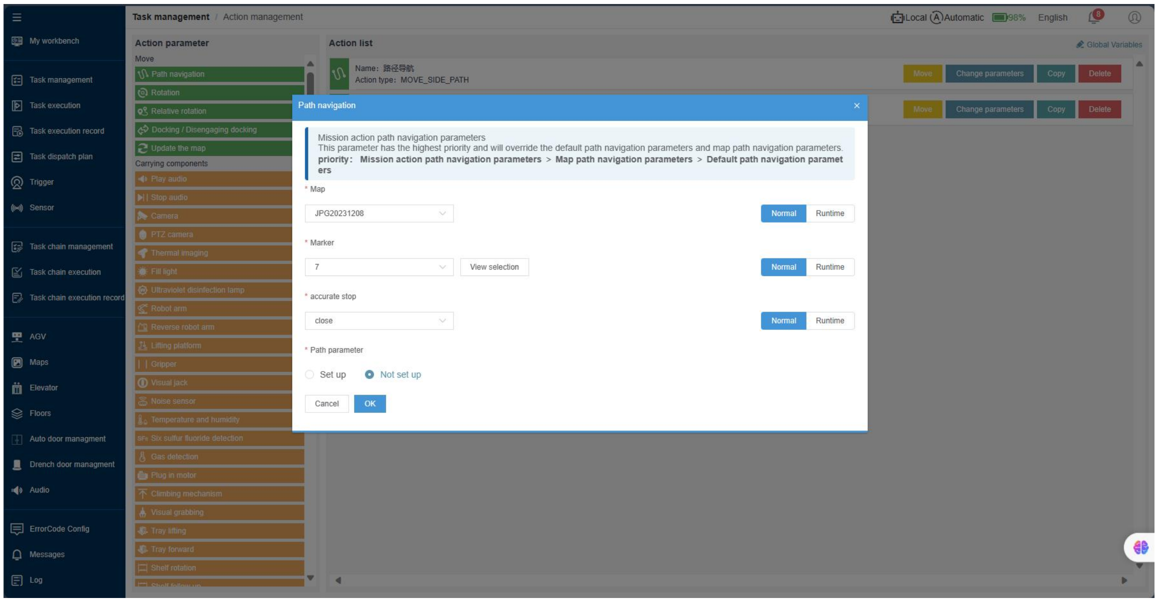


After the task is successful created, click 'Action' in the task bar to jump to the action programming dashboard.

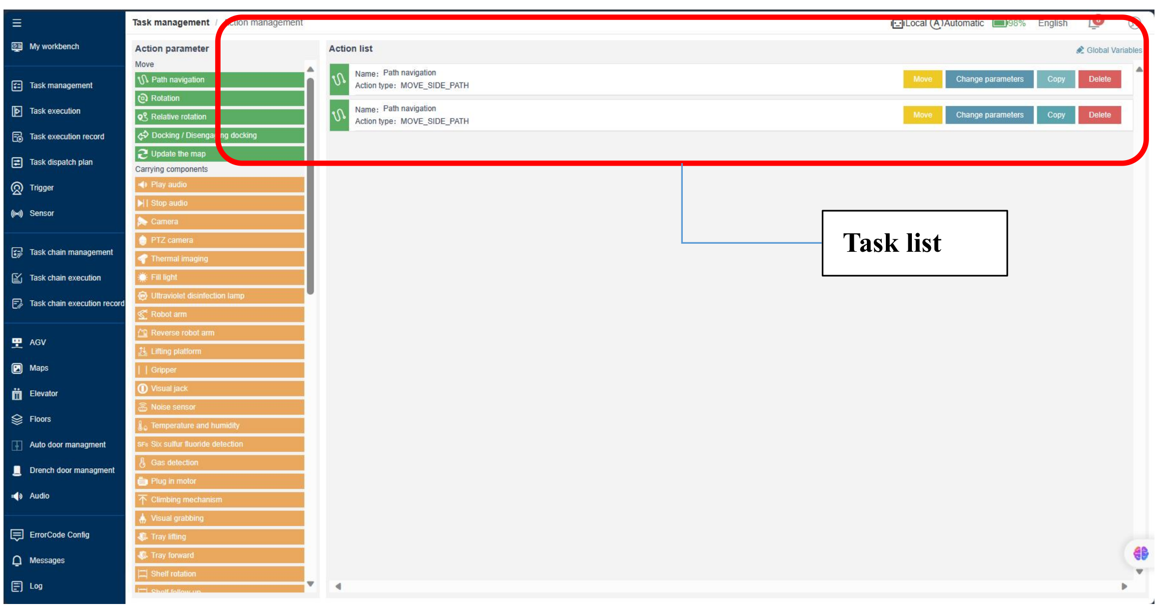


In the action editing window, select the corresponding map and the task point, and click 'OK' to save and exit back to the task list page.


Task management / Action management
Action parameter
Action list
Path navigation
Mission action path navigation parameters
This parameter has the highest priority and will override the default path navigation parameters and map path navigation parameters.
priority: Mission action path navigation parameters > Map path navigation parameters > Default path navigation parameters
Map
Select Map
Marker
Select Marker ID
accurate stop
close
Normal
Runtime
Path parameter
Set up
Not set up
Cancel
OK
Move
Change parameters
Copy
Delete



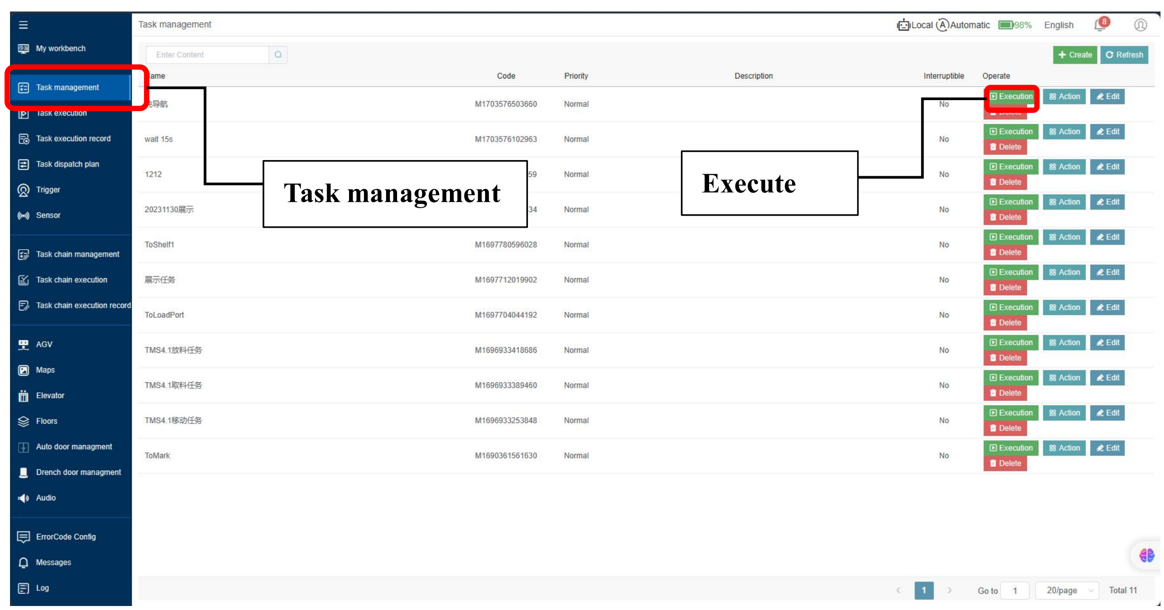
Create other tasks in the same way.

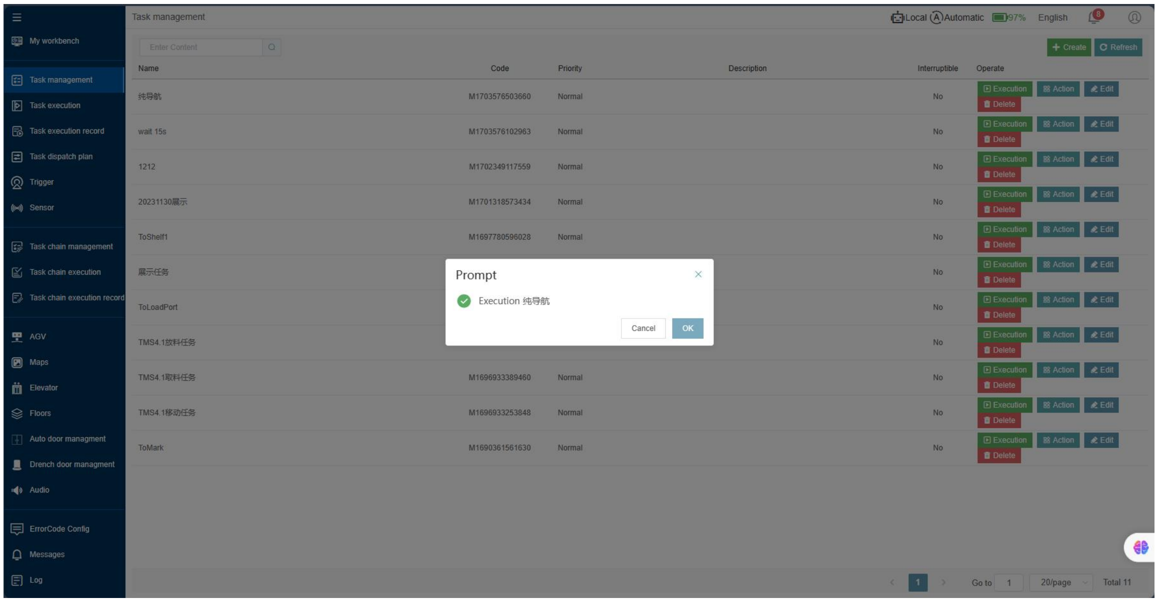


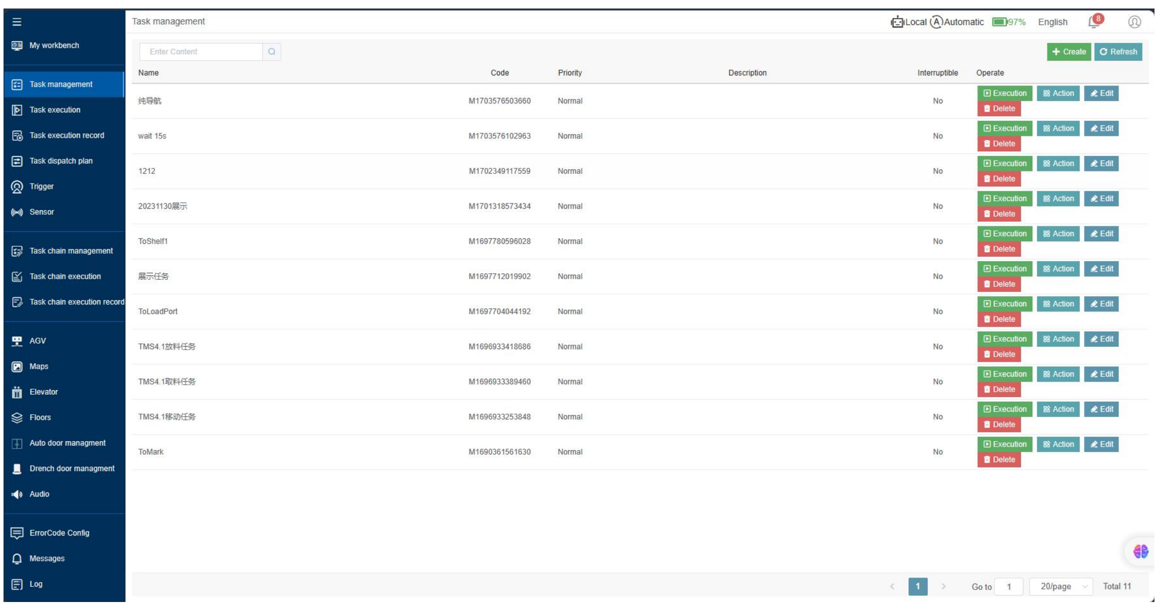
4.1.4.2 EXECUTE TASK

Click ‘AGV’ to switch to robot management page. Use the remote controller in manual mode to move the robot to move to the task point. Click ‘Control Mode’ to switch to automatic mode in the menu.

Click ‘Task Management’ to switch to the task management page, and click the ‘execute” button to start executing the preset task;

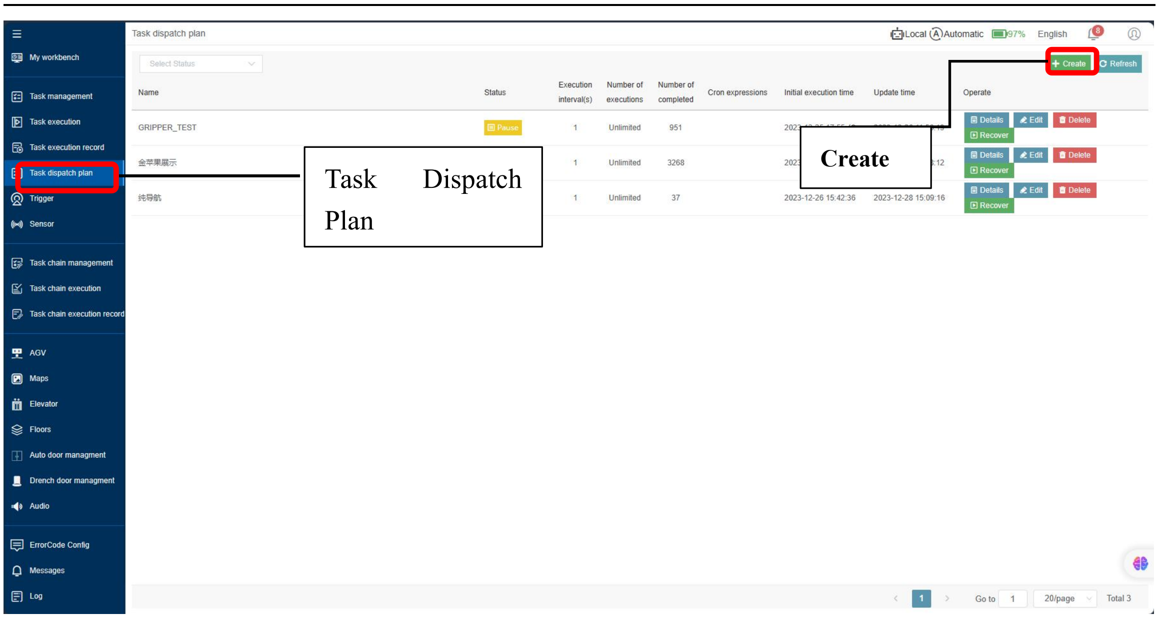


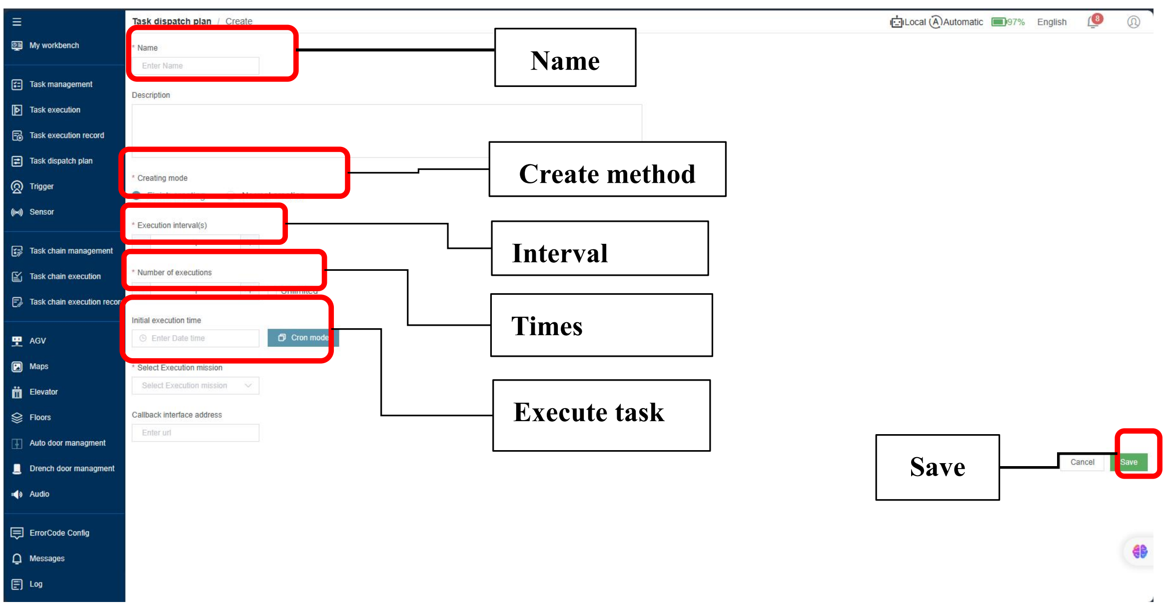


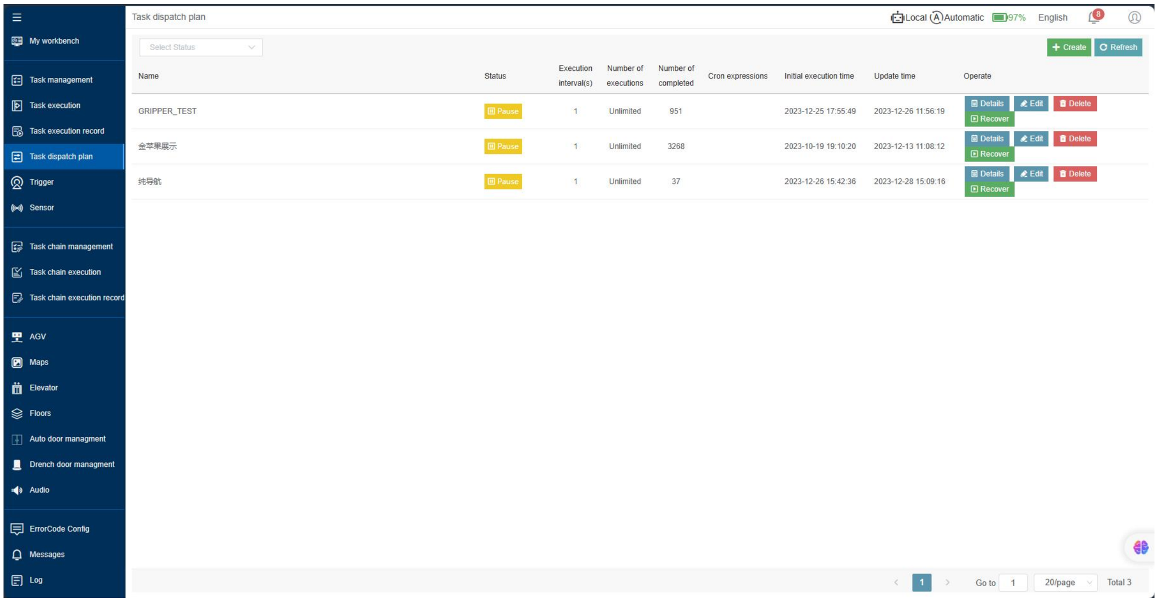


4.1.4.3 TASK DISPATCH PLAN

The task dispatch plan is mainly used to create a series of recurring tasks. Click ‘Task Dispatch Plan’ to switch to the dispatch dashboard. Click ‘Create’ to switch to the scheduling plan creation page, fill in the page as shown in the figure, and then click ‘Save’ to start executing the dispatch plan.


Task dispatch plan
Task Dispatch Plan
Create
Refresh
Name
Status
Execution interval(s)
Number of executions
Number of completed
Cron expressions
Initial execution time
Update time
Operate
GRIPPER_TEST
Pause
Unlimited
Details
Edit
Delete
Recover


Task dispatch plan / Create
Name
Description
Creating mode
Create method
Execution interval(s)
Interval
Number of executions
Times
Initial execution time
Cron mode
Select Execution mission
Execute task
Callback interface address
Cancel
Save

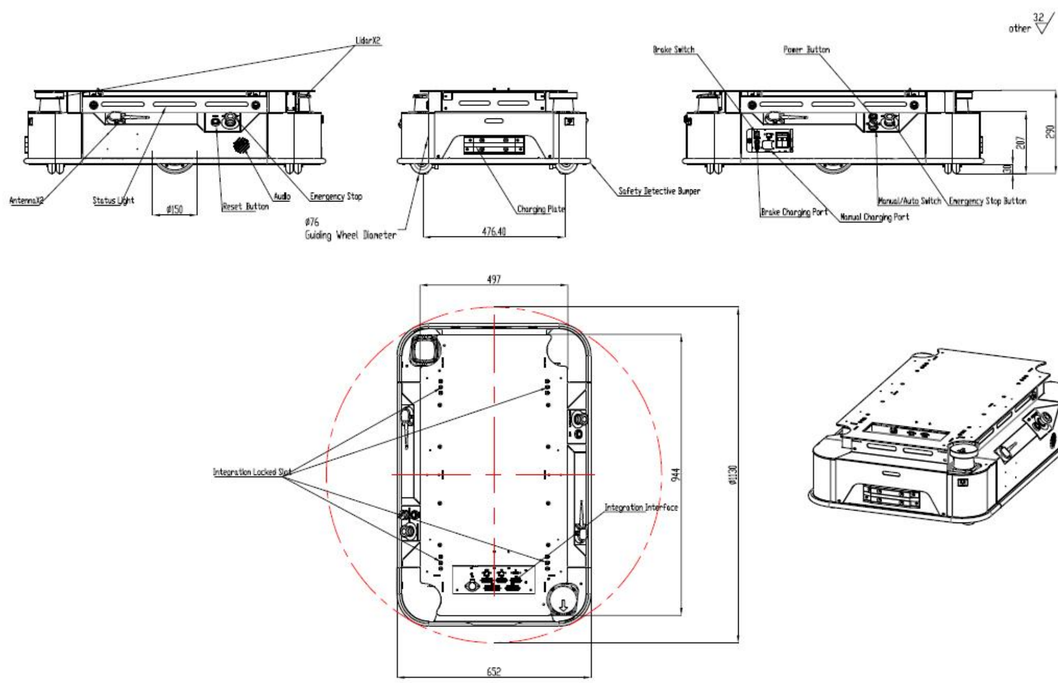


LIDAR CONFIGURATION OPERATION

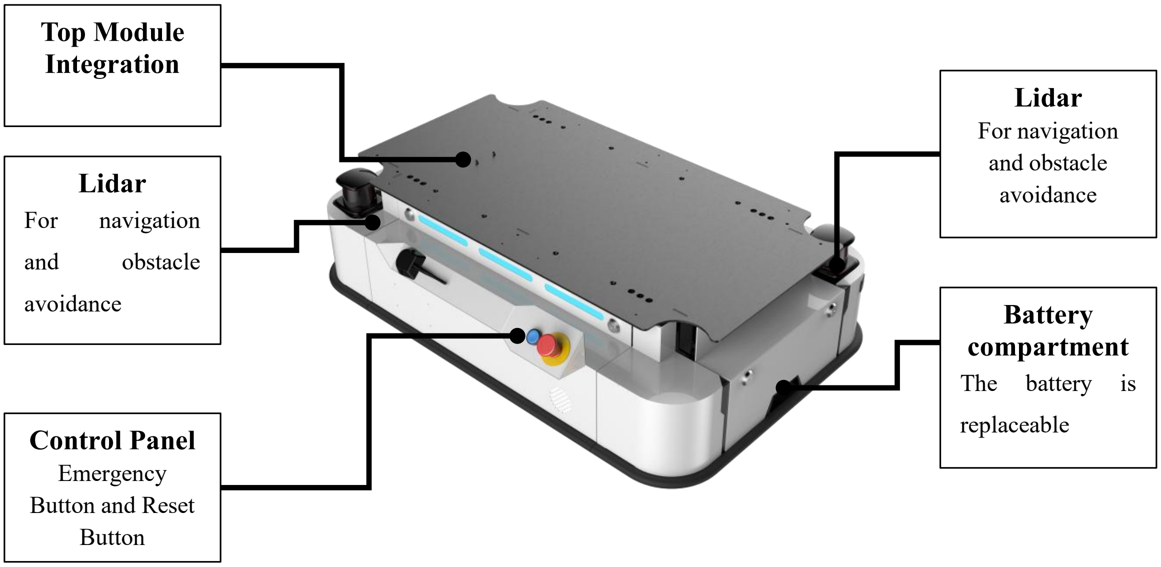
The P200 lidar area needs to be configured by using the Toolkit tool. Please consult YOUIBOT engineers for specific operations.

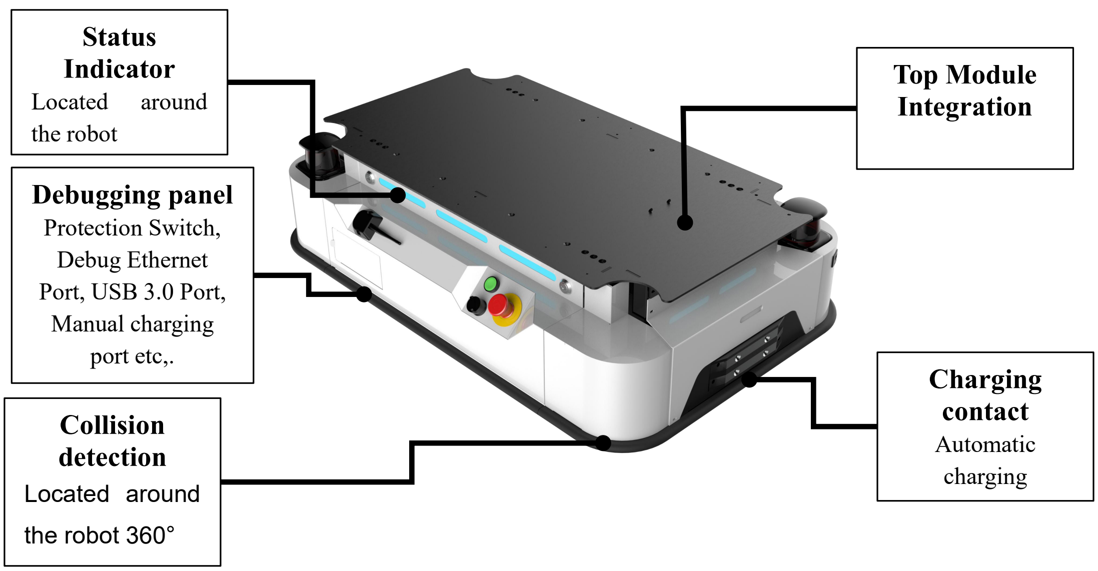
CHAPTER5
HARDWARE DESIGN

5.1 STRUCTURE



5.2 HARDWARE STRUCTURE





5.3 LIDAR

Lidar is a radar system that emits laser beam to detect the target's position, speed and other characteristics. The mechanism is by sending a probe to the target signal (laser beam) and then receives the signal reflected back from the target (target echo). By comparing the emission and receipt of signals, scanner can obtain information about its distance, altitude, speed, and even shape parameters relative to target detection. Therefore, realizing the function of tracking and recognition of external objects.

The P200 robot uses LiDAR as an important positioning sensor. It is necessary to maintain cleanliness of LiDAR surface, and be aware of objects that are lower than LiDAR or protruding to the scanner in the operational space of the robot to prevent any damages caused by navigational errors.

5.4 DEBUGGING INTERFACE COMPARTMENT

The cover debugging interface compartment on the side of the robot can be easily opened to access to the functional debugging.

It is not recommended for untrained personnel to operate and access the debug interface compartment without supervision.

Figure 5.1 Debug Interface Compartment

5.5 DEBUG INTERFACES

The debug interface compartment includes release brake, release brake power interface, Ethernet interface, power protection switch, manual charging socket and USB port, as shown in Figure 5.2.

The integrated interface component on the top of the robot is arranged with IO configuration connector, communication protocol, network port, USB port, top module's security control connector and top module's power supply interface, as shown in Figure 5.2.

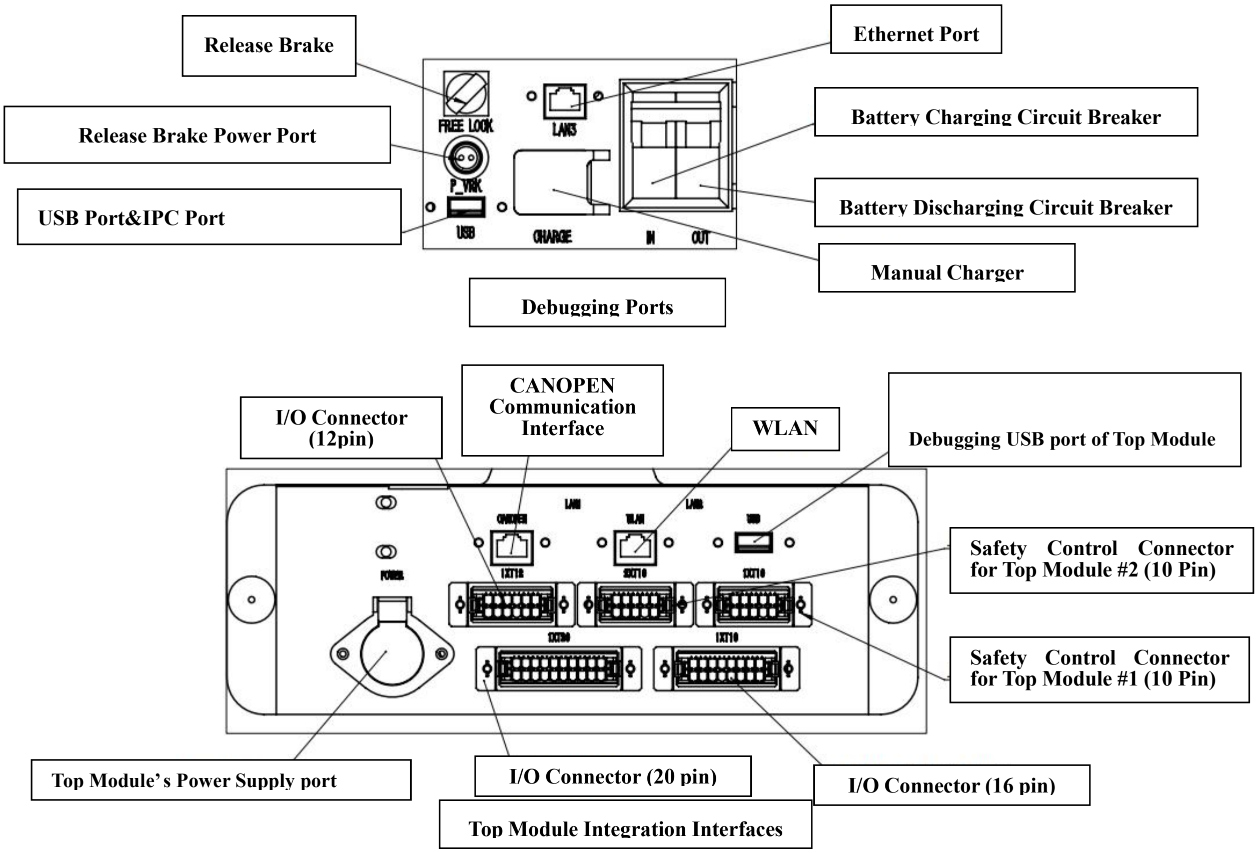


Figure 5.2 Interface schematic diagram

CHAPTER6
ROBOT POWER SUPPLY

6.1 CHARGING

6.1.1 CHARGING STATION DESCRIPTION

The automatic charging station is designed for automatic charging of the robot. The robot will automatically identify the charging station and dock with the charging station for automatic charging. After the robot receives the command for docking, the charging contact plate of the robot will connect with the battery and dock with the charger. After the docking, the charger detects that the charging contact of the robot is connected, and switch on the power supply to start charging session. During this time, please do stay clear of the connector panel on the charging station and the charging contact plate on robot. When the robot's charging contact plate disconnects from the charger, the charger immediately switches off power. During the charging situation, if robot is disconnected manually the charging contact plate might still be charged. Please stay clear from it.

Robot supports automatic charging and manual charging. There is a manual charging socket on the side of the robot, which can be connected to a manual charger if required. When manual charging is plugged in, the connector panel of the charger will also be charged. Manual charging is not recommended under general working conditions.

There is a voltage and current display on the top of the charging station. The red number is the power supply voltage of the charging contact plate, and the blue number is the power supply current of the charging contact plate. During the charging process, the charging station will automatically adjust the power supply current of the robot, with a range of (5-15A).

6.1.2 VOLTAGE AND CURRENT SETTING

There will be a blinking dot as an indication in the default interface when the charger is set up, when the dot is blinking in the red column, long press the power down key to enter the voltage setting, short press the up and down keys to adjust the voltage, short press the confirmation key to confirm the setting. When the dot is blinking in the blue field, long press the power down key to enter the current setting, short press the up and down keys to adjust the current, short press the confirmation key to confirm the setting.

6.1.3 PRECAUTIONS FOR CHARGING

- Do not place reflective objects (such as mirrors) near the charger.
- In the process of docking between the robot and the charger, there should be no interference or interruption of the session. Workers should stay clear of the charger and the robot. The people are not allowed to walk around between the robot and the charger. The robot is not to be interrupted or deviated from its path with external forces.
- Before ending a charging session properly, the charging station shall not be powered off. Robot and the charger shall not be separated manually.
- The best distance between charging station and charging point is 50cm-120cm.

6.2 QUICK BATTERY SWAP

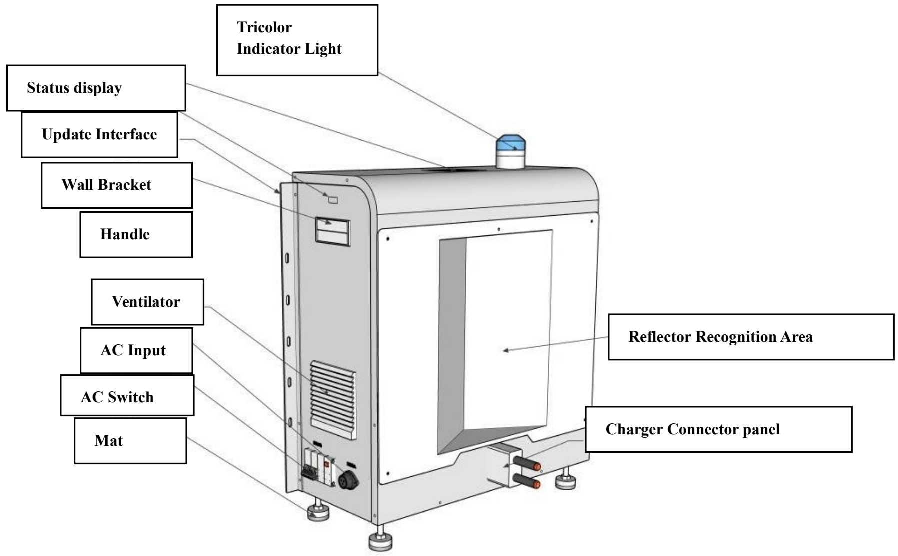
The P200 robot has a quick battery replacement function.

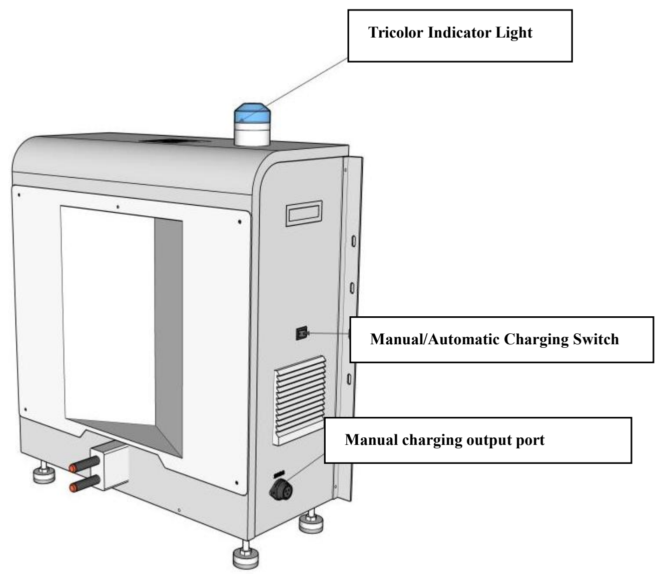
The procedure for replacing the P200 robot battery is as follows:

1. Long press the power button of the robot to turn off the robot;
2. Open the debugging interface compartment to switch off the power circuit breaker;
3. Open the cover panel on the battery compartment at the rear side of the robot;
4. Disconnect the power cable.
5. Disconnect the BMS signal cable.
6. Retrieve the battery and replace it with a new one.
7. Connect BMS signal cables.
8. Connect power cables.
9. Close the robot battery replacement window, switch on the power circuit breaker from debug interface compartment, and close the cover shut.

6.3 BATTERY OVERDISCHARGE RECOVERY

When the battery is over discharged, the power supply of the robot cannot be turned on, and there is no current or voltage using the manual charging cable on the automatic charging station. Please follow the battery unlocking process to recover:





1. Switch the forced charging switch button to MANUAL, the DC output interface outputs no-load voltage, and the front charging brush plate is not charged.

2. The no-load voltage is the screen setting voltage value, and the no-load output current is 5A.

3. When the charger detects current at the output port in forced mode, the charger will charge according to the normal curve.

4. When the charging completion conditions are met, the charger waits for the battery to leave and is in standby mode.

5. After completing charging, please turn off the forced charging switch.

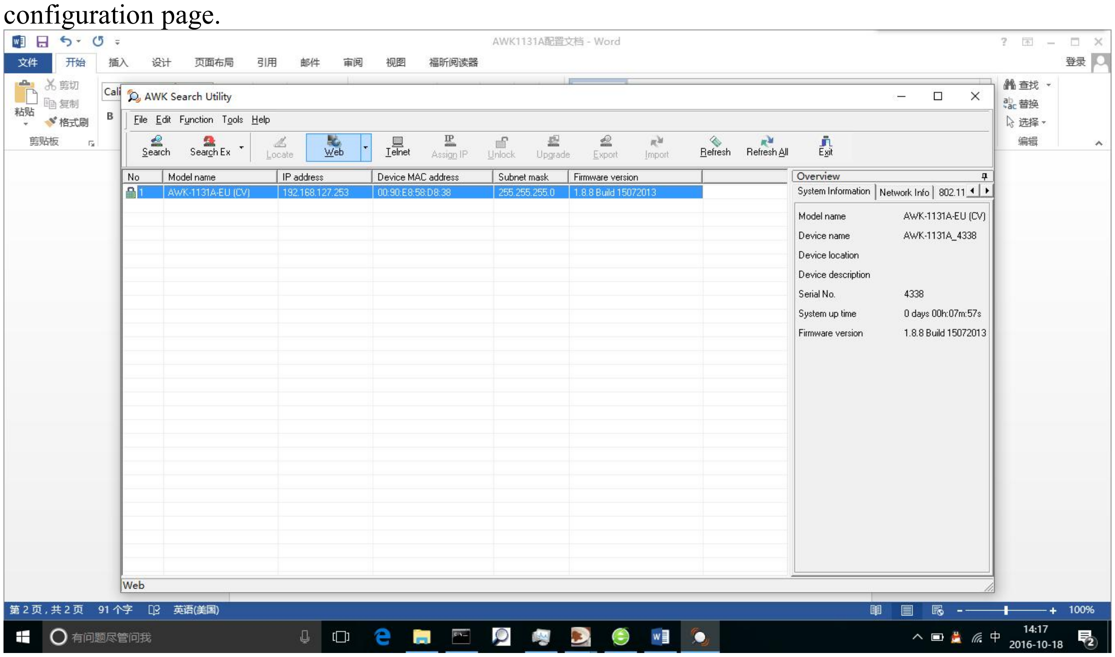
6. (When switching to AUTO, the charger detects the voltage of the charging brush block and realizes automatic charging).

CHAPTER7
NETWORK

The current section introduces the steps to set up the robot wireless AP. This section takes the Moxa brand AP as an example. The settings of other brands of wireless AP are similar. If you have any questions during use, you can ask Youibot technical staff for help.

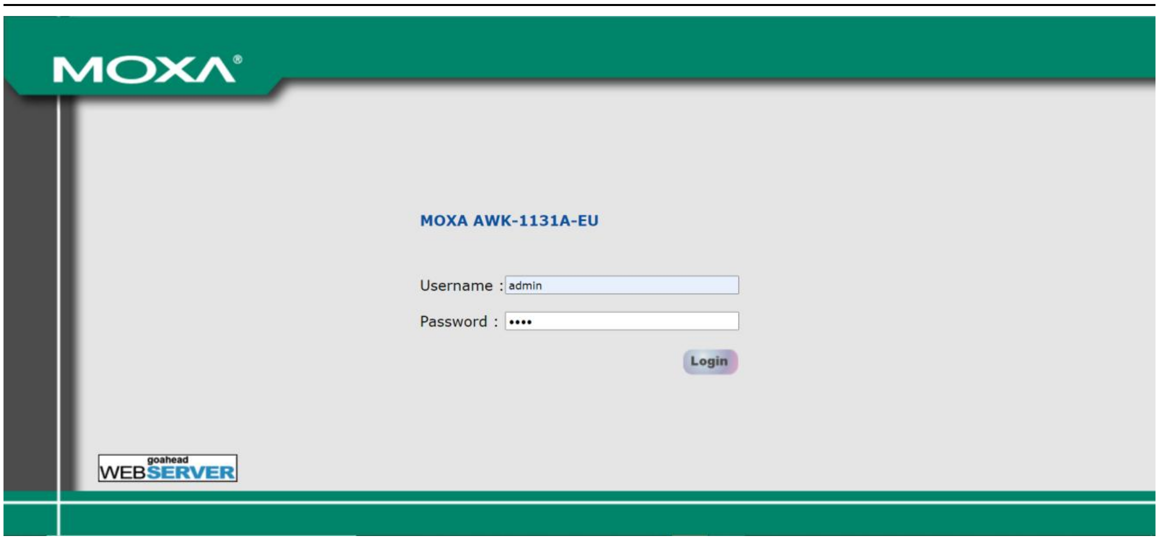
7.1.1 CONNECT TO THE CLIENT

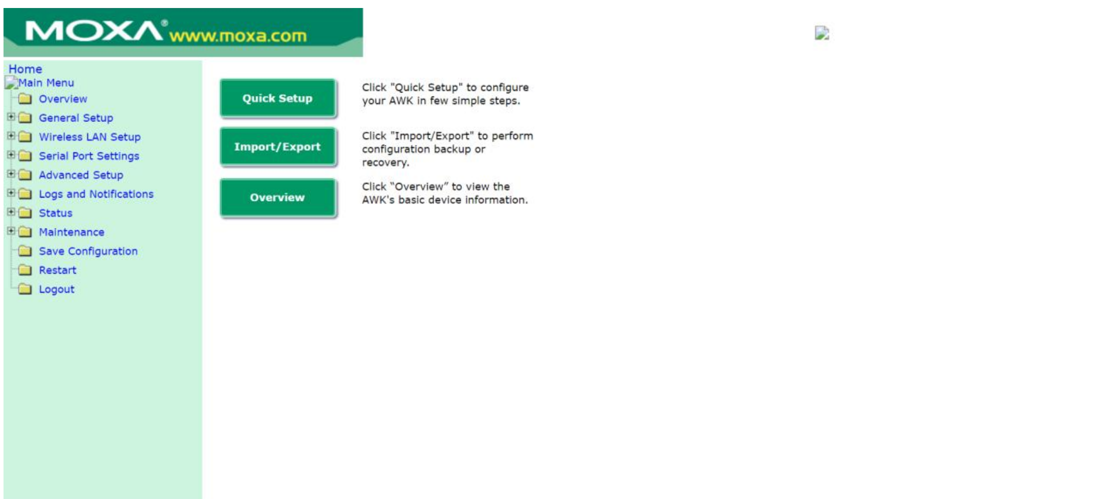
1. Use a network cable to connect the MOXA client and the mobile PC, and turn off the WiFi on the PC.
2. Open MOXA tool ‘Wireless Search Utility’ and click ‘Search’. The network starting with ‘AWK’ in the list is the MOXA client we need to configure.
3. Select the network in the list, click ‘Web’, and the software will automatically go to the configuration page.



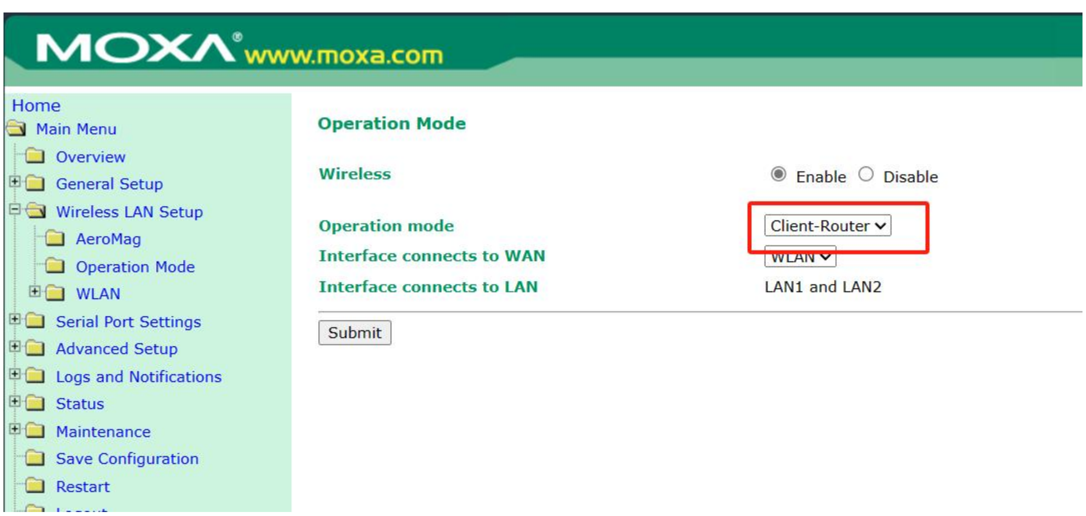
7.1.2 CONFIGURE MOXA

1. Log in to the system with the username ‘admin’ and password ‘moxa’.

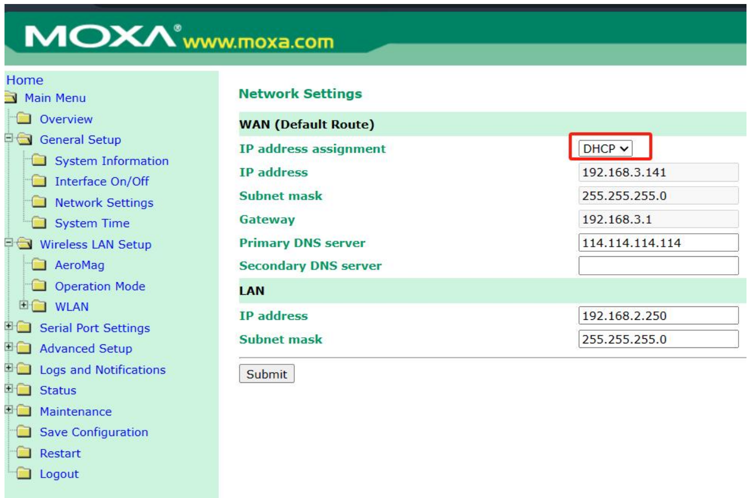




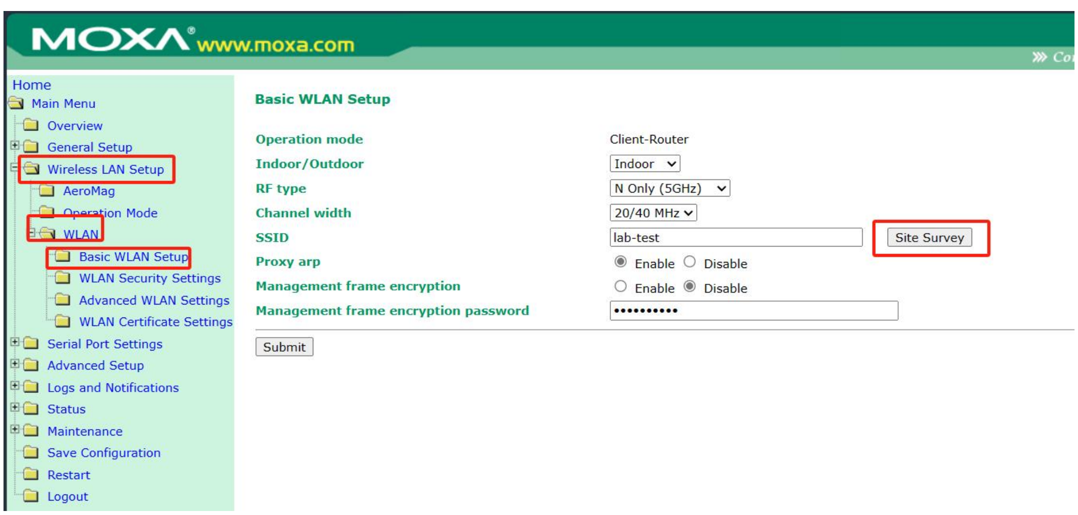
2. Turn to “wireless LAN Setup’ and then enter the “Operation Mode”, select the Operation mode to “Client-Router” mode and click “Submit” to save the setting.

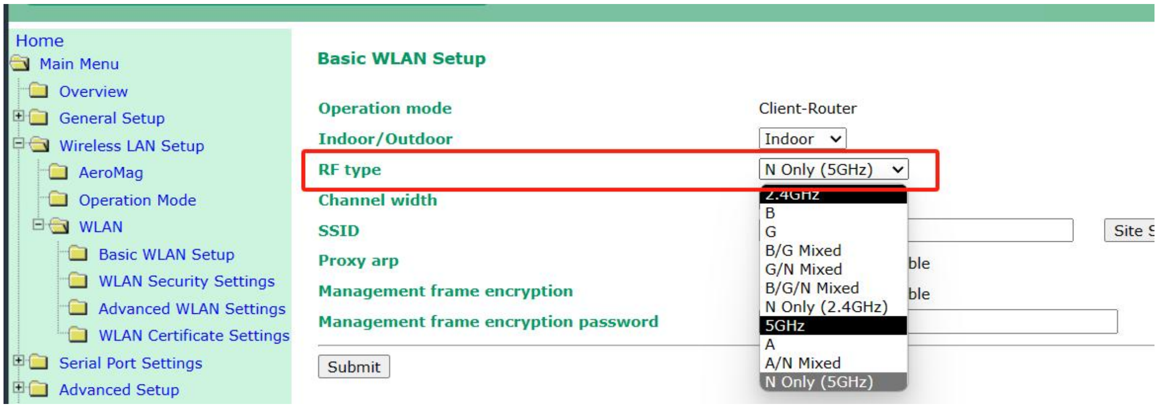


3. Turn to “General Setup” and then enter the “Network Settings”, select the “IP address assignment” to “DHCP” mode and click the “Submit” to save the setting.

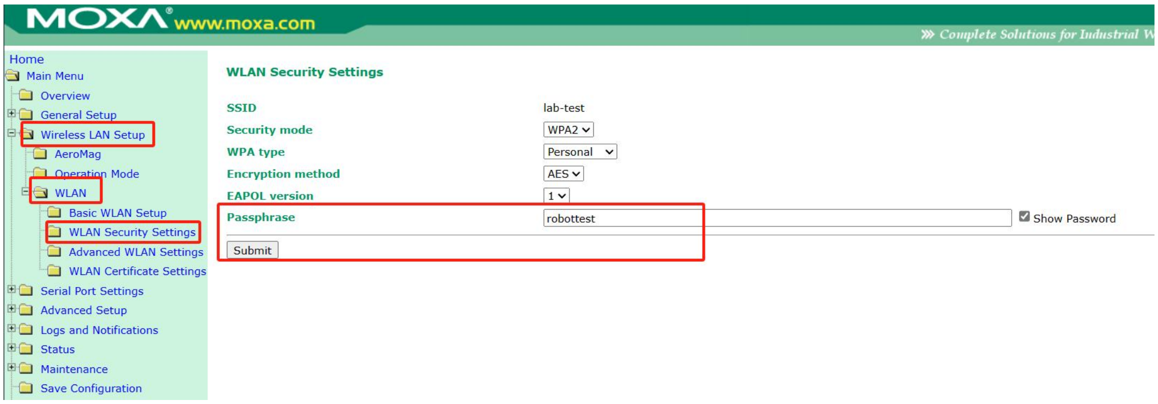


4. Turn to “Wireless LAN Setup’ and then enter the “WLAN” then “Basic WLAN Setup”, click the “Site Survey” to find the WIFI which user wants robot connect and then select the RF type of the WIFI. After finishing the connection, click the “Submit” to save the setting.

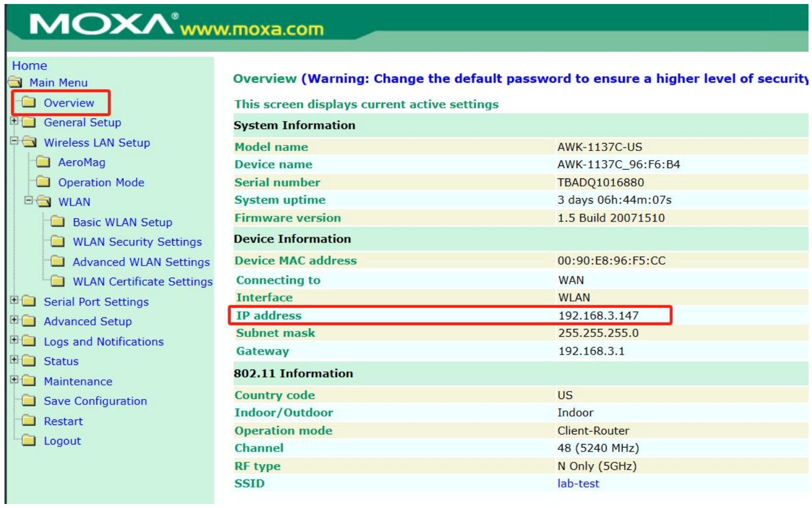




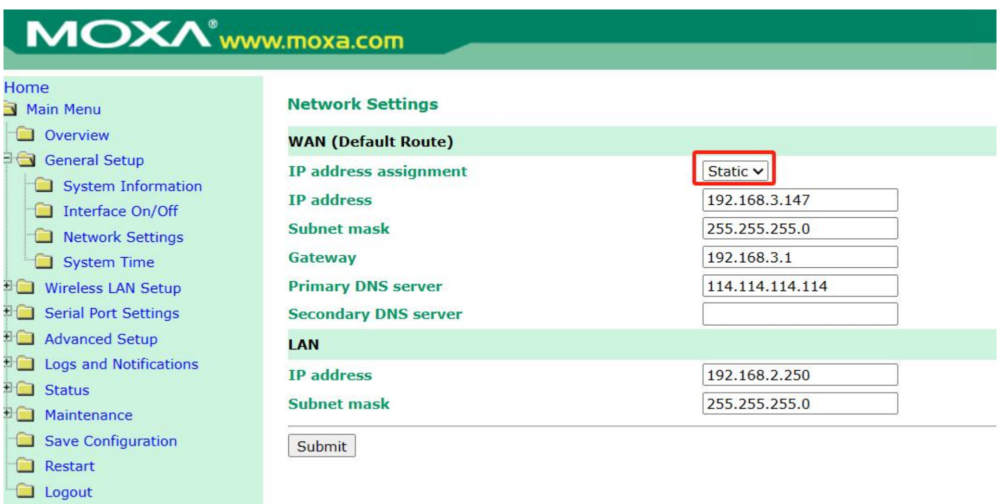
5. Turn to "Wireless LAN Setup' and then enter the "WLAN" then "WLAN Security Settings", enter the WIFI password in the "Passphrase" bar, click the "Submit" to save the setting and restart the MOXA module. **Make sure the password is correct or the robot cannot connect to the WIFI successfully.**



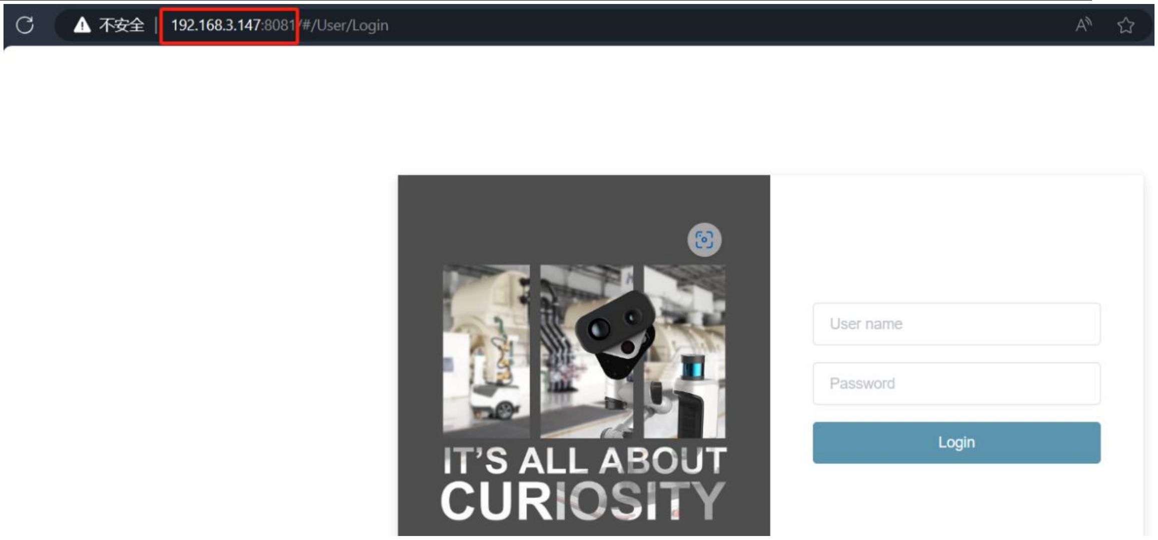
6. After hearing a "didi" sound, the MOXA module restart is done, and then log in to the MOXA module again check the "Overview" and check the IP assigned by the local router.



7. Turn to “General Setup” and then enter the “Network Settings”, select the “IP address assignment” to “Static” mode input the “IP address” checked in step 6, and change the Gateway to the same network segment. click the “Submit” to save the setting and restart the MOXA module.



8. After finishing the MOXA module, the user can log in to the YOUICOMPASS system by inputting the IP address in step 6 and adding the port number (8081).


不安全
192.168.3.147:8081/#/User/Login
IT'S ALL ABOUT
CURIOSITY
User name
Password
Login

CHAPTER8 TOP MODULE INTEGRATION INSTRUCTIONS

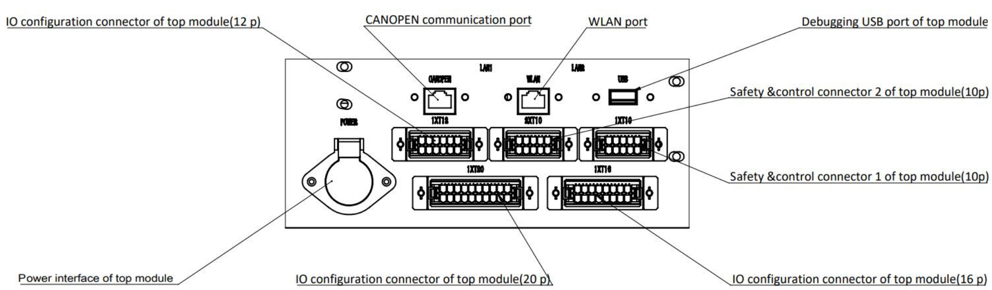
8.1 LIMITATION

8.1.1 WEIGHT AND SIZE LIMIT

The weight of top module should not exceed 200KG as the payload of robot base is limited and whole robot safety should be considered. And its size should not exceed 650mm(width) and 1014mm(length). In terms of height, it is not recommended to make the top module quite high to avoid the accidents such as tipping and overturning etc., caused by the swing of the center of gravity. It is recommended that the height of top module is not more than 1800mm and the height of the center of gravity is not greater than 600mm, which is one third of the while robot height.

8.1.2 POWER LIMIT

The robot base has external power extension interface, which can provide1 channel 48V DC, the power supply current is not greater than 15A, the power is not greater than 720W;1 channel 24V DC, the power supply current is less than 3A, and the power is less than 72W.



Top module ports 2

8.2 TOP MODULE FUNCTION INSTRUCTION

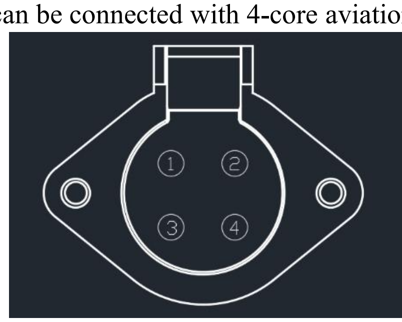
The top module can be integrated into the robot system through extension interface to realize safety&protection, status display, collaborative control and other functions.

8.3 THE WIRING INSTRUCTION OF TOP MODULE

This section introduces the functions and definitions of the ports and these interfaces. If the specific port number is not listed when it introduces detailed wiring method, it defaults to corresponding port number of the current section.

8.3.1 POWER SUPPLY OF TOP MODULE

There is power supply port that can be connected with 4-core aviation plug.



Pin definitions are shown in the following table.

Pin number	Symbol	Project	Description
1	2P24	24 V DC positive pole	[Port Description] Supply 24V DC power for the top module [Recommended wire] Copper wire, cross sectional area≥0.5mm² Load parameters: Current ≤3A, power ≤72W
2	2N24	24 VDC negative pole	
3	111B+	Connect to battery directly DC positive pole	[Port Description] Supply the power for the top module. The voltage changes as the battery level changes. Max 58.4V, Min 51.2V(Battery protection system is activated) [Recommended wire] Copper wire, cross sectional area≥0.5mm² 【 Load parameters 】 Current ≤15A, power ≤720W
4	B-	Connect to battery directly DC negative pole	

8.3.2 NETWORK PORT OF TOP MODULE

The integration extension ports of top module include 3 network ports which contain two internal network segments (LAN1 and LAN2) and one external network segment (WLAN). The default internal network segment is 192.168.0.***, and the default external network segment is 192.168.2.***. The internal network segment is used for communication between the internal devices of the robot base, while the external network is used for communication between the scheduling system and the upper system, etc. During usage, the device that needs to use the external network segment and the external network segment should not be set up to 192.168.0.*** otherwise, the robot will crash. The performance and definition of the two network ports are the same.

When using WLAN port, please take the top module as Client and use MODBUS TCP protocol to connect to YOUIPilot 192.168.2.***+ 3001.

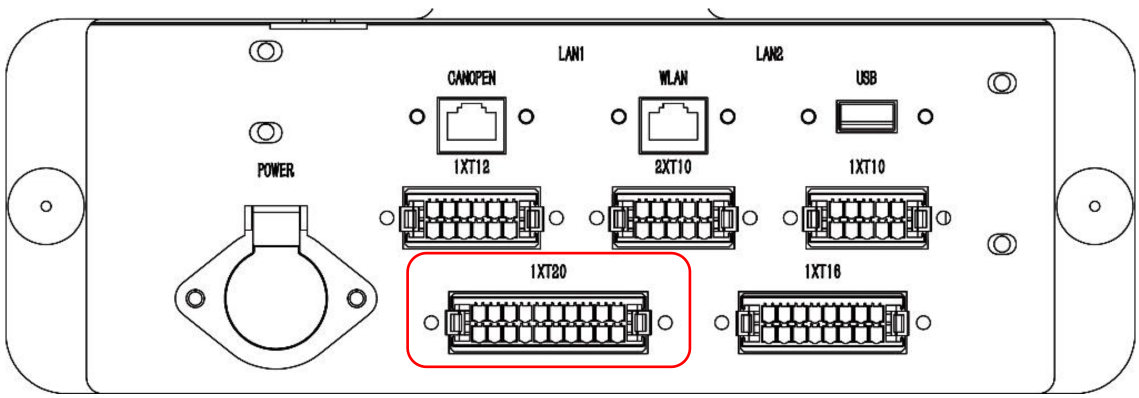
Please find the port definitions as follows:

<table>
<tr><th>Port Name</th><th>Network segment</th><th>Network segment IP</th><th>Description</th></tr>
<tr><td>WLAN</td><td>The external network</td><td>192.168.2. * * *</td><td rowspan="3">Port Description: 100 MBIT/s network communication port
[Recommended cables] Category 5 cable (Cat 5) and cable that is better than category 5 cable
Wiring Connection Mode: T56A or T56B</td></tr>
<tr><td>LAN1</td><td>The internal network</td><td>192.168.0. * * *</td></tr>
<tr><td>LAN2</td><td>The internal network</td><td>192.168.0. * * *</td></tr>
</table>

8.3.3 USB PORT OF TOP MODULE

The extension ports of the top module include a USB extension port. This port is directly connected to the industrial personal computer, so that it is not recommended to use it as a power supply. At the same time, this port does not support extension. This port can be connected to the wireless controller receiver when the robot box is unpacked at the first time, or for debugging.

8.3.4 WIRING CONNECTION INSTRUCTIONS OF PORT 1XT20(REGISTER)



The integration interface of top module offers the related **functions of holding registers**. This function is integrated in the 1XT20 port. The operator is able to use the read/write register in YOUICompass to read and write the value, or use other controllers for this kind of operation. When in use, connecting the A1 pin to 24V- is for PNP input, and connecting the A1 pin to 24V+ for NPN input. 24V- and 24V+ can be provided by A1 pin and A2 pin in 1XT12, or provided by pin 1 and pin 2 in the Power socket or provided by the stabilized power supply of top module. The pin definitions in A1 are shown in the following table:

Pin number	Symbol	Address	Description
Input extension point, MODBUS_TCP server, IP: 192.168.0.50, port: 502, when using the YOUICompass to operate, select function code 03 to read, the standard current is 1.5A@24V DC, and the inrush current is Max7. 5A@24V DC			
A1	DXCOM	/	Common port of input module
A2	DX0	4196	High level 1， low level 0 (high level refer to high voltage status while low level refers to low voltage)
A3	DX1	4197	High level 1，low level 0
A4	DX2	4198	High level 1，low level 0
A5	DX3	4199	High level 1，low level 0
A6	DX4	4200	High level 1，low level 0
A7	DX5	4201	High level 1，low level 0
A8	DX6	4202	High level 1，low level 0

A9	DX7	4203	High level 1，low level 0
A10	Reserved	/	
Output extension point, MODBUS_TCP server, IP: 192.168.0.50, port: 502, when using YOUICompass to operate, select function code 06 to write, the maximum load is 1.5A@24V DC			
B1	DYCOM	/	Common port of output module
B2	DY0	4206	High level 1，low level 0
B3	DY1	4207	High level 1，low level 0
B4	DY2	4208	High level 1，low level 0
B5	DY3	4209	High level 1，low level 0
B6	DY4	4210	High level 1，low level 0
B7	DY5	4211	High level 1，low level 0
B8	DY6	4212	High level 1，low level 0
B9	DY7	4213	High level 1，low level 0
B10	Reserved	/	/

The instructions for YOUICompass to read/write the register

This section only introduces how to use the register. For detailed operations and content descriptions such as "action management", "task management", "global variables", and "sequence number" involved in the operation process, please refer to the relevant information in the "YOUICOMPASS User Manual", which will not be introduced repeatedly in this section.

There are three types of registers as check register, read register, write register in the task list of task management page on YOUICompass as shown in figure 8.1. Check register is used for reading the current value and putting it into global variable so that it is convenient to use the value for logical judgement in action flow. The read register is used for reading the register while the write register is used for writing the register.

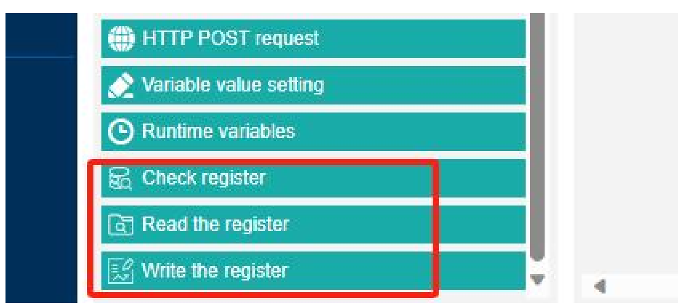


Figure 8.1 Register function

【Check register】

After clicking the “check register” on the left action list, please enter the action editing page as shown in Figure 8.2. Please select the remote mode and enter IP address 192.168.0.50, port 502, function code 03-read holding register, slave ID 1. Then please enter the register address that needs to be read as per actual wiring connection.

Please set the selected register value（usually 1）in “End of Action”. Please set the selected register value representing exception (usually 0) in “task exception”. Please set up the time that is determined as timeout (unit: second) in “task timeout”. Finally click “confirm” to finish the action creation.

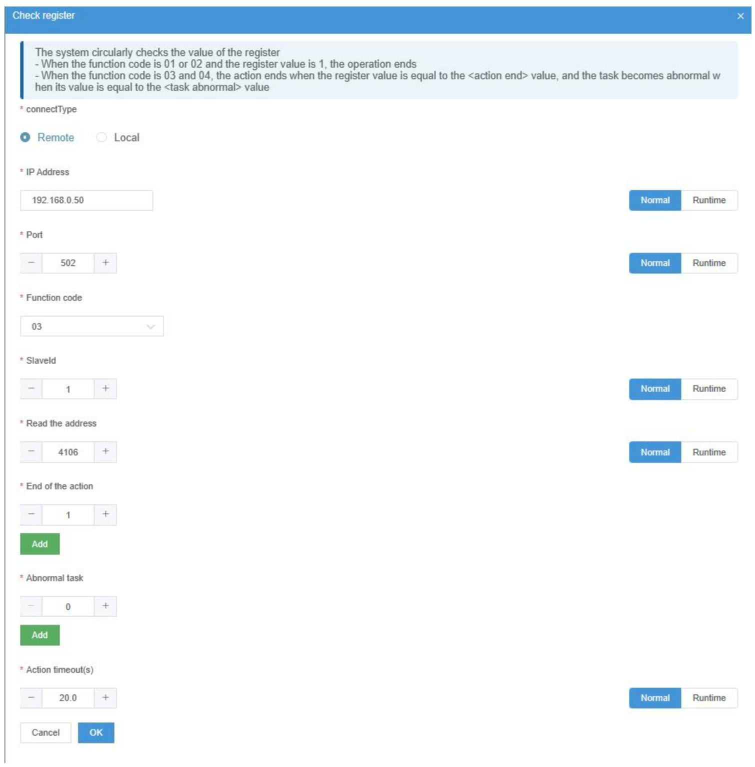
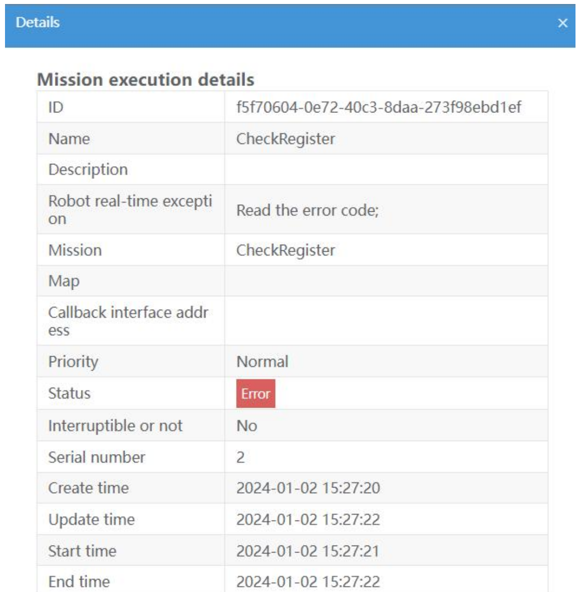


Figure 8.2 Check register editing page

Please click "Execution" in task management list to judge the value of the register according to the task status. For example, if the task ends normally, the value is 1; if the task is abnormal and the error code is read and displayed on the detail page, the value is 0, as shown in Figure 8.3. Please note do not set "End of Action" and "task " to the same value, otherwise, it will cause unnecessary risks!



Details

Mission execution details

ID	f5f70604-0e72-40c3-8daa-273f98ebd1ef
Name	CheckRegister
Description	
Robot real-time exception	Read the error code;
Mission	CheckRegister
Map	
Callback interface address	
Priority	Normal
Status	Error
Interruptible or not	No
Serial number	2
Create time	2024-01-02 15:27:20
Update time	2024-01-02 15:27:22
Start time	2024-01-02 15:27:21
End time	2024-01-02 15:27:22

Figure 8.3 The error information of check register

【Read register】

To read the register is similar to check the register. The difference is that the read register only completes the operation of reading the value but not change the task status as per the value. Normally, reading the register is used for checking the current electric potential of the input register and coordinating with global variables and HTTP POST requests.

In the upper right corner of the action editing page, click Global Variables to open the global variable editing page, enter the variable name and default value, and click the Save button. If the operator needs more global variables, click the Add button, and then set up the name and default value respectively, as shown in Figure 8.4. **Please note that the names of different variables cannot be the same.**

Click the "Read Register" action on the left action list to enter the action editing page, as shown in Figure 8.5. Select remote mode, input IP as 192.168.0.50, port as 502, select function code as 03-read holding register, slave id as 1, and then input the address of the register to be read according to the actual wiring connection, the input register address is from 4196 to 4203. Select the previously saved global variable in "Read Result" to use in other actions.

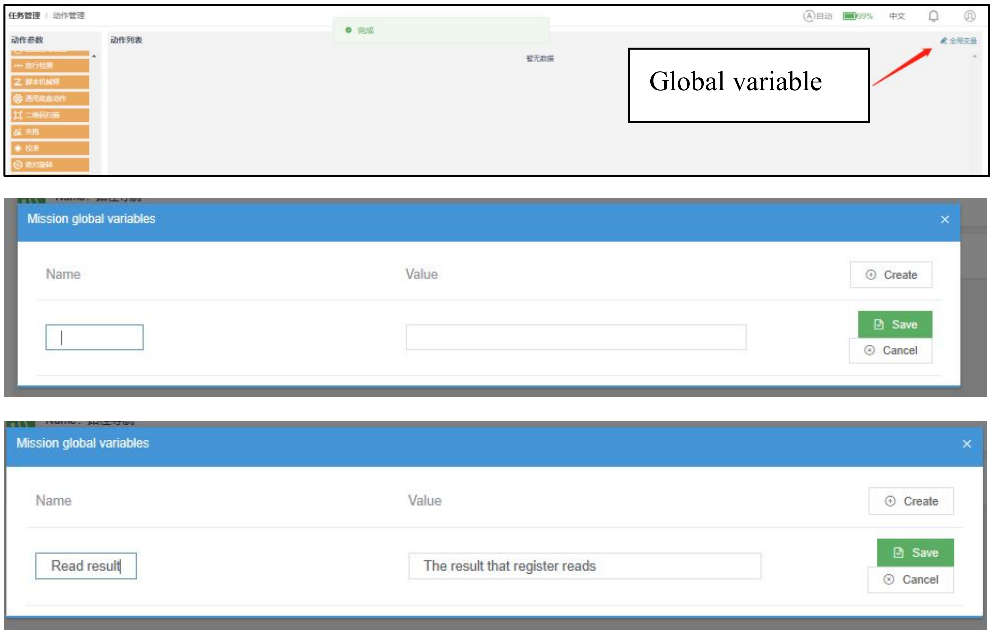


Figure 8.4 Global variable editing page

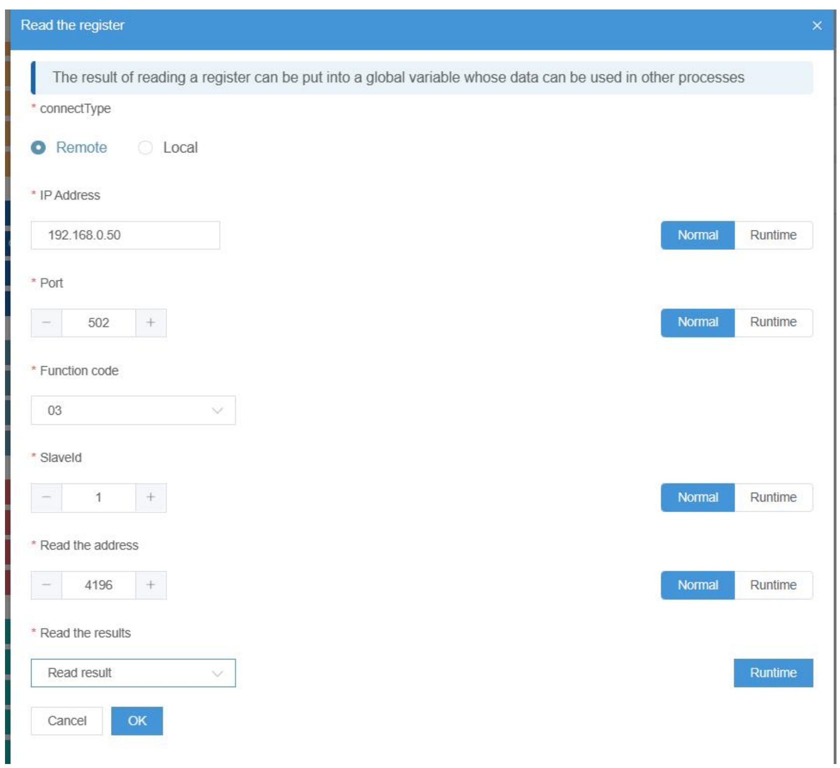


Figure 8.5 Read register action edit page

Normally, the reading result can be sent to the specified interface using HTTPPOST request action, as shown in Figure 8.6, in httpBody (send content), select the runtime mode on the right side, and then select "read result" the global variable, enter the URL of the input port, and click OK. The overall action sequence is shown in Figure 8.7. The execution result of the whole actions of Figure 8.7 is that read the specified register address's current status and then send the result to the interface http://localhost:8787/result.

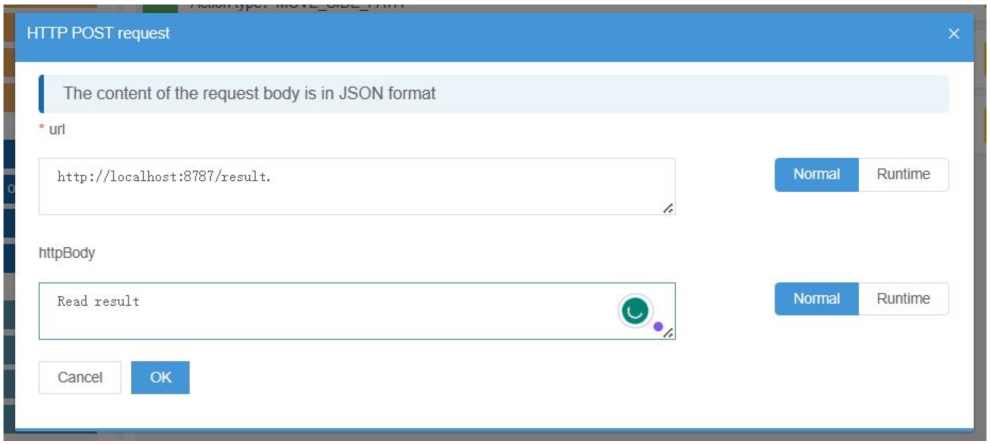


Figure 8.6 HTTPPOST request action edit page

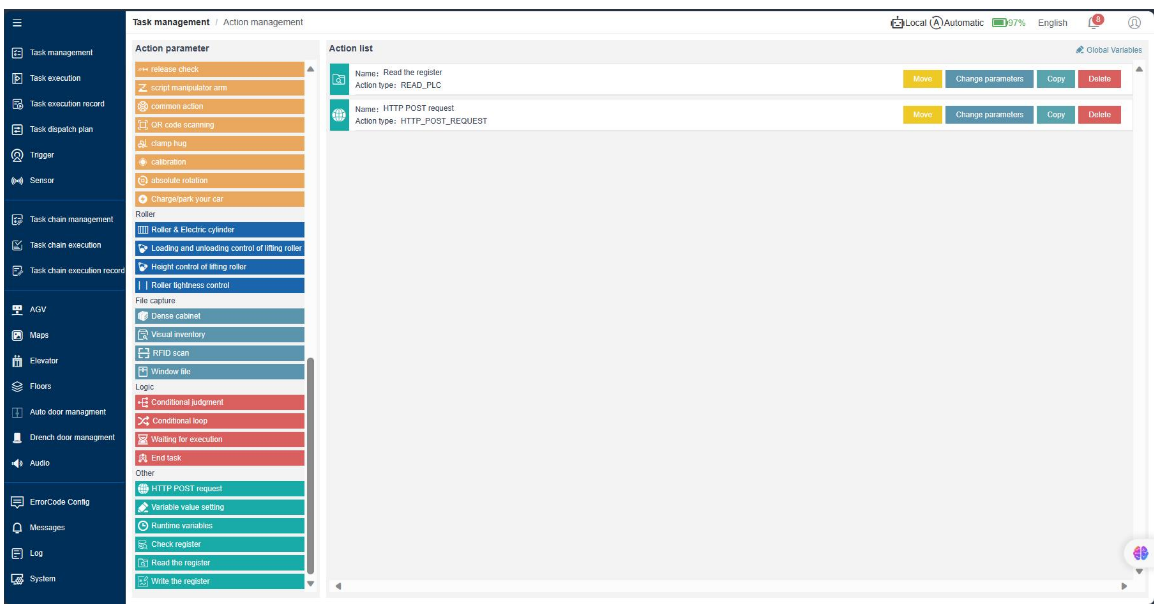


Figure 8.7 Result read&send action list

【Write register】

The write register supports writing the single holding register or multiple holding registers. Among them, function code 06 corresponds to "06H - write the single holding register" in the MODBUS function code, and 16 corresponds to "10H - write the multiple holding registers" in the

MODBUS function code. Please select remote mode, IP address 192.168.0.50,port number 502,slave ID 1.

When selecting the function code 06(06H), please enter the register address that needs to be written as per the actual wiring connection. The output register address is from 4206 to 4213. It will change the corresponding value of this register after the required value that also can be written through global variable is entered as shown in Figure 8.8. When selecting the function code 16(10H),the register address is start bit address for required operation and the number of values that have been filled in written value will be operated as many bits as possible, and the maximum number can't exceed 8bits.Global variables cannot be used at this time, as shown in Figure 8.9.

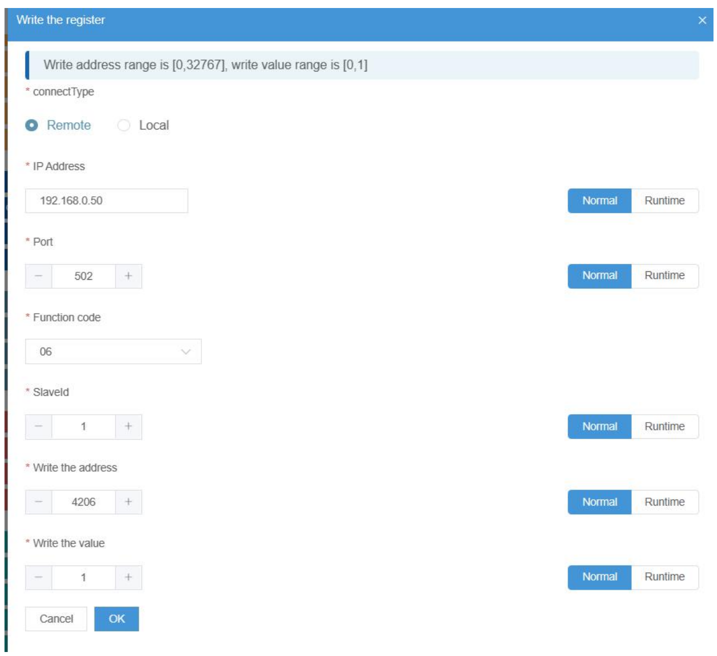


Figure 8.8 Write the single register value

8.3.5 WIRING CONNECTION INSTRUCTIONS OF PORT 1XT16

1XT16 ports contain function extension ports such as function button, status button and so on, which can extend the functions of tricolor status indicator, power on/off of cobot, reset button, manual/auto switch knob and light strip.

The pin definitions are shown in the following table:

Pin number	Symbol	Effective Electrical potential	Description
A1	Y0	1N24	The drive signal of red status indicator (tricolor indicator)
A2	Y1	1N24	The drive signal of green status indicator (tricolor indicator)
A3	Y2	1N24	The drive signal of yellow status indicator (tricolor indicator)
A4	Y3	1N24	Undefined
A5	C3	Defined as per wiring connection	The output common port of the cobot control cabinet. Determine the effective electrical potential of this pin
A6	Y11	Defined by A5 pin	Power-on signal of cobot control cabinet
A7	Y12	Defined by A5 pin	Power-off signal of cobot control cabinet
B1	X21	2N24	Reset button extension collection point
B2	X22	2N24	Manual/automatic switch knob collection point. Normally open contact corresponds to manual mode, normally close contact corresponds to automatic mode.
B3	Y20	2N24	indicator drive of reset button extension
B4	2TR485A	/	extension light strip controller 485-A terminal
B5	2TR485B	/	extension light strip controller 485-B terminal
B6	Undefined	/	
B7	Undefined	/	
B8	Undefined	/	

Wiring connection instructions of extension tricolor indicator

The tricolor status indicator can be extended for top module to indicate the current indicator status. Please refer to 3.6.4 indicator effect &status table for the status indicators' definitions. Please connect the red control terminal of tricolor status indicator to Pin A1, green control terminal to A2 and yellow control terminal to A3, and common port is connected to A2 pin of 1XT12 directly, which completes tricolor status indicator extension.

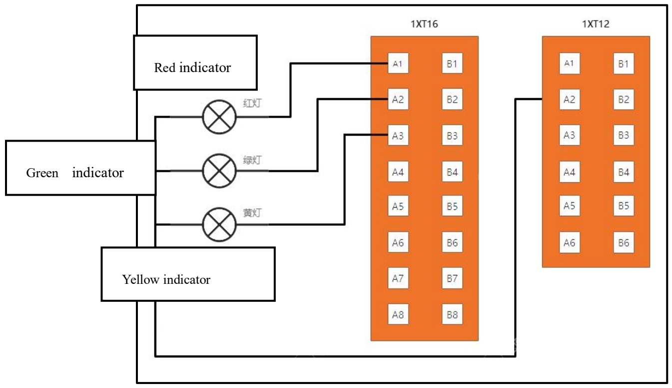


Figure 8.10 Extension tricolor status indicator circuit diagram

Wiring Connection instructions of cobot power-on/off

Except that POWER port is connected to power supply and WLAN port is connected to network cable, the control cabinet of cobot can receive the ON signal of robot base through A6 pin and OFF signal of robot base through A7 pin to power on or off the robot base and cobot at the same time. After the output common terminal of the cobot's control cabinet is connected to pin A5, the control cabinet of cobot can receive the on/off signal from pin A6/A7 when robot base is powered on or off.

In this way, it realizes robot base ON/OFF linkage.

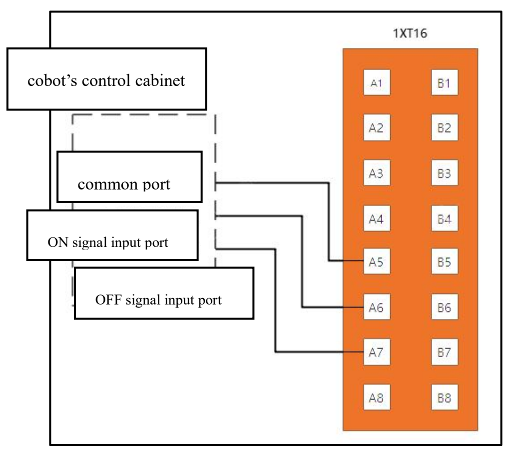


Figure 8.11 Cobot power-on/off extension---circuit diagram

Wiring Connection instructions of extension reset button

To facilitate the reset operation, additional reset buttons can be added on the top module. Please connect the normally open contact 1.a of reset button to Pin 2 of the Power port and 1.b to pin B1. Please connect 1.a to Pin A2 of 1XT12 and 1.b to pin B3 to complete the reset button extension. Please refer to circuit diagram figure 8.12.

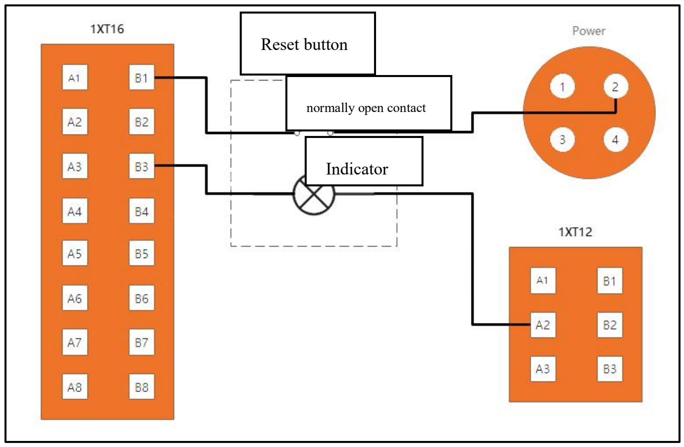


Figure 8.12 Reset button extension circuit diagram

Wiring Connection instructions of manual/auto switch knob extension

To facilitate the manual/automatic switch operation, additional manual/auto switch knob can be installed on the top module. Please connect the common port of manual/auto switch knob to pin 2 of power port and connect the normally closed terminal to Pin B2 to realize the manual/auto switch knob extension.

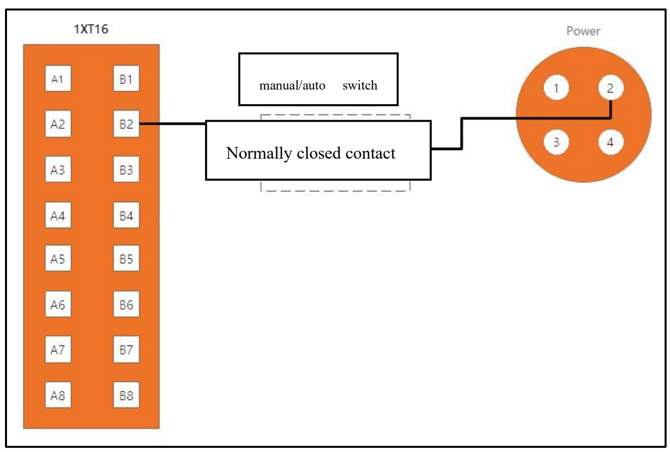
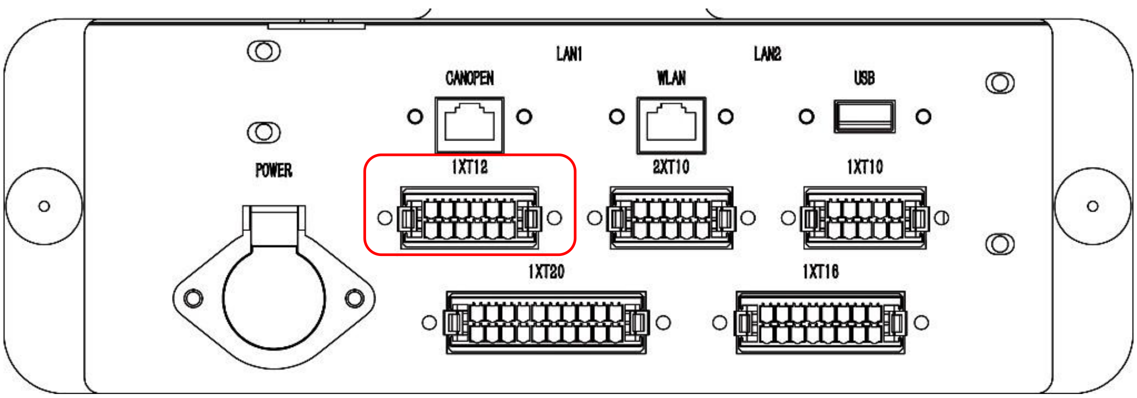


Figure 8.13 manual/auto switch knob extension

Wiring connection instructions of light strip extension

More indicator lights like turn light, multicolor status light etc., can be integrated into the top module and managed by the multicolor status light controller of top module which is optional. When the multicolor status light controller is powered on, please connect the A terminal of 485 communication port to Pin B4 and the B terminal to Pin B5.

8.3.6 WIRING CONNECTION INSTRUCTIONS OF PORT 1XT12



The 1XT12 port contains the start button of external power supply, signal collection port of anti-collision bumper. The 1XT12 supplies the extension connection of power start button, power button and front/rear anti-collision bumper of robot base for top module.

Pin definitions are shown as follows:

Pin number	Symbol	Effective Electrical potential	Description
A1	1P24	/	Start signal drive of top module's extension device - positive pole
A2	1N24	/	Start signal drive of top module's extension device - negative pole
i	**NOTE!**	**Pin A1 and Pin A2 is only used for the contact signals like drive button and switch etc., and not allowed to drive other devices.**	
A5	X2	1N24	Signal collection point of power start button
A6	X3	1N24	The signal collection point of the front anti-collision bumper is connected in series with the front anti-collision bumper of the robot base.
B1	X4	1N24	The signal collection point of rear anti-collision bumper is connected in series with rear anti-collision bumper of robot base.
B2	X11	/	Normally open contact 1.a of power start button of robot base
i	NOTE!	Pin B2 has the electricity of battery positive pole(＞48V） which is dangerous.	

B3	X14	/	Normally open contact 1.b of power start button of robot base
B4	AF21	2N24	Emergency stop signal output point of the robot base. When the robot base is under the emergency stop status, this point outputs high level (2P24) but low level(2N24) when in normal status.
B5	undefined		
B6	undefined		

Wiring Connection instructions of extension power start button

In order to facilitate the start operation, the additional power start button can be installed on the top module. Please connect the normally open contact of power start button to Pin A2 and 1.b to Pin A5 to realize the extension of power start button. Please refer to circuit diagram 8-14.

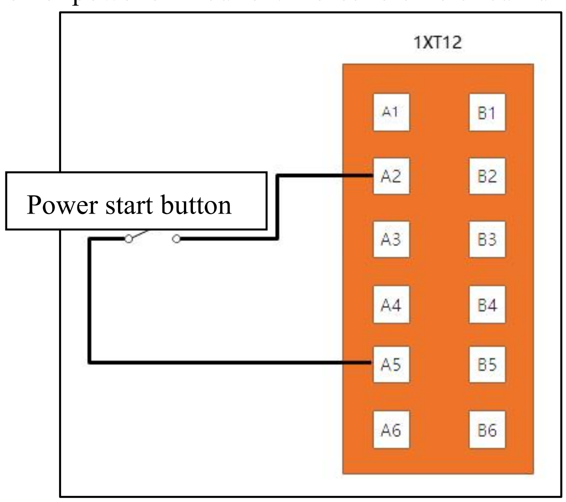


Figure 8.14 Power start button extension circuit diagram

Wiring Connection instructions of extension anti-collision bumper

To improve the safety of top module, the anti-collision bumper's function can be extended on the top module. The anti-collision bumper can be classified as rear and front anti-collision bumper as per the installation position to distinguish the signal source. Please connect the normally closed contact pair 1.a of the front anti-collision bumper to Pin A2 and 1.b to Pin A6. Please connect the normally closed contact pair of the rear anti-collision bumper to Pin A2 and 1.b to Pin B1 to complete its extension. Please refer to circuit diagram 8-15.

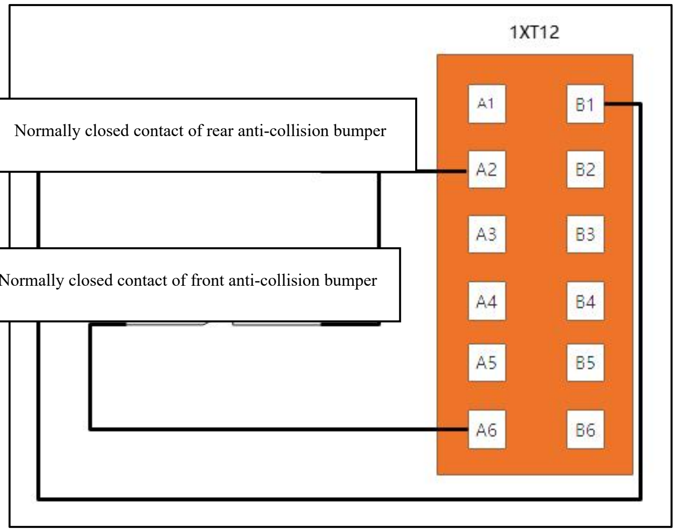


Figure 8.15 Anti-collision bumper extension circuit diagram

Wiring connection instructions of power button extension

In order to control the robot base's power-on/off, the power button of robot base can be set up on the top module. Please connect the normally open contact of power on button to pin B2 and 1.b to Pin B3 to power on the robot base and the extended battery of top module. Please refer to figure 8.16.

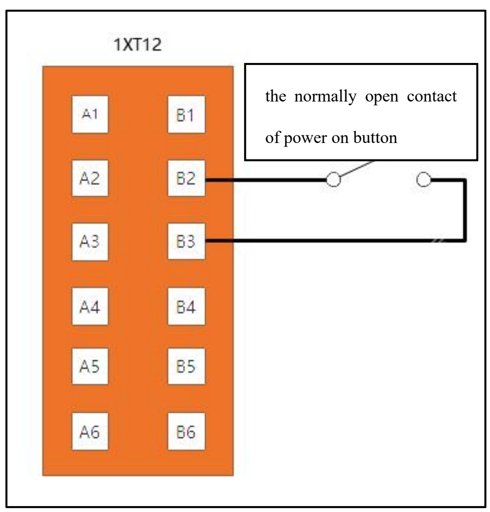
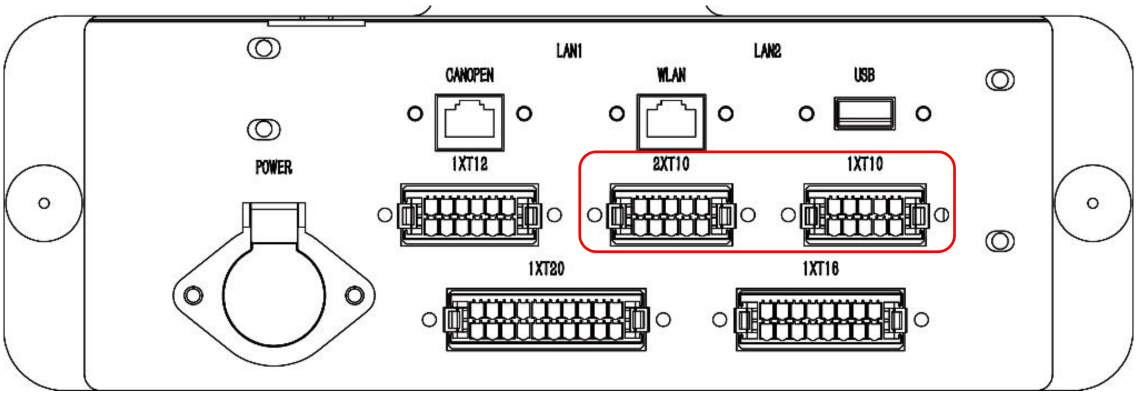


Figure 8.16 Circuit diagram of extended power-on off switch

8.3.7 PORT 1XT10 AND 2XT10 WIRING CONNECTION INSTRUCTION



1XT10 and 2XT10 ports contain extension ports of the emergency stop button and lidar which can extend the emergency stop button and safety obstacle avoidance lidar to improve the safety or make it easier to operate. Meanwhile, in order to facilitate troubleshooting, the emergency stop button signal sources are classified to robot base serial signal and extension signal of top module. As per lidar installation position, the safety obstacle avoidance lidar signal is classified to front and rear obstacle avoidance signals.

Pin definitions are shown in the following table:

Pin number	Symbol	Electrical potential	Description
1XT10 port, if there is no extension related component, please connect the A1 of 1XT10 to A1 of 2XT10			
A1	AX0	Electrical potential 1N24	The Pin A1 of 1XT10 connects to emergency button and then emergency button connects to A1 of 2XT10
A2	AX1	Electrical potential 1N24	The Pin A2 of 1XT10 connects to emergency button of top module and then emergency button of top module connects to A2 of 2XT10
A3	AX6	Electrical potential 1N24	When the top module needs to integrate with Lidar, it works with relay. The normally closed contact of relay connects to 1XT10-A3 while another terminal of the normally closed contact of relay connects to 2XT10-A3. A3 is the inner space signal (emergency stop area) of front obstacle

			avoidance lidar. When the inner space signal of front lidar is not triggered, the normally closed contact of relay is still normally closed, the robot works normally. When the inner space signal of front lidar is triggered, the normally closed contact of relay turns to normally opened, the robot stops moving.
A4	AX7	Electrical potential 1N24	When the top module need to integrate with Lidar, it works with relay. The normally closed contact of relay connects to 1XT10-A4 while another terminal of the normally closed contact of relay connects to 2XT10-A4. A4 is the middle space signal (stop area) of front obstacle avoidance lidar. When the middle space signal of front lidar is not triggered, the normally closed contact of relay is still normally closed, the robot works normally. When the middle space signal of front lidar is triggered, the the normally closed contact of relay turns to normally opened, the robot stops moving.
A5	AX10	Electrical potential 1N24	When the top module needs to integrate with Lidar, it works with relay. The normally closed contact of relay connects to 1XT10-A5 while another terminal of the normally closed contact of relay connects to 2XT10-A5. A5 is the outer space signal (deceleration area) of front obstacle avoidance lidar. When the outer space signal of front lidar is not triggered, the normally closed contact of relay is still normally closed, the robot works normally. When the outer space signal of front lidar is triggered, the normally closed contact of relay turns to normally opened, the robot decelerates to move until the middle space signal of front lidar is triggered.
B1	AX11	Electrical potential 1N24	When the top module needs to integrate with Lidar, it works with relay. The normally closed contact of relay connects to 1XT10-B1 while another terminal of the normally closed contact of relay connects to 2XT10-B1. B1 is the error signal of front obstacle avoidance lidar. When the error signal of front obstacle avoidance is not triggered, the normally closed contact of relay is still normally closed, the

			robot works normally. When the error signal of front lidar is triggered, the normally closed contact of relay turns to normally opened, the robot stops moving.
B2	AX12	Electrical potential 1N24	When the top module needs to integrate with Lidar, it works with relay. The normally closed contact of relay connects to 1XT10-B2 while another terminal of the normally closed contact of relay connects to 2XT10-B2. B2 is the inner space signal (emergency stop area) of rear obstacle avoidance lidar. When the inner space signal of the rear lidar is not triggered, the normally closed contact of relay is still normally closed, the robot works normally. When the inner space signal of rear lidar is triggered, the normally closed contact of relay turns to normally opened, the robot stops moving.
B3	AX13	Electrical potential 1N24	The Pin A1 of 1XT10 connects to emergency button and then emergency button connects to A1 of 2XT10
B4	AX14	Electrical potential 1N24	The Pin A2 of 1XT10 connects to emergency button of top module and then emergency button of top module connects to A2 of 2XT10
B5	AX15	Electrical potential 1N24	When the top module needs to integrate with lidar, it works with relay. The normally closed contact of relay connects to 1XT10-A3 while another terminal of the normally closed contact of relay connects to 2XT10-A3. A3 is the inner space signal (emergency stop area) of front obstacle avoidance lidar. When the inner space signal of front lidar is not triggered, the normally closed contact of relay is still normally closed, the robot works normally. When the inner space signal of front lidar is triggered, the normally closed contact of relay turns to normally opened ,the robot stops moving.
2XT10 port.If there is no extension related component, please connect the A1 of 2XT10 to A1 of 1XT10			
A1	X0	\	Robot's emergency stop signal collection point.
A2	X1	\	The top module's emergency stop signal collection point.

A3	X6	\	Inner space trigger signal collection point of robot base’s front obstacle avoidance lidar.
A4	X7	\	Middle space trigger signal collection point of robot’s front obstacle avoidance lidar.
A5	X10	\	Outer space trigger signal collection point of robot’s front obstacle avoidance lidar.
B1	X11	\	Error trigger signal collection point of robot base’s front obstacle avoidance lidar.
B2	X12	\	Inner space trigger signal collection point of robot base’s rear obstacle avoidance lidar.
B3	X13	\	Middle space trigger signal collection point of robot base’s rear obstacle avoidance lidar.
B4	X14	\	Outer space trigger signal collection point of robot base’s rear obstacle avoidance lidar.
B5	X15	\	Error trigger signal collection point of robot base’s rear obstacle avoidance lidar.

Wiring Connection instructions of extension emergency stop button

Please connect one terminal of two groups normally closed contacts of the emergency stop signal in parallel, and connect the terminal1.a in parallel to the A1 pin of the 1XT10 port, and 1.b to the A1 pin of the 2XT10 port to complete the emergency stop button extension of robot base. Please connect the 1.a in parallel to pin A2 to 1XT10,1.b to pin A2 of 2XT10 to complete the emergency stop button extension of top module. Please refer to the circuit diagram 8.17.

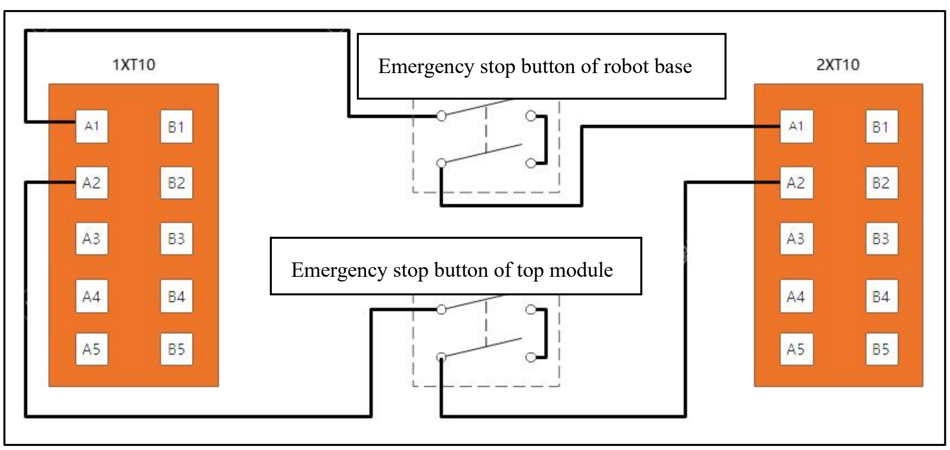


Figure 8.17 Emergency stop button extension circuit diagram

Wiring Connection instructions of extension safety obstacle avoidance lidar

The normally closed contact 1.a triggered by inner space of front lidar to 1XT10-A3 and 1.b to 2XT10-A3. The normally closed contact 1.a triggered by middle space of front lidar connects to

1XT10-A4 and 1.b connects to 2XT10-A4. The normally closed contact 1.a triggered by outer space connects to 1XT10-A5 while another terminal 1.b connects to 2XT10-A5. The normally closed contact 1.a triggered by error signal connects to 1XT10-B1 and 1.b connects to B1 of 2XT10. Then the extension obstacle avoidance of front lidar can be completed. The rear lidar's wiring connection works in the similar way connecting to B2, B3, B4 and B5 pins respectively. Please refer to circuit diagram 8.18.

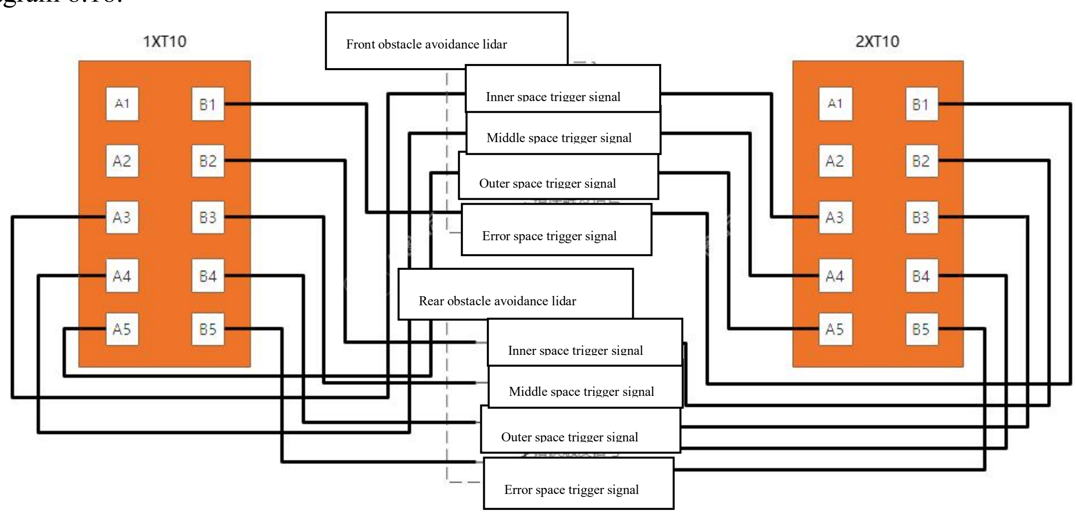


Figure 8- 18 Safety obstacle avoidance lidar extension circuit diagram

CHAPTER9 TOP MODULE INTEGRATION EXAMPLE

9.1 COBOT INTEGRATION

Cobot integration is that the cobot, vision, effector and other components are integrated on the autonomous mobile robot. This chapter introduces electrical components wiring and control methods including the cobot and its control unit, extension power switch, emergency stop button and obstacle avoidance sensor and other components. The top module structure design will not be illustrated here as there are various of top module designs. Please refer to network topology diagram 9.1 and power supply topology diagram 9.2.

Notice! The power of the top module extension port is limited. Therefore, it is recommended that the power of 24V switching power supply is not greater than 240 watts, and the power of the 12V switching power supply is not greater than 60 watts.

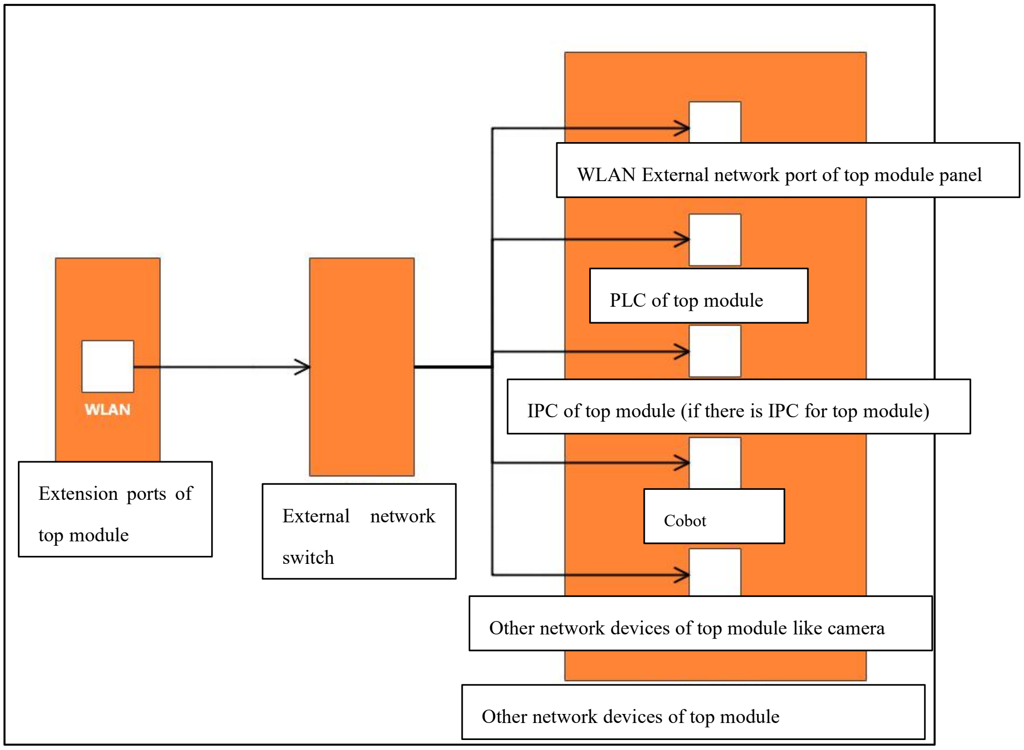


Figure 9.1 Network topology diagram of top module

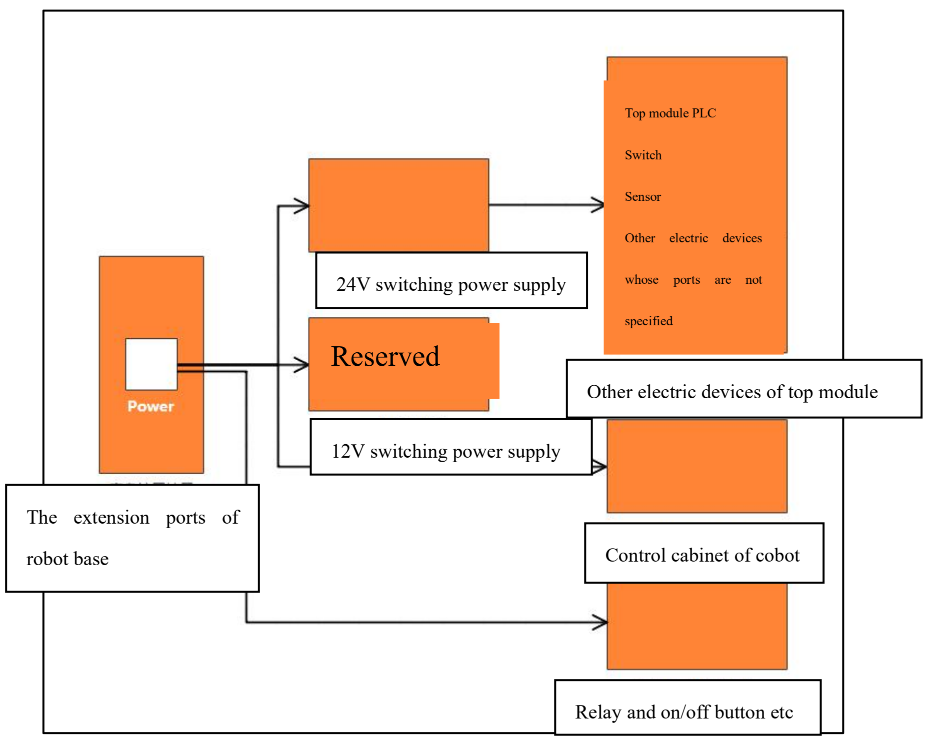


Figure 9.2 Power supply topology diagram of top module

9.1.1 COBOT CIRCUIT EXAMPLE

In order to realize the cobot control and ensure the safety of the whole robot, the cobot needs to be connected to power, network, remote on/off, protective stop and emergency stop signals of robot base etc., Please refer to the **Wiring Connection instructions of extension emergency stop button** for its connection method. The remote on/off signal can power on the cobot and the robot base at the same time. Please refer to 8.3.5《Remote on/off cobot wiring connection》.

The protective stop signal of top module can interlock the cobot with the robot base. The interlock function can be realized through IO signal output by the corresponding function codes defined by the user. The IO signal communicates between the PLC of top module and the robot base's registers. Safety interlocking function refers that the robot base locks with the cobot. When the cobot enters the movement status, its output signal makes the robot base enter the protective stop state, so as to avoid collision and other accidents caused by the movement of the robot base or the movement of the cobot when the robot base is still in motion.

Please refer to the following detailed method and figure 9.3 of how to realize the safety interlocking.

(1) Please configure the input and output points of the protective stop signal of top module's PLC.

(2) Please configure the cobot's protective stop drive points of top module's PLC.

(3) Please configure the protective stop signal input and output points of robot base's 1XT20 ports.

(4) The cobot communicates with top module's PLC through the network. When the cobot isn't finishing the movement, it informs the top module's PLC to output the protective stop signal and then robot base informs the top module's PLC that the robot base stops moving after the robot base enters the protective stop status. The PLC cancels the cobot protective stop drive to allow the cobot to move after the PLC receive the robot base's stop signal.

(5) When the cobot finishes the movement, it informs the top module's PLC to release the protective stop signal. The top module's PLC informs the robot base's PLC to release the protective stop status. After the robot base informs the top module's PLC the protective stop status has been released, the top module's PLC drives the cobot to enter into protective stop status, which cancels the interlocking.

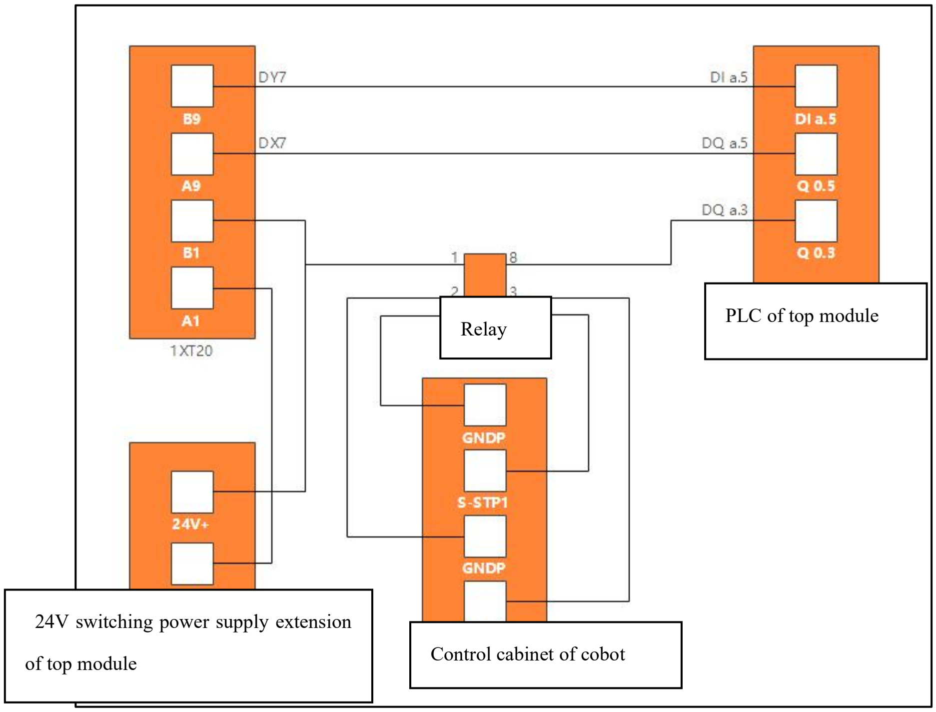


Figure 9.3 Circuit diagram of Safety interlocking function

Please note that the pin number marked on the circuit diagram except 4 pin numbers of 1XT20 ports can't be changed but other pin numbers should correspond to actual pin numbers of right functions.

The top module's PLC is Siemens-S7-1200, relay is IDEC- SJ2S-05B, 24V switching power supply is MEANWELL-DDR-240C-24, cobot is Elite-EC66Mfor reference in the circuit diagram.

9.1.2 EMERGENCY STOP CIRCUIT EXAMPLE

In order to ensure the safety of the top module, top module needs to extend the emergency stop switch or be installed with the sensors such as obstacle avoidance lidar and ultrasonic waves etc. This section takes the emergency stop switch as an example for reference. For the wiring methods of other safety components such as obstacle avoidance lidar and ultrasonic sensor, please refer to the circuit diagram example in this section and the previous interface introduction content as connecting to the PLC, and then to the extension ports of robot base, which realizes the unified and efficient linkage control. Meanwhile, the obstacle avoidance lidar can be classified to front and rear lidar as per installation positions and connected to the PLC of top module.

When top module extends many emergency stop switches, please connect the normally closed contacts of emergency stop button in series and then connect to the corresponding ports of 1XT10 and 2XT10. Meanwhile, the circuit can be separated and connected to top module's PLC and cobot through relay to realize linkage control. Please refer to circuit diagram 9.4. Top module's PLC is siemens-S7-1200, relay is Schneider-RXM4GB2BD,24V switching power supply is MEANWELL-DDR-240C-24, and cobot is Elite-EC66Mfor reference in the circuit diagram.

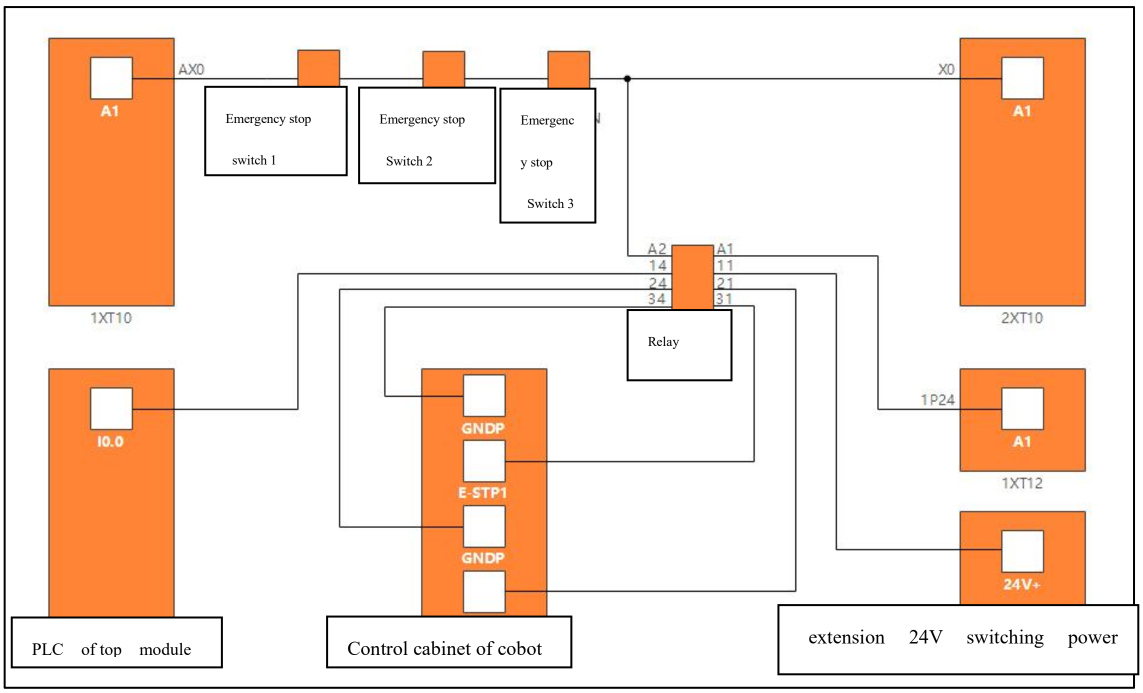


Figure 9.4 Emergency stop circuit diagram of top module

Please note the pin number marked in this circuit diagram except that the 1P24 has to be connected into, other pin numbers can be replaced with the pin numbers with same functions, especially that the pin numbers are marked for PLC and cobot's control cabinet because the pin numbers are marked differently as per different brands cobot and their control cabinets. Different PLC brands, specification and control programming determines that the PLC's pin number are marked differently. Therefore, the pin number should be adjusted as per actual situation.

CHAPTER10
DAILY USAGE AND MAINTENANCE

10.1 DAILY CHECK AND MAINTENANCE

When the system is shut down and restarted, the following safety inspections need to be conducted. (at least once every 24 hours for the 24-hour continuously running system).

Number	Component	Maintenance task
1	Emergency stop button	Normally, when the button is pressed, the robot will stop running immediately.
2	Anti-collision bumper	Normally, when the anti-collision gets touched, the robot will stop running immediately.
3	Lidar	Normally, when an obstacle enters the lidar's obstacle avoidance area, the robot will stop running.
4	Lighting system	When testing the safety device, the status and function of the robot lighting system and speaker system should be checked at the same time.

10.2 WEEKLY CHECK AND MAINTENANCE

Please perform the following maintenance tasks once a week

Number	Component	Maintenance task
1	Top plate and two sides of robot base	The wet cloth should be used to clean the outside of the robot. Please don't use the compressed air to clean the robot
2	Laser scanner	Please clean the optical outer cover of the scanner for optimal performance. Please use approved detergent. Please avoid using corrosive or abrasive detergent.
3	Castor	Please wipe the dirt with a wet cloth and make sure there are no debris on the castors.
4	Driving wheel	Please wipe the dirt with a wet cloth and make sure there are no debris on the wheels.

10.3 COMMONLY DAMAGED PART LIST OR CONSUMABLE ITEM LIST

Some parts of the robot are commonly damaged parts or consumables, and the parts name and service cycle are as follows:

Number	Component	Service life and replacement cycle
1	Universal wheels of robot base	Please check the wheels once every six months as per wheel wear status. Please replace the wheels as per its actual requirements.
2	Driving wheels of robot base	Please check the wheels once every six months as per wheel wear status. Please replace the wheels as per its actual requirements.
3	Robot battery	Cycle life ≥2000 times, battery capacity 80% after 2000 times Replacement cycle is 2 years
4	Laser scanner	Please check the scanner once a month and replace it as required.
5	Emergency stop button	Please check whether the button is working properly and check the mechanical safety - emergency stop function once every 3 to 4 months according to EN/ISO 13850.
6	Charging contact plate	Please check the charging contact plate once every six months to see if it is in good condition. Please replace it as required.

The above commonly damaged parts or consumables need to be replaced whenever the service life or replacement cycle reaches the working life.

10.4 OPERATION TIME AND MAINTENANCE REQUIREMENTS

The robot needs to be maintained and inspected regularly, and after a certain period of operation it needs to be inspected or maintained:

Number	Component	Maintenance cycle	Maintenance task
1	Universal wheel	Two months	Please remove dust and debris
2	Driving wheel	Two months	Please remove dust and debris

CHAPTER11 MAINTENANCE AND DISPOSAL

11.1 MAINTENANCE

All safety instructions in this manual must be strictly followed in maintenance service work.

The maintenance, calibration and repair work must be carried out according to the latest service manual.

The maintenance must be performed by an authorized system integrator or Shenzhen Youibot Robotics Co., Ltd. When the parts need to be returned to Shenzhen Youibot Robotics Co., Ltd., they should be operated in accordance with the provisions of the user manual (see carrying and cautions for details).

It should ensure the safety level specified for maintenance and repair work, comply with valid national or regional work safety regulations, and must test whether all safety functions can work properly.

The purpose of maintenance work is to ensure that the system is working properly, or assist in restoring it to a normal state if the system fails. The maintenance includes troubleshooting and actual maintenance.

11.2 DISCARD DISPOSAL

The P200 must be disposed of in accordance with applicable national laws and regulations and national standards.

CHAPTER12
QUALITY ASSURANCE

12.1 PRODUCT QUALITY GUARANTEE

The P200 robot has a limited warranty period of 12-month.

If the new equipment and its components are found to be defective due to manufacturing or material defect within 12 months (if including transportation time, the longest time is not more than 15 months), Shenzhen Youibot Robotics Co., Ltd. shall provide necessary spare parts to replace or repair the relevant parts.

The equipment or components that are replaced or returned to Shenzhen Youibot Robotics Co., Ltd. shall be owned by Shenzhen Youibot Robotics Co., Ltd.

If the product is no longer within the warranty period, Shenzhen Youibot Robotics Co., Ltd. reserves the right to charge the customer for replacement or maintenance costs.

If the equipment is defective out of the warranty period, Shenzhen Youibot Robotics Co., Ltd. shall not be liable for any damage or loss arising thereof, such as loss of production or damage to other manufacturing equipment.

12.2 DISCLAIMER

If the equipment defect is caused by improper usage or failure to follow the relevant information set out in the user manual, then the “product quality assurance” becomes invalid. The malfunction caused by the following conditions is not covered by this warranty:

- It does not meet industrial standards or does not install, wire or connect other control equipment as required as per the user manual;
- It exceeds the specifications or standards listed in the user manual;
- The product is used for purposes instead of those specified one;
- The storage mode and working environment exceeds the specified range in the user
- manual (such as pollution, salt damage, moisture condensation, etc.);
- The product damage due to improper transportation;

- Damage caused by the accident or collision;
- The assembled parts and accessories are not the original;
- Damage caused by the modification, debugging or maintenance of the original parts
- by a third party other than Shenzhen Youibot Robotics Co., Ltd. or its designated
- integrator;
- Natural disasters such as fires, earthquakes, tsunamis, lightning strikes, high winds
- and floods. In addition to the above, the malfunction is not caused by Shenzhen Youibot
- Robotics Co., Ltd., including but not limited to the following conditions which are not
- covered by the warranty:
- The date of manufacture or the start date of warranty cannot be identified;
- Changes to the software or internal data;
- The malfunction cannot be reproduced or the malfunction cannot be identified by
- Shenzhen Youibot Robotics Co., Ltd.;
- The product can’t be used where there is the radioactive equipment, biological test equipment or for dangerous purpose judged by Shenzhen Youibot Robotics Co., Ltd.;

According to the Product Quality Assurance Agreement, Shenzhen Youibot Robotics Co., Ltd. undertakes the quality assurance only for the flaws and defects in the products and parts sold to distributors. And Shenzhen Youibot Robotics Co., Ltd. shall not be liable for any other express or implied assurances or liabilities, including but not limited to any implied assurance of merchantability or for a particular purpose. Besides, Shenzhen Youibot Robotics Co., Ltd. shall not be liable for any form of indirect damage or consequences arising from the product.

CHAPTER13
CONTACT US

Youibot is committed to helping you achieve rapid success in the field of robotics through P200. If you have any comments or technical questions about the improvement of P200, please leave a message on our official website www.youibot.com or send an email to the official email address info@youibot.com for feedback.

If you need to contact sales for information about P200 or other Youibot products, please log on to our official website for sales contact information.

Thank you for your patience and support!